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INTRODUCTION TO MICROBIOLOGY

CHAPTER NO. 01

Dr. Qazi Adnan

INTRODUCTION TO MICROBIOLOGY

Microbiology:

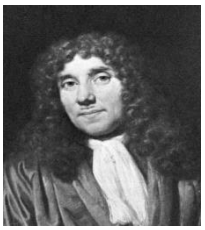
"Study of any organism too small to be seen with the human eye is called as microbiology."

- Microorganisms are Oldest life form on earth. (3.5 billion years ago)

History of Microbiology
1500's -1900's







1500's:

- Before the 1500's most theories of disease were based on superstition.
- No knowledge of microorganisms.
- Thought that you can get disease from another person who was sick but didn't know why.
- Early Observations & Experiments:
  - Microscopes
  - van Leeuwenhoek & Hooke
  - Spontaneous Generation Controversy
  - Germ Theory of Disease & Robert Koch

1600's:

- Robert Hooke and Antoni van Leeuwenhoek started using crude microscopes
- Van Leeuwenhoek observed what we now call bacteria and protists, he called them "animalcules". Called "Father of Microbiology."
- Francesco Redi 1668 Disproved Spontaneous Generation.
- It was thought that organisms arose from inorganic or rotting organic material.

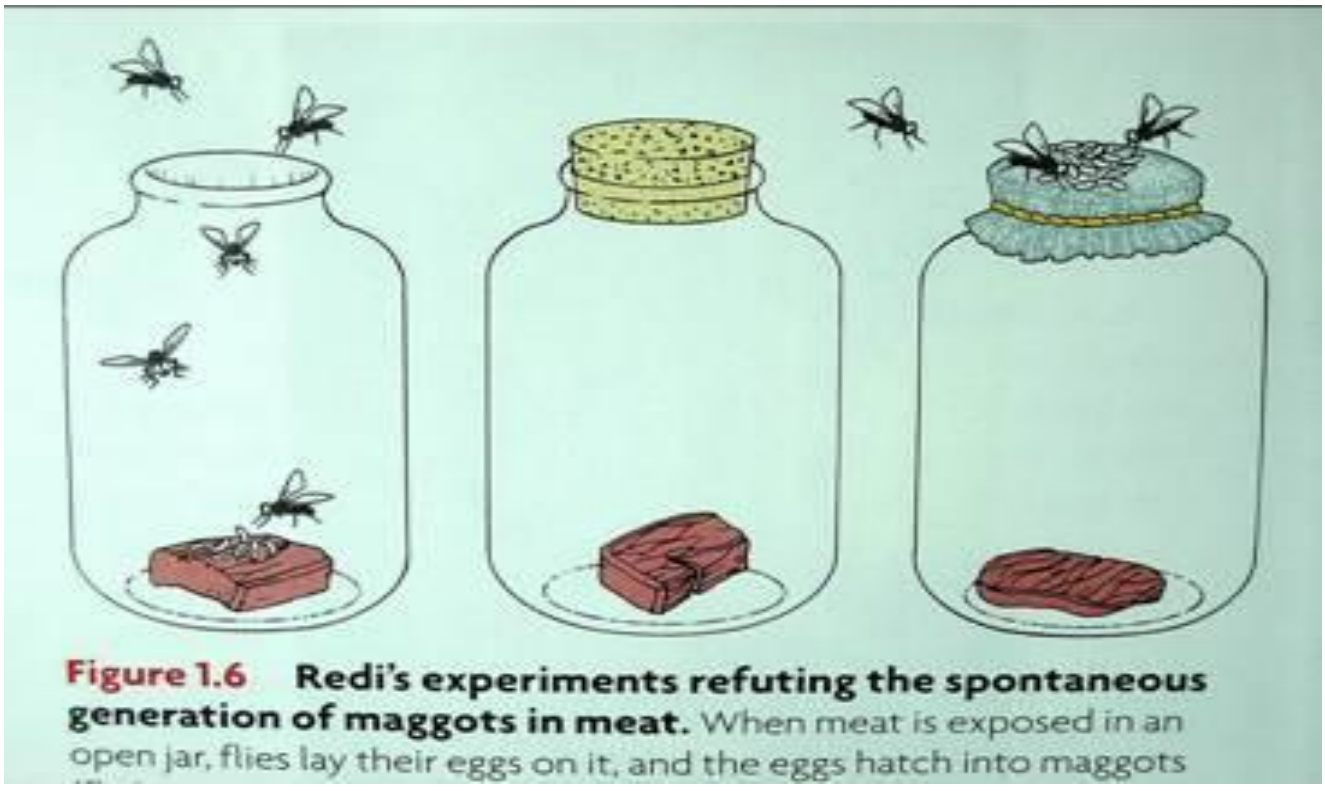


Figure 1.6 Redi's experiments refuting the spontaneous generation of maggots in meat. When meat is exposed in an open jar, flies lay their eggs on it, and the eggs hatch into maggots

1700's

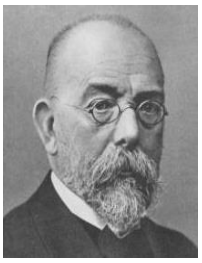


- **Edward Jenner** in 1789- Developed small pox vaccination by using a milder disease, cowpox. He took liquid from the sores a patient with cowpox and put it into a healthy person. Then gave that healthy person liquid from a patient with small pox and discovred that the healthy person didn't get sick. Risky experiment gave us the first way to prevent disease.

1800's More prevention



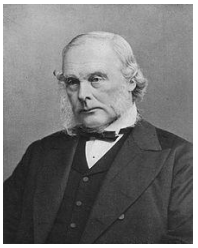
- **Ignaz Semmelweis**, hand washing before surgery could prevent "childbirth fever" i.e. Strep. Drs. would deliver babies w/o washing hands, or after performing autopsies on women who had died from childbirth fever.
- **Louis Pasteur** (1864)-demonstrated that microorganisms are present in air not created by the air- further disproving spontaneous generation. Helped in development of germ theory of disease (microorganisms may be the cause of some or all disease).



- **Joseph Lister:** Father of antiseptic surgery concept, sanitation/hygene procedure (food handlers, water). Connected work of Semmelweis and Pasteur to develop and popularize the chemical inhibition of infection during surgery. (washed surgical wounds with phenol carbolic acid)
- **Germ Theory of Disease**
  - It was proposed by **Robert Koch** in 1876.
  - Bacillus antracis- caused anthrax- could take the blood of infected animals and injected blood to healthy sheep and healthy sheep got the disease.
  - Koch's postulates prove specific bacteria cause a specific disease.
  - Koch's postulates:



1. Microorganism must be present in every case of the disease.
2. The microorganism must be isolated from the diseased host and grown in pure culture.
3. The disease must be reproduced when the pure culture of the microorganism is inoculated into a healthy animal.
4. The organism must be recoverable from the experimental infected host.



1500-1800's

- Disease was caught from someone who was sick.
- Microorganisms exist!
- Disease was caused by a microorganism, that can be transferred from another person.
- **Viruses:**
  - **Dimitri Iwanowski** (1890-1900) first to discover viruses.
  - Tobacco mosaic virus



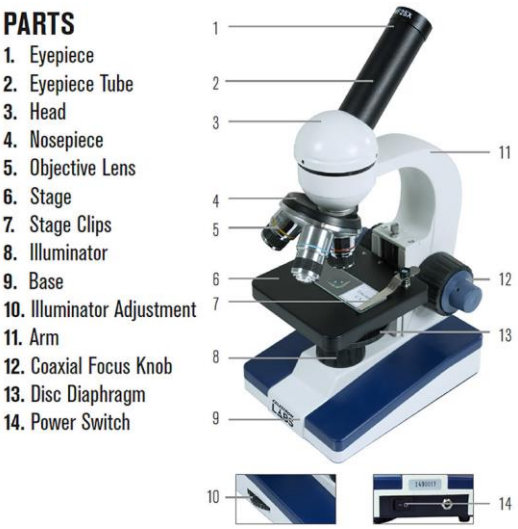
1900's:

- Alexander Fleming 1920's- First antibiotics (penicillin)
- Fleming discovered that a mold accidentally growing on one of his petri dishes had anti-bacterial activity.



MICROSCOPE:

A microscope is an instrument used to see objects that are too small to be seen by the naked eye.



MICROSCOPY

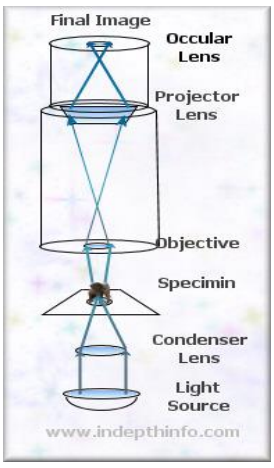
"Microscopy is the science of investigating small objects and structures using such an instrument."

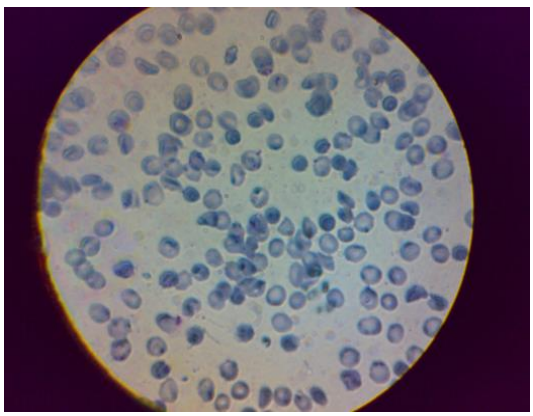
Types of Microscopy:

- Following are the major types of microscopy
  1. Bright field microscopy
  2. Dark field microscopy
  3. Phase contrast microscopy and DIC microscopy
  4. Fluorescent microscopy
  5. Electron microscopy
     - i. Scanning electron microscopy
     - ii. Transmission electron microscopy

1. Bright Field Microscopy:

- produces a dark image against a brighter background
- has several objective lenses
- Commonly used microscopes in labs are bright field microscopes

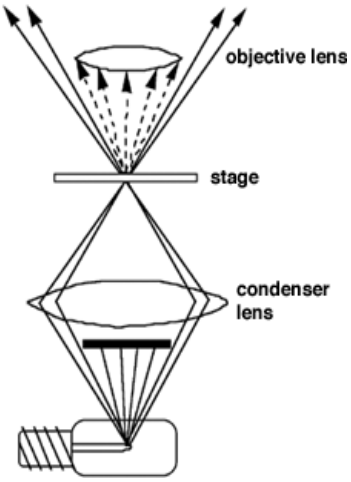


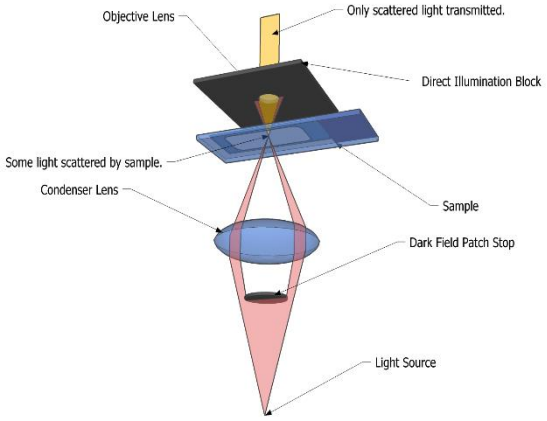


2. Dark Field Microscopy:



- Light enters the microscope for illumination of the sample
- A specially sized disc, the patch stop blocks some light from the light source, leaving an outer ring of illumination.
- A wide phase annulus can also be reasonably substituted at low magnification
- The condenser lens focuses the light towards the sample
- The light enters the sample. Most is directly transmitted, while some is scattered from the sample
- The scattered light enters the objective lens, while the directly transmitted light simply misses the lens and is not collected due to a direct illumination block
- Only the scattered light goes on to produce the image, while the directly transmitted light is omitted





3. Phase Contrast Microscopy:

- Works on the principle of interference of two light rays
- One is the undifracted
- Other is diffracted
- Let S (red) be light passing through medium surrounding sample and D (blue) light interacting with specimen. S and D typically interfere to yield P (green), which is what we can usually detect.
- P will be phase shifted compared to S, but our eyes cannot detect phase shifts.
- Phase contrast microscopy effectively converts this phase shift into an intensity difference we can detect

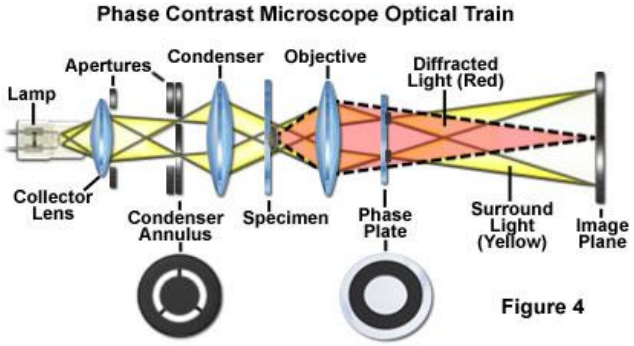


Figure 4

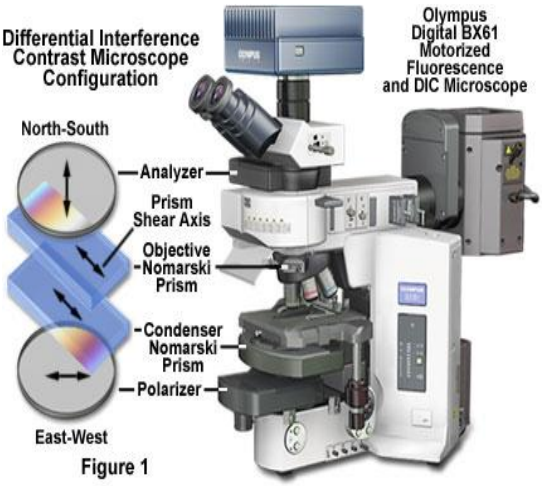


Figure 1

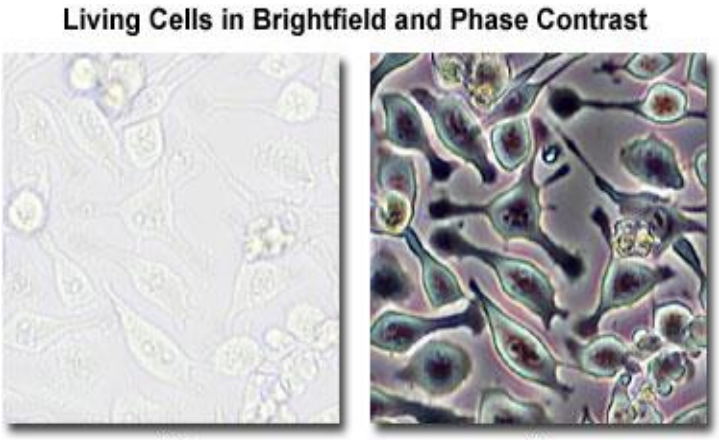


Figure 2

4. **Florescent Microscopy:**
   - Works on the principle of excitation potential of the electrons of a florescent dye
   - Energy is absorbed by the atom which becomes excited.
   - The electron jumps to a higher energy level.
   - Soon, the electron drops back to the ground state, emitting a photon (or a packet of light) - the atom is fluorescing.
   - Florescent dyes called florochromes are used for this purpose

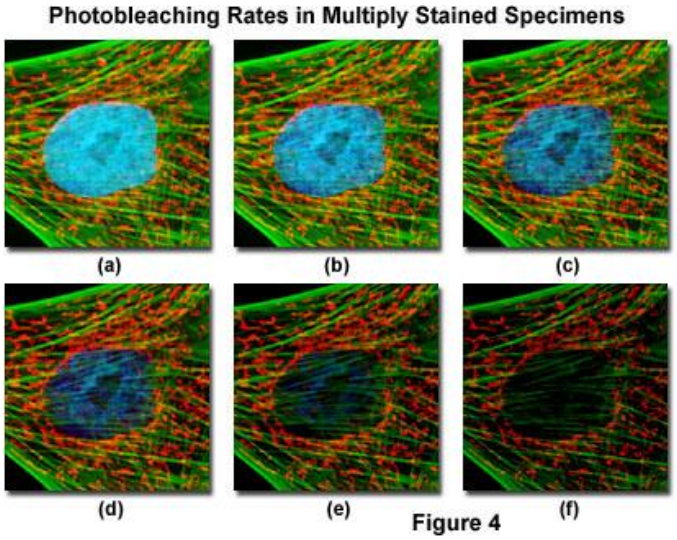


Figure 4

5. **Electron Microscopy:**
   i. Scanning electron microscopy
   ii. Transmission electron microscopy

BACTERIA

CHAPTER NO. 02

Dr. Qazi Adnan

BACTERIA

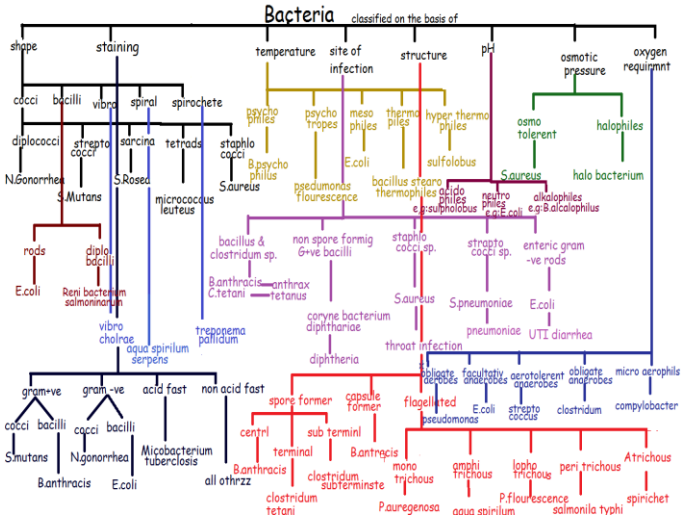
"Bacteria are microscopic single-celled organisms that thrive in diverse environments."

Introduction:

Bacteria are prokaryotes. The entire organism consists of a single cell with a simple internal structure. Unlike eukaryotic DNA, which is neatly packed into a cellular compartment called the nucleus, bacterial DNA floats free, in a twisted thread-like mass called the nucleoid.

Bacterial cells also contain separate, circular pieces of DNA called plasmids. Bacteria lack membrane-bound organelles, specialized cellular structures that are designed to execute a range of cellular functions from energy production to the transport of proteins.

On the outside, bacterial cells are generally surrounded by two protective coverings: an outer cell wall and an inner cell membrane. However, certain bacteria, like the mycoplasmas do not have a cell wall at all. Some bacteria may even have a third, outermost, protective layer called the capsule. Lastly, bacterial surfaces can be covered by whip-like extensions: flagella or pili. According to the authors of "Mims Medical Microbiology, 5th Ed" (Saunders, 2013), long flagella aid in motility while short pili help bacteria to attach to host surfaces.



Classification of Bacteria

Bacteria can be classified on the following basis:

- On the Basis of Oxygen Requirement
- On the Basis of Staining
- On the Basis of pH
- On the Basis of Temperature
- On the Basis of Structure
- On the Basis of Osmotic Pressure
- On the Basis of Shape

ON THE BASIS OF O_2 REQUIREMENT

Bacteria are classified on the basis of O_2 requirement as:

- **Obligate aerobes**
  Bacteria which utilize oxygen for producing ATP are called obligate aerobes
  e.g. *Pseudomonas, Mycobacterium*
- **Facultative anaerobes**
  Bacteria which can produce ATP both by using oxygen or without using oxygen are called as facultative anaerobes
  e.g. *E.coli*
- **Aerotolerant anaerobes**
  Bacteria which don't utilize oxygen for producing ATP but they can protect themselves from reactive oxygen molecules.
  e.g. *Streptococcus pyogenes*
- **Obligate anaerobes**
  Bacteria which don't utilize oxygen for producing ATP and are destroyed in the presence of reative oxygen molecules.
  e.g. *Clostridium, bacteriodes*
- **Microaerophiles**
  Bacteria which require oxygen to survive but requires environment containing lower levels of oxygen that are present in the atmosphere
  e.g. *Campylobacter*

ON THE BASIS OF STAINING

Bacteria are classified on the basis of staining as:

- **Gram positive**
  Gram positive bacteria are the bacteria which are stained dark violet or purple with stain (Safranine or Fuschine)
  e.g *Streptococcus mutans, Staphylococcus aureus, Mycobacterium tuberculosis, Bacillus anthracis, Bacillus subtilis*
- **Gram negative**
  Gram negative bacteria are the bacteria which are stained pink or red with stains (Safranine or Fuschine)
  e.g *Escherichia coli, Salmonella typhi, Shigella dysenteriae*

ON THE BASIS OF TEMPERATURE

On the basis of temperature bacteria are classified as:

- **Psychrophiles**
  Psychrophiles are the bacteria which grow at low temperature
  e.g *Bacillus psychrophilus*
- **Psychrotrophs** e.g *Listeria monocytogenes*
- **Mesophiles**
  Mesophiles are the bacteria which grow at middle or normal temperature range.
  e.g. *E.coli*
- **Thermophiles**
  Thermophiles are the bacteria which grow at high temperature (i.e 40°C-70°C)
  e.g *Bacillus stearothermophilus*
- **Hyperthermophiles**
  Hyperthermophiles ae bacteria which live in extreme hot environment (i.e above 70°C)
  e.g *Sulpholobus*

ON THE BASIS OF STRUCTURE

Bacteria are classified on the basis of structure as:

- **Spore former**
  Bacteria which form spores are called spore forming Bacteria. These are further classified as:
  (a) Centre spore former e.g *Bacillus cereus*
  (b) Terminal spore former e.g *Clostridium tetani*
  (c) Subterminal spore former e.g *Clostridium subterminale*
- **Capsule former**
  Bacteria which form capsule are called capsule forming bacteria.
  e.g *Bacillus anthracis, Streptococcus mutans*
- **Flagella**
  (a) Monotrichous => Bacteria having only single flagella
  e.g *Pseudomona auregenosa*
  (b) Amphitrichous => Bacteria having tuft of flagella on both poles of bacteria
  e.g *Aquaspirillum serpens*
  (c) Lophotrichous => Bacteria having a tuft of flagella on one pole of bacteria
  e.g *Pseudomona flouresence*
  (d) Peritrichous => Bacteria having flagella covering its whole body
  e.g *Salmonella typhi*
  (e) Atrichous => Bacteria having no flagella on their body
  e.g *spirochetes*

ON THE BASIS OF pH

Bacteria can be classified into following types on the basis of pH:

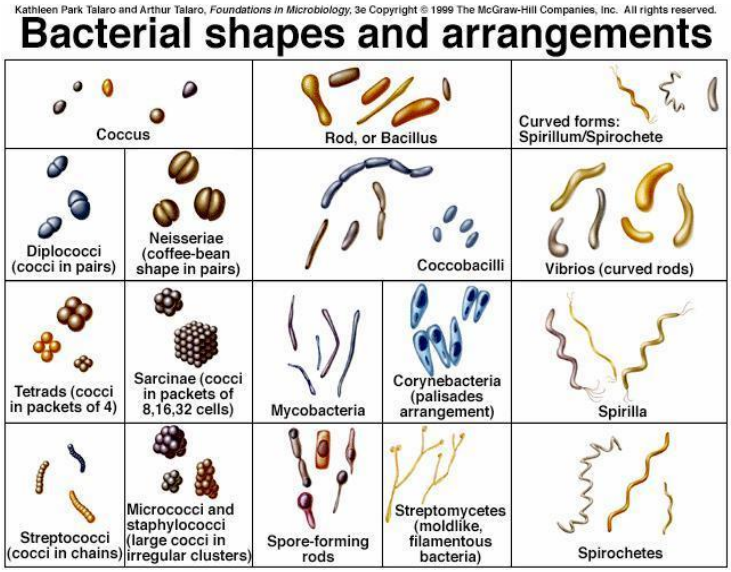
- **Acidophiles**
  Bacteria which live in acidic conditions with pH less than 3
  e.g *Sulpholobus*
- **Neutrophiles**
  Bacteria which live in neutral environment.
  e.g *Escherichia*
- **Alkalophiles**
  Bacteria which are capable of surviving in alkaline (8.5-11) environments
  e.g *Bacillus alkalophiles*

ON THE BASIS OF OSMOTIC PRESSURE

Bacteria can be classified into following types on the basis of osmotic pressure

- **Osmotolerant**
  Bacteria which can tolerate high osmolarity
  e.g *S.aureus*
- **Halophiles**
  Bacteria which live in high salt concentrations
  e.g *Helicobacter pylori*

ON THE BASIS OF SHAPE:



- **Cocci**
  Cocci are round shaped bacteria.
  These are further classified as:
  (a) Diplococci
  e.g. *Neisseria gonorrhoreae*
  (b) Streptococci
  e.g. *Streptococcus mutans*
  (c) Sarcina
  e.g. *Sarcina rosea*
  (d) Tetrad
  e.g. *Micrococcus luteus*
  (e) Staphylococci
  e.g *Staphylococcus aureus.*
- **Bacilli**
  Baccill are rod shaped bacteria. These are further classified as:
  (a) Diplobacilli e.g *Moraxella lacunata*
  (b) Coccobacilli e.g *Bordtella pertussis*
- **Vibrio**
  Vibrio are curved rod shaped (comma shaped) bacteria
  e.g *Vibrio cholerae*
- **Spiral**
  These are twisted shaped bacteria
  e.g *Aquaspirillum serpens*
- **Spirochete**
  These are long and slender bacteria and are tightly coiled and seem like miniature springs or telephone cords
  e.g *Treponema pallidum*

MICROBIAL STAINING

Why we Stain Microbes?

We stain microbes for following reasons:

- To study their structure, shape and other structural characteristics.
- Structures have to be contrasted from their environment so that they can be easily seen.
- To divide microbes in to specific groups for diagnostic purpose, biological stains and staining procedure in conjugation with light microscopy have become major tools in microbiology.

STAIN

Stains or dyes are chemicals with two parts

- **Chromophore**
  The part that has the color.
- **Auxochroem**
  The other part that helps in the coloring properties.

Types of STAIN:

The chromophore of a stain maybe charged either positive or negative and are classified as:

- **Acidic stains**
  A negatively charged chromophore dye (**acidic stain**)
- **Basic stains**
  A positively charged chromophore stain (**basic stain**)

ACIDIC STAIN

- Eosin red
- Nigrosin
- INDIAN INK
- Picric acid yellow
- Congo red

BASIC STAINS

- Malachite green
- crystal violet
- methylene blue
- safranin
- basic fuschin (magenta dye)

STAINING:

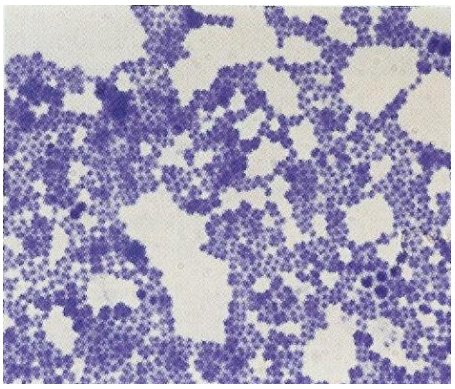
Staining is one such contrasting process where in chemicals (stains) are used to impart their color to the cells structures with which they come in contact.

Types of staining

There are following types of staining:

- Simple staining
- Differential staining
- Special staining
- General purpose staining

➢ Simple Staining:



- It is the most valuable stain for staining bacteria.
- It is excellent for the genus *Corynebacterium*
- In sporing bacilli stained with this stain the spore appear as unstained bodies within blue cells.

Procedure:

1- Stain for 1 min.
2- Rinse with water.
3- Dry it

Simple Staining of Yeast:



Cultures

- Yeast culture

Reagents

- Methylene blue (**1 to 2 minutes)**
- crystal violet (**20 to 60 seconds)**
- carbol fuschin (**15 to 30 second)**

YEAST:

- *Candida albicans*
- *Sacchromyces cerevisiae*
- *S. anamensis*
- *Schizosaccharomyces pombe*

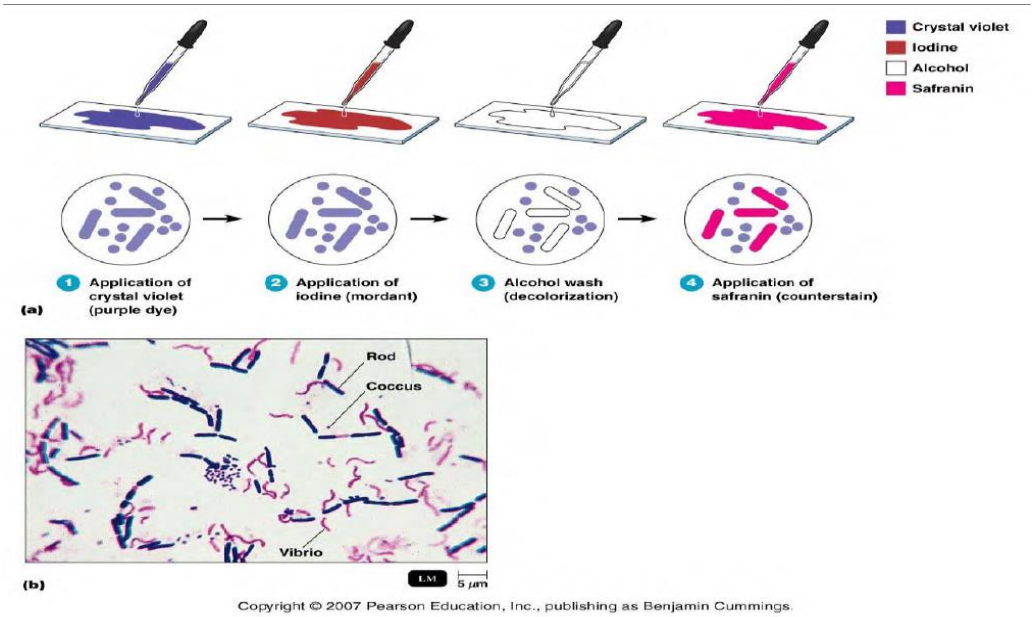
➢ Differential Staining

- Staining procedure which differentiates or distinguishes between two or more types of bacteria is termed as differential staining technique.

There are two types of differential staining;

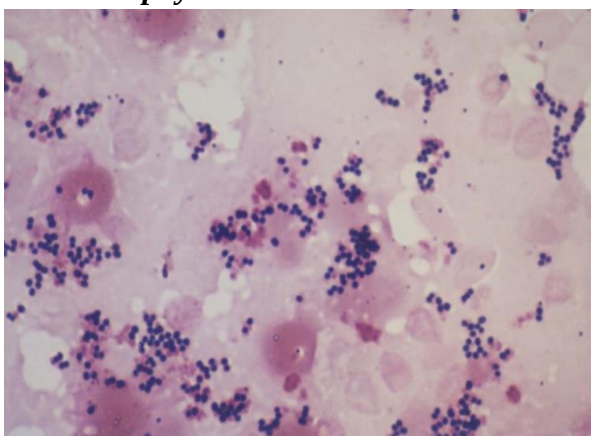
- ✓ Gram staining
- ✓ Acid fast staining

- ✓ **Gram Stain:**
  1. Crystal Violet
  2. Grams Iodine (mordant)
  3. Alcohol
  4. Safranin

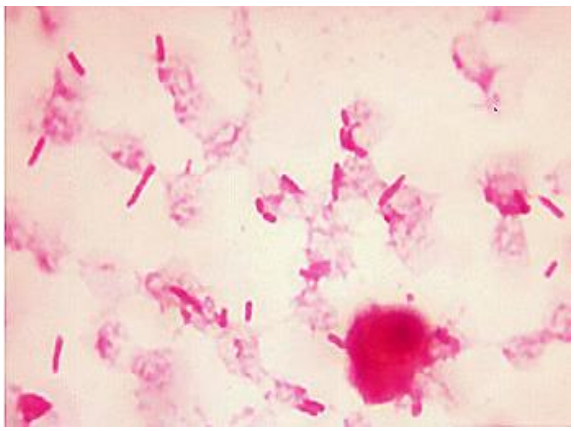


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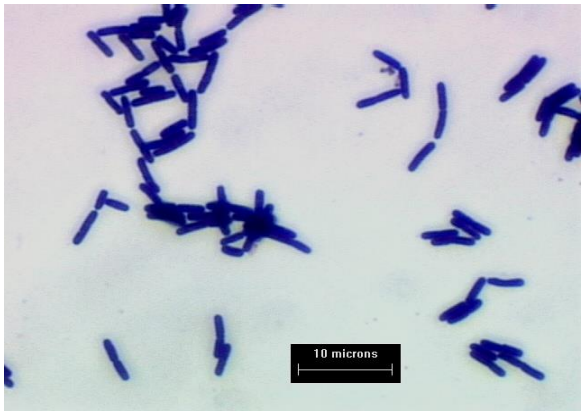
Staphylococcus aureus



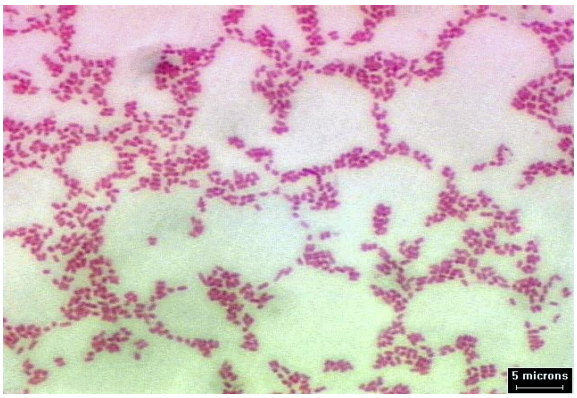
Escherichia coli



Bacillus cereus

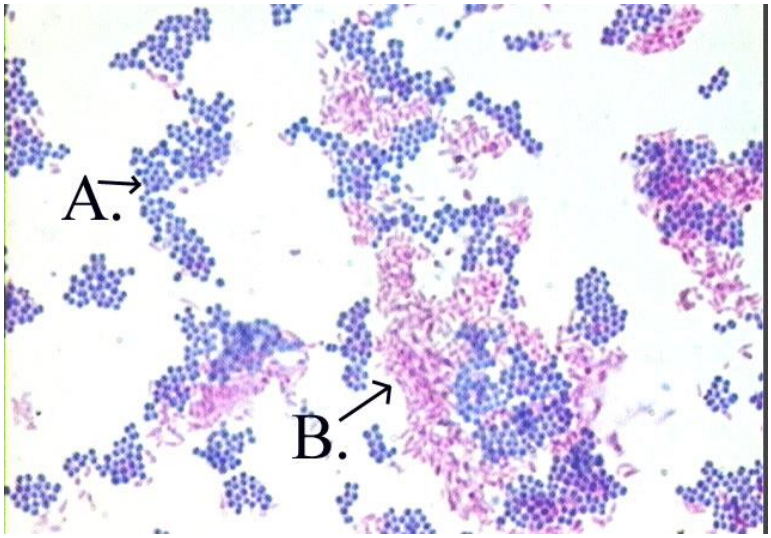


Citrobacter



✓ **ZIEHL-NEELSEN'S METHOD (Acid Fast Stain)**
It is used for examination of Mycobacterium.

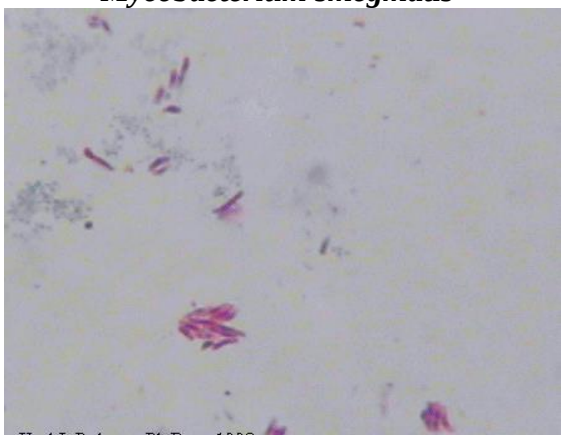
Acid fast and non-acid fast bacteria



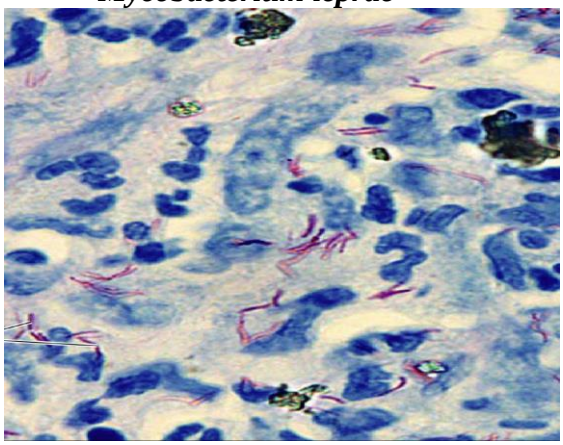
Procedure:

- Flood the slide with strong carbol fuchsin and heat until steam rises (but do not boil)
- Decolourize in acid-alcohol
- counter stain with Methylene blue
- Acid-fast organisms are red, other organisms are blue

Mycobacterium smegmatis



Mycobacterium leprae

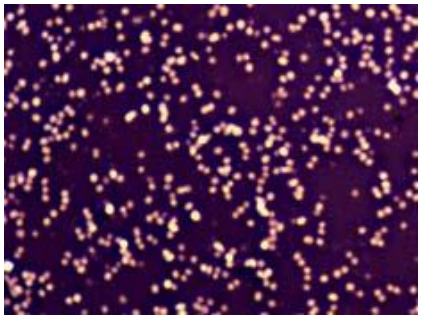


➢ Special Straining:

- Negative staining
- Capsule
- Endospore
- Flagellum
- Pili

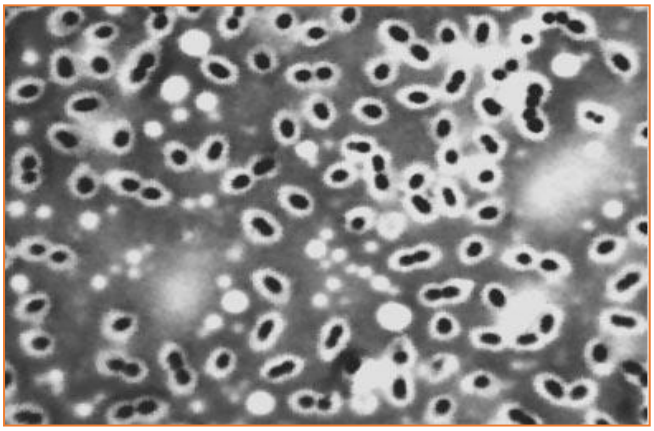
- **Negative Staining:**

In negative staining technique the background is stained, and bacteria remains colorless. When this preparation is examined microscopically with low intensity of light a phase like effect is observed. Capsules appear as halo or light ring around each cell against dark background.
Nigrosine stain is used for negative stain. E.g. Negative stain of cocci.

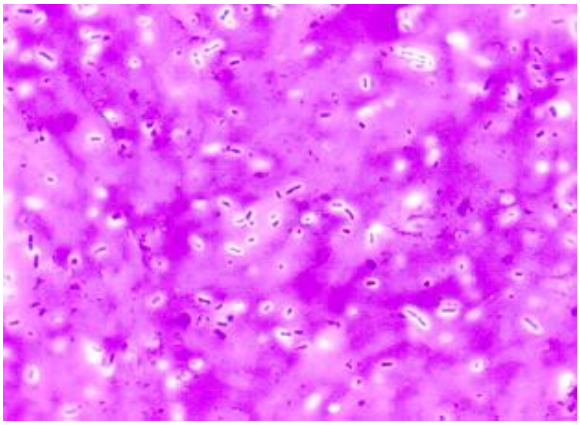


- **Capsule Staining (MUIR'S METHOD)**
  - Make smear
  - Apply crystal violet
  - Rinse with copper sulphate

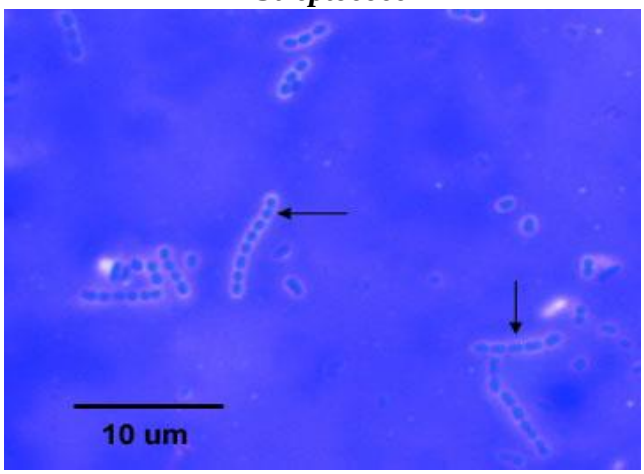
Transparent Capsule around Cells



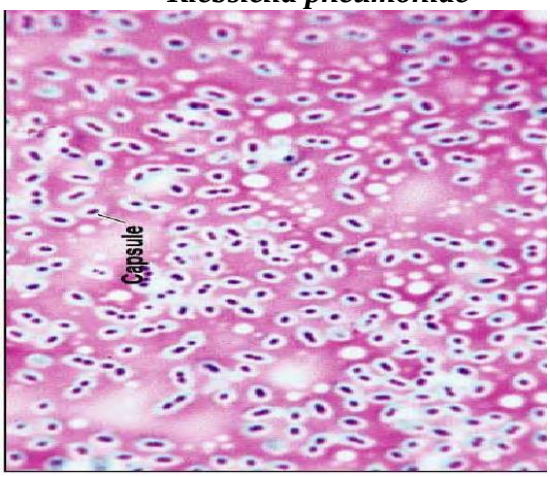
Streptococcus lactis



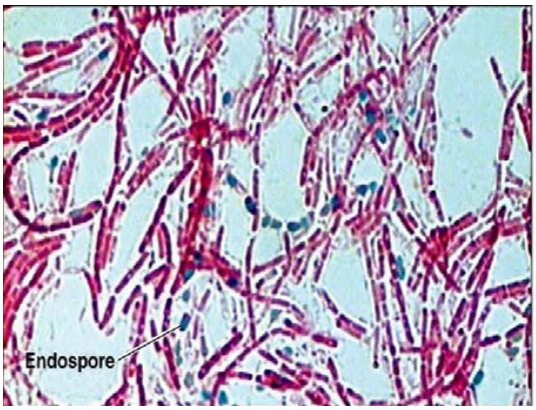
Streptococci



Klebsiella pneumoniae



✓ **Spore Staining:**
- Make smear
- Apply Malachite green
- Apply steam
- Counterstain with safranin





Bacillus subtilis

Morphology of Bacterial Cell

The Structure of a bacterial cell consists of:

1- Capsule or Glycocalyx
2- Flagella
3- Fimbriae and Pili
4- Bacterial Endospore
5- Bacterial Cell Wall

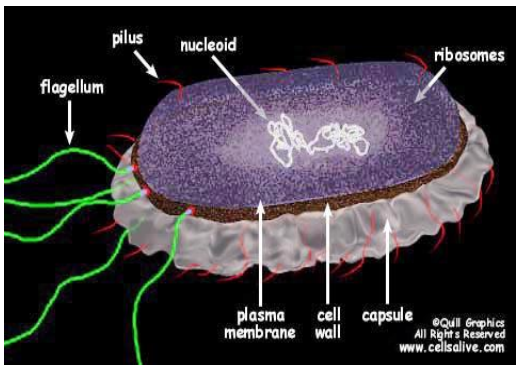
1- Capsule or Glycocalyx:

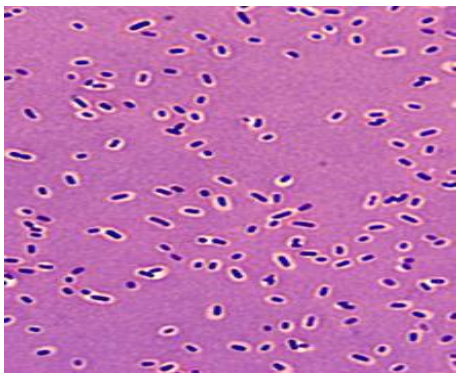
- Some bacterial cells have the capability of secreting a substance around their surface in the form of coat called glycocalyx.
- If this substance is organized and firmly attached to the bacterial cell wall then this glycocalyx is called as a capsule.
- If the substance is lossely attached to cell wall then this glycocalyx is called as slime layer.

Composition:

Capsule is composed of two types of sugars

- **Homopolysaccharides:** If sugar subunits are same then it is homopolysaccharide e.g, capsule of *Streptococcus mutans* contain glucan from sucrose
- **Heteropolysaccharides:** If the sugar base is not identical then it is heteropolysaccharide e.g, capsule of *Klebssiella pneumoniae*

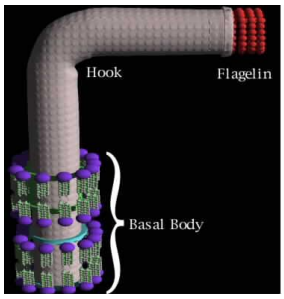




Functions of Capsule

1- Prevents from drying of cell
2- May block the attachment of bacteriophage
3- Prevent phagocytosis
4- Help in attachments
5- Prevents from aggregation with other bacteria
6- They also exclude bacterial viruses and most hydrophobic toxic materials such as detergents.

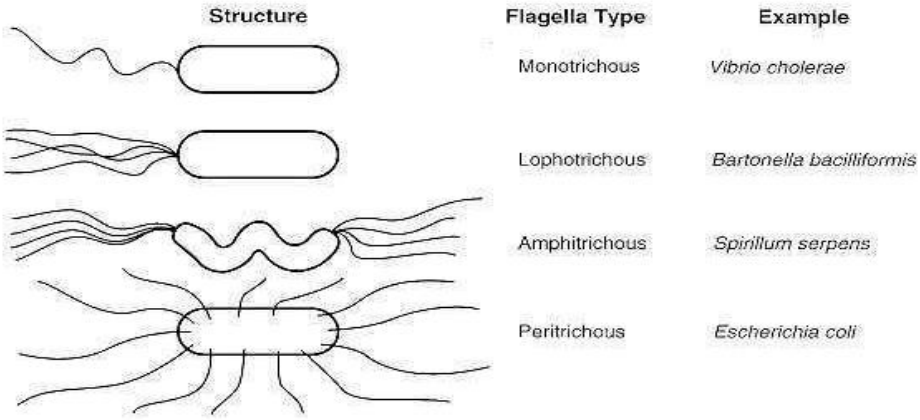
2- Flagella:



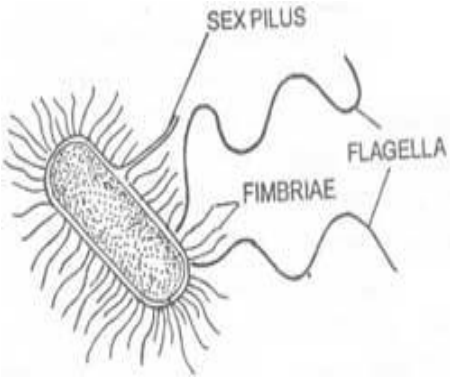
- The word *flagellum* in Latin means whip
- The role of the flagellum is locomotion but it also often has function as a sensory organelle, being sensitive to chemicals and temperatures outside the cell
- Structurally, bacterial flagella are long, filamentous surface appendages about 12 to 30 nm in diameter
- Length: 3 to 12 µm

Composition:

- A flagellum consists of three parts:
  - The long filament, which lies external to the cell surface
  - The hook structure at the end of the filament
  - The basal body, to which the hook is anchored and which imparts motion to the flagellum.
- Chemically, flagella are constructed of a class of proteins called flagellins
- The hook and basal-body structures consist of numerous proteins. Mutations affecting any of these gene products may result in loss or impairment of motility



3- Fimbriae and Pili:

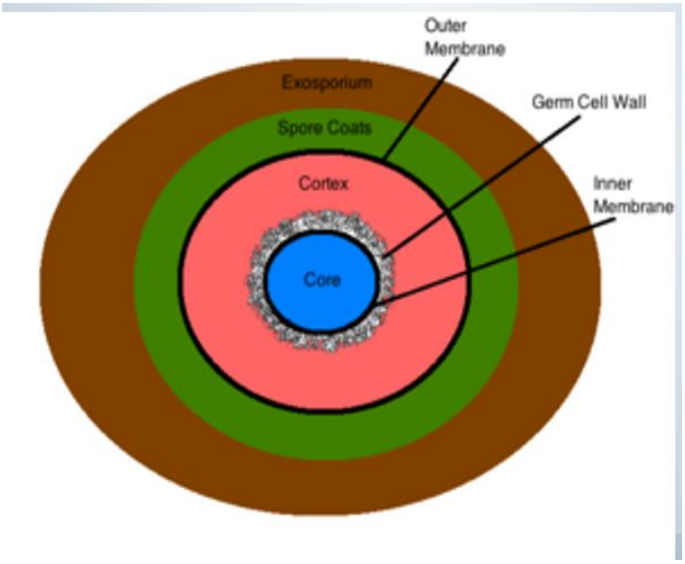


- Many gram-negative bacteria contain hairlike appendages that are shorter,straighter,and thinner than flagella and are used for attachment rather than for motality.
- These structures consist of a protein called *pilin* arranged helically around a central core,are devided into two types.
- Fimbriae and pili having very different functions.
- Fimbriae(*singular:fimbria*) can occur at the poles of the bacterial cell,or they can be evenly distributed over the entire surface of the cell.
- They can number anywhere from a few to several hundred per cell.
- Like the Glycocalyx,fimbriae enable a cell to adhere to surfaces,including the surface of other cell.
- For example,fimbriae attached to the bacteria *Neisseria gonorrhoeae* help the microbe colonize the mucos membranes.
- Once colonization occurs,the bacteria can cause disease.

- When fimbriae are removed (*genetic mutation*) then colonization cannot happen, and disease is not caused.
- Pili (*singular:pilus*) are usually longer than fimbriae and number only one or two per cell.
- Pili join bacterial cells in preparation for transfer of DNA from one cell to another.

4- Endospore:

- Bacterial endospores are special tough, dormant and resistant spores produced by some Gram-positive bacteria of Firmicute family during unfavorable environmental conditions.
- Endospores are developed within the vegetative cells (hence the name, endo = inside).



5- Bacterial Cell Wall:

- It is necessary to explain the cell wall of Gram +ve bacteria and Gram –ve bacteria

i. Cell wall of Gram +ve Bacteria:

- It is composed of maily two components
  - Peptidoglycan
  - Techoic acid
- 95% of the gram +ve cell wall is composed of peptidoglycan
- **E.g.** *Staphylococcus aureus, Streptococcus pneumoniae, Staphylococcus epidermidis, Clostridium perfringens, Streptococcus mutans, Micrococcus luteus, Clostridium botulinum, Clostridium tetani, Streptococcus mitis, Streptococcus salivarius* etc.

ii. Cell wall of Gram –ve bacteria:

- The cell wall of gram –ve bacteria is composed of
  - Peptidoglycan 5-10%
  - Lipopolysaccharide (Outer membrane)
- **E.g.** *Escherichia coli, Klebsiella pneumoniae, Haemophilus influenzae, Salmonella enterica, Neisseria gonorrhoeae, Enterobacter cloacae, Legionella pneumophila, Proteus vulgaris, Helicobacter pylori, Treponema pallidum* etc.

Peptidoglycan:

- Peptidoglycan, also known as murein, is a polymer consisting of sugars and amino acids
- The sugar component consists of alternating residues of, *N*-acetylglucosamine and *N*-acetylmuramic acid.
- Amino acid chain mostly consists of L-alanine, D-glutamic acid and D-alanine.
- **Functions:**
    - It provides the streangth to the bacterial cell
    - Some antibacterial drugs such as penicillin interfere with the production of peptidoglycan by binding to bacterial enzymes known as penicillin-binding proteins or transpeptidases

Teichoic acid:

- Teichoic acid is bacterial polysaccharide of glycerol phosphate or ribitol phosphate.
- **Functions:**
    - The main function of teichoic acids is to provide rigidity to the cell-wall by attracting cations such as magnesium and sodium

Lipopolysaccharide:

- It is further composed of three components
    a. O antigen (or O polysaccharide)
    b. Core oligosaccharide
    c. Lipid A

a. O antigen (or O polysaccharide):

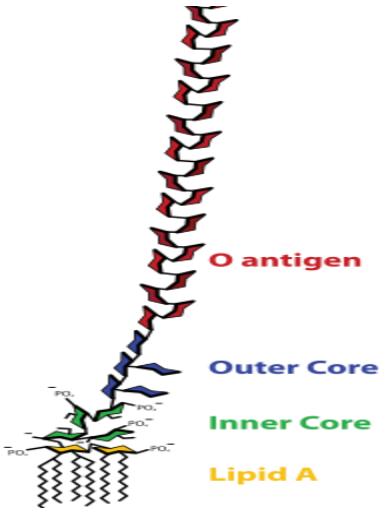
- A repetitive glycan polymer contained within an LPS is referred as the O antigen, O polysaccharide, or O side-chain of the bacteria. The O antigen is attached to the core oligosaccharide and comprises the outermost domain of the LPS molecule. The composition of the O chain varies from strain to strain. For example, there are over 160 different O antigen structures produced by different *E. coli* strains.

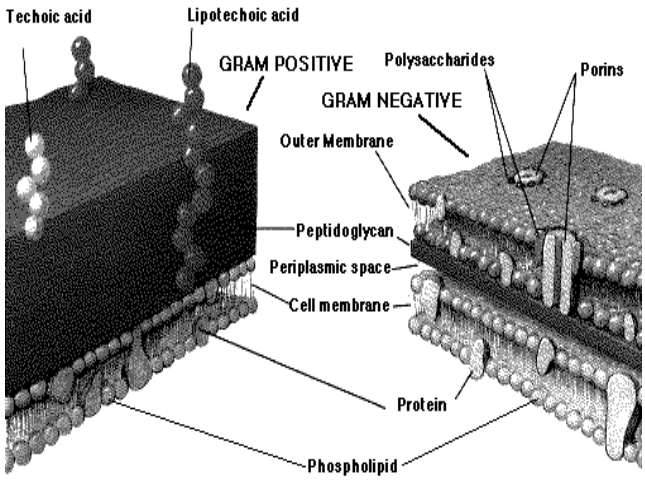
b. Core oligosaccharide:

- The Core domain always contains an oligosaccharide component that attaches directly to lipid A and commonly contains sugars

c. Lipid A:

- Lipid A is, in normal circumstances, a phosphorylated glucosamine disaccharide decorated with multiple fatty acids.





MICROBIAL NUTRITION

In Microbial nutrition we will look into the aspects of microbial nutrition related to the nutrient requirements of microorganism and the nutritional types. To obtain energy and construct new cellular components, organisms must have supply of raw materials or nutrients.

Nutrients:

"Nutrients are substances used in biosynthesis and energy production.

1. NUTRITIONAL REQUIREMENTS:

Microbial cell composition shows that 95% of cell dry weight is made up of a few major elements (carbon, oxygen,hydrogen, nitrogen, Sulphur, phosphorous, potassium, calcium, magnesium and iron)

These are further classified as:

- The common Nutrient Requirements
- Growth Factors

i. The Common Nutrient Requirements

a) Macronutrients

"The nutrients which are required in large amounts are called macronutrients."

- These are the components of carbohydrates, lipids, proteins, and nucleic acids
- Potassium, calcium, magnesium and iron function as cations and part of enzymes and cofactors.

  e.g. carbon, oxygen, hydrogen, nitrogen, sulfur, phosphorus

b) Micronutrients

"The nutrients which are required in very small amounts are called as micronutrients"

- These are referred to as trace elements
- Most are essential for activity of certain enzymes, usually as cofactors.

  e.g iron, copper, molybdenum, and zinc;s contaminants in water, glassware, and regular media components often are adequate for growth

ii. Growth Factors

Other than micronutrients and macronutrients, microbes also require three main components or precursors for growth which are as under:

1- Vitamins

Role of vitamins as growth factors is given in the table below:

Vitamins	Functions	Examples
Biotin	Carboxylation of CO_2 fixation & one carbon metabolism	*Saccharomyces cerevisiae*
Cyanocobalamin (B_{12})	Molecular rearrangement	*Lactobacillus species*
Folic Acid	One carbon metabolism	*Enterococcus faecalis*
Lipoic Acid	Transfer of acyl group	*Lactobacillus casei*
Pantothenic Acid	Precursor of coenzyme	*Proteous morganii*
Pyridoxine (B_6)	Amino acid metabolism	*Lactobacillus species*
Niacin (Nicotinic Acid)	Precursor of NAD and NADP	*Haemophilus influenza*
Riboflavin (B_2)	Precursor of FAD	*Bacillus anthracis*
Thiamine (B_1)	Aldehyde group transfer	

2- Amino acids

Amino acid required for synthesis of protein

3- Purines and Pyrimidines

Required for the synthesis of nucleic acids (DNA & RNA)

2. NUTRITIONAL TYPES OF MICROORGANISMS:

Nutrients can be of two types depending upon the source:

- **Carbon Source**
  Carbon source refers to any carbon containing molecule (carbohydrates, amino acids, fatty acids, CO_2) used by an organism for the synthesis of its organic molecules.
  These are further divided into following types:
  - **Autotrophs** − Those who utilize carbon dioxide directly as a source of carbon
    e.g *Cyanobacteria*
  - **Heterotrophs** − Those who utilize organic molecule as source of carbon
    e.g Purple non-Sulphur bacteria

- **Energy source**
  A source from which useful energy can be extracted or recovered either directly or by means of a conversion or transformation.
  - **Phototrophs** − Those who utilize light as energy source.
    e.g *Rhodobacter capsulatus, Chromatium chlorobium*
  - **Chemotrophs** − Those who utilize organic and inorganic source as an energy source.
    e.g *Hydrogenovibrio crunogenus*

Classification of Nutritional Types Microorganisms

On the basis of types of nutritions microorganisms can also be converted into following types

1- Photoautotrophs:
Organisms that carry out photosynthesis using energy from sunlight, Carbon Dioxide and water.
e.g sulphur bacteria, cyanobacteria

2- Photoheterotrophs:
These are heterotrophic phototrophs. These are organism which use light for energy but can not use CO_2 as their sole carbon source.
e.g green and purple non-sulphur bacteria

3- Chemoautotrophs:
Organisms which create their own energy and biological materials form inorganic chemicals.
e.g Nitrifying bacteria

4- Chemoheterotrophs:
Organisms which derives its energy from chemicals and needs to consume other organisms in order to live.
e.g mostly pathogens

PHYSICAL FACTORS AFFECTING BACTERIAL GROWTH

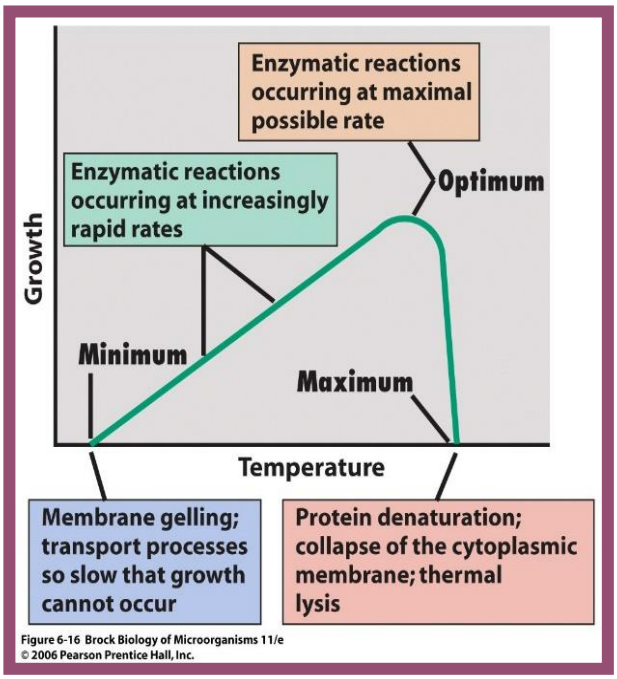
Bacterial growth is affected by various factors given as under:

1- Temperature
2- pH
3- Osmotic pressure
4- Oxygen

1- Temperature:

All bacteria grow in a specific range of range of temperature. An increase or decrease in that temperature affects the bacterial growth which is described as follows:

- **Minimum Temperature**
  Temperature below which growth ceases, or lowest temperature at which microbes will grow.
- **Optimum Temperature**
  Temperature at which its growth rate is the fastest.
- **Maximum Temperature**
  Temperature above which growth ceases, or highest temperature at which microbes will grow.



Classification of Bacteria on the basis of Temperature

On the basis of temperature bacteria can be classified into following types:

- **Psychrophiles (0-20ºC)**
  - Bacteria having maximum growth rate at cold temperature.
  - Most extreme representatives inhabit permanently cold environments
  - **e.g** *Bacillus psychrophilus*
- **Mesophiles (20 – 45ºC)**
  - Bacteria having maximum growth rate at midrange temperature
  - These are found in warm-blooded animals and in terrestrial and aquatic environments in temperate and tropical latitudes.
  - **e.g** *Escherichia coli*
- **Thermophiles (50- 80ºC)**
  - Bacteria having maximum growth rate at temperature in between 45ºC and 80ºC.
  - **e.g** *Bacillus stearothermophilus*
- **Hyperthermophiles**
  - Bacteria having growth rate maximum at temperature greater than 80°C
  - These organisms inhabit hot environments including boiling hot springs, as well as undersea hydrothermal vents that can have tessmperatures in excess of 100ºC.
  - **e.g** *Sulpholobus*

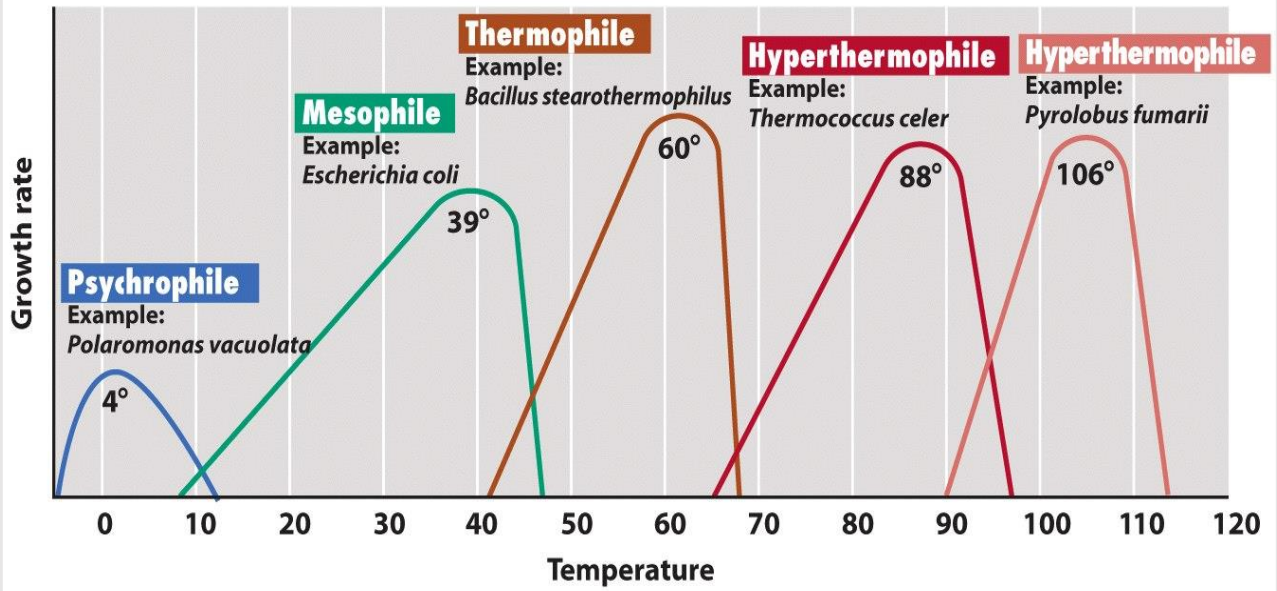


Figure 6-17 Brock Biology of Microorganisms 11/e
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Temperature Ranges for Microbial Growth			
		Cardinal Temperatures	
Microorganism	**Minimum**	**Optimum**	**Maximum**
Non-Photosynthetic Prokaryotes			
Pseudomonas florescence	4	25-30	40
Staphylococcus aureus	6.5	30-37	46
Enterococcus faecalis	0	37	44
Escherichia coli	10	37	45
Neisseria gonorrhea	30	35-36	38
Thermoplasma acidphilum	45	59	62

2- pH

- pH is the measure of H+ ion concentration.
- Each organism has a pH range and a pH optimum
- Human blood and tissues has pH 7.2±0.2

Examples:

Thiobacillus thiooxidans – 2.2-2.8
fungi – 4-6

- The acidity or alkalinity of an environment can greatly affect microbial growth. Most organisms grow best between pH 6 and 8, but some organisms have evolved to grow best at low or high pH. The internal pH of a cell must stay relatively close to neutral even though the external pH is highly acidic or basic.
- On the basis of pH bacteria are classified as follows;
  - **Acidophiles :** organisms that grow best at low pH (*Helicobacter pylori*, ***Thiobacillus thiooxidans***)
  - **Alkalophiles** : organisms that grow best at high pH (*Vibrio cholera*)
  - **Neutrophiles** : organisms that grow best at neutral pH (*Escherichia coli*)

3- Osmotic Pressure:

"Pressure which needs to be applied to a solution to prevent the inward flow of pure solvent across the semi-permeable membrane."

Osmotic pressure depends on the surrounding solute concentration and water availability.

On the basis of osmotic pressure bacteria can be classified as:

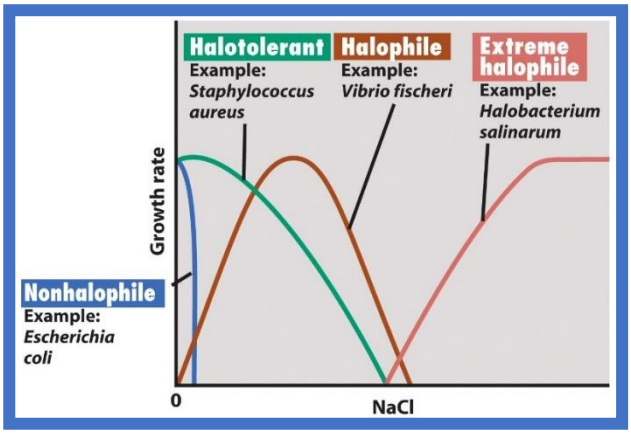
- **Osmophiles**
  Organisms which thrive in high solute. **e.g** *Saccharomyces rouxii*, etc
- **Osmotolerant**
  Organisms which tolerate high solute. **e.g** *Staphylococcus aureus*, etc
- **Halophiles**
  Organisms which thrive in high salt. **e.g** *Halobacterium* etc
- **Halotolerant**
  Organisms which tolerate high salt
- **Barophiles**
  Organisms which thrive in high pressure. **e.g** *Halomonas Salaria*, etc
- **Barotolerant**
  Organisms which tolerate high pressure

Water Activity:

"Water activity or a_w is the partial vapor pressure of water in a substance divided by the standard state partial vapor pressure of water."

Water activity of various bacteria is given in the table as follows:

Water Activity of Several Substances		
Water activity (a_w)	**Material**	**Example Organisms**
1.000	Pure Water	*Caulobacter, Spirillum*
0.995	Human Blood	*Streptococcus, Escherichia*
0.980	Sea Water	*Pseudomonas, Vibrio*
0.950	Bread	*Most Gram Positive rods*



4- Oxygen:

- **Aerobes**
  Bacteria which require oxygen.
  - **Obligate** : require oxygen to grow
    e.g. *Pseudomonas, Mycobacterium, Bordetella pertussis*
  - **Microaerphiles :** require reduced level of oxygen
    e.g. *campylobacter*

- **Anaerobes**
  Bacteria which can grow without oxygen too.
  - **Facultative** : can live with or without oxygen but grow better with oxygen
    e.g. *E.coli, enteroccoci*
  - **Aerotolerant anaerobes :** can tolerate oxygen but grow better without oxygen.
    e.g. *Clostridium perfringens, Treponema pallidum*
  - **Anaerobes :** do not require oxygen. Obligate anaerobes are killed by oxygen.
    e.g. *Clostridium tetani*

TABLE 6.1 The Effect of Oxygen on the Growth of Various Types of Bacteria

	a. Obligate Aerobes	b. Facultative Anaerobes	c. Obligate Anaerobes	d. Aerotolerant Anaerobes	e. Micro-aerophiles
Effect of Oxygen on Growth	Only aerobic growth; oxygen required.	Both aerobic and anaerobic growth; greater growth in presence of oxygen.	Only anaerobic growth; ceases in presence of oxygen.	Only anaerobic growth; but continues in presence of oxygen.	Only aerobic growth; oxygen required in low concentration.
Bacterial Growth in Tube of Solid Growth Medium					
Explanation of Growth Patterns	Growth occurs only where high concentrations of oxygen have diffused into the medium.	Growth is best where most oxygen is present, but occurs throughout tube.	Growth occurs only where there is no oxygen.	Growth occurs evenly; oxygen has no effect.	Growth occurs only where a low concentration of oxygen has diffused into medium.
Explanation of Oxygen's Effects	Presence of enzymes catalase and superoxide dismutase (SOD) allows toxic forms of oxygen to be neutralized; can use oxygen.	Presence of enzymes catalase and SOD allows toxic forms of oxygen to be neutralized; can use oxygen.	Lacks enzymes to neutralize harmful forms of oxygen; cannot tolerate oxygen.	Presence of one enzyme, SOD, allows harmful forms of oxygen to be partially neutralized; tolerates oxygen.	Produce lethal amounts of toxic forms of oxygen if exposed to normal atmospheric oxygen.

CULTURE MEDIA

"A culture media is a special medium used in microbiological laboratories to grow different kinds of microorganisms."

A growth or a culture medium is composed of different nutrients that are essential for microbial growth. Culture media is formed because of the following reasons:

- Bacteria have to be grown or cultured for their identification.
- By appropriate procedures they have to be grown separately on culture media and obtained as pure for study.
- ***Colony*** – macroscopically visible collection of millions of bacteria originating from a single bacterial cell.

History

- The original media used by Louis Pasteur was meat broth
- Liquid medium – diffuse growth
- Solid medium – discrete colonies.
- Cooked cut potato by Robert Koch – earliest solid medium
- Gelatin – not satisfactory
  - liquefy at 24°C

Agar

- Frau Hesse
- Used for preparing solid medium
- Obtained from seaweeds.
- No nutritive value
- Not affected by the growth of the bacteria.
- Melts at 98°C & sets at 42°C
- 2% agar is employed in solid medium

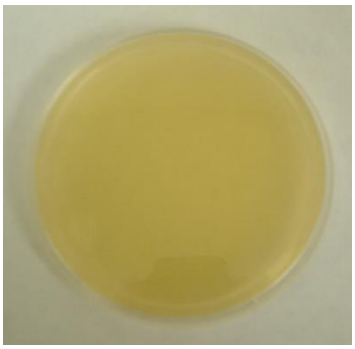
TYPES OF CULTURE MEDIA

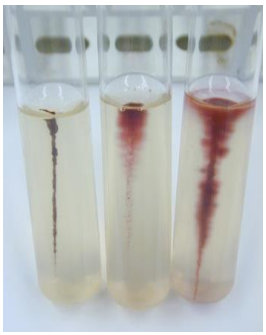
Culture media can be classified into several type on the basis of their consistency, their constituents and their oxygen requirements.

I. Based on their consistency

On the basis of their consistency culture media can be divided into following types:

- **Solid media** – contains 2% agar
  Colony morphology, pigmentation, hemolysis can be appreciated.
  e.g. Nutrient agar, Blood agar
- **Liquid media** – no agar.
  For inoculum preparation, Blood culture, for the isolation of pathogens from a mixture.
  e.g. Nutrient broth
- **Semi solid medium** – 0.5% agar.
  e.g. Motility medium







II. Based on the constituents/ ingredients

On the basis of their constituents or ingredients bacteria can be divided into following types:

- **Simple media / basal media**
  - e.g: NB, NA
  - NB consists of peptone, meat extract, NaCl,
  - NB + 2% agar = Nutrient agar

- **Complex media**
  - Media other than basal media.
  - They have added ingredients.
  - Provide special nutrients
  - Its exact chemical composition is not known
- **Synthetic or defined media**
  - Media prepared from pure chemical substances and its exact composition is known
  - Eg: peptone water – 1% peptone + 0.5% NaCl in water
- **Special media**
  - **Enriched media**
    - Substances like blood, serum, egg are added to the basal medium.
    - Used to grow bacteria that are exacting in their nutritional needs and are used to grow mostly fastidious bacteria.
    - A **fastidious organism** is any **organism** that has a complex nutritional requirement. In other words, a **fastidious organism** will only grow when specific nutrients are included in its diet.
    - Eg: Blood agar, Chocolate agar



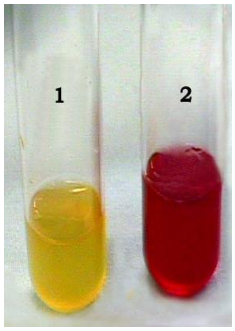
Blood Agar



Chocolate Agar



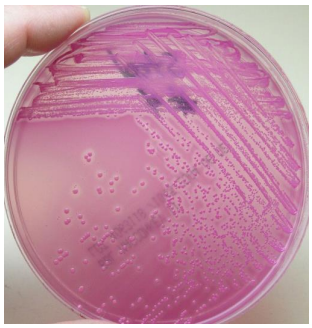
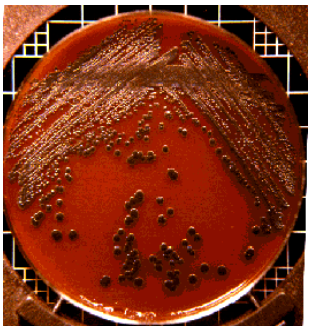
  - **Enrichment media**
    - Liquid media used to isolate pathogens from a mixed culture.
    - Media is incorporated with inhibitory substances to suppress the unwanted organism.
    - e.g.
      i. Selenite F Broth – for the isolation of Salmonella, Shigella
      ii. Alkaline Peptone Water – for Vibrio cholerae



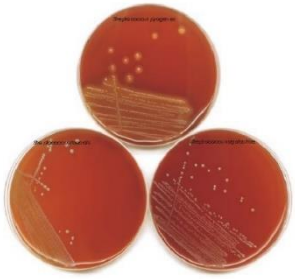
  - **Sugar media**
    - Media containing any fermentable substance.
      Eg: glucose, arabinose, lactose, starch etc.
    - Media consists of 1% of the sugar in peptone water.
    - Contain a small tube (Durham's tube) for the detection of gas by the bacteria



- **Selective media**
  - The inhibitory substance is added to a solid media.
  - e.g.
    i. TCBS – for V.cholerae
    ii. LJ medium – M.tuberculosis
    iii. Wilson and Blair medium – S.typhi
    iv. Potassium tellurite medium – Diphtheria bacilli

			
TCBS	Mac Conkey's Medium	Pot. Tellurite Medium	LJ Media

- **Indicator media**
  - These media contain an indicator which changes its color when a bacterium grows in them.
  - e.g.
    i. Blood agar
    ii. MacConkey's medium
    iii. Christensen's urease medium

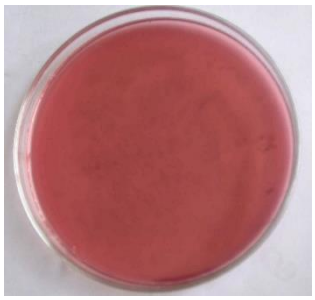
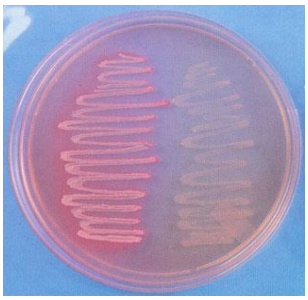






- **Transport media**
  - Media used for transporting the samples.
  - Delicate organisms may not survive the time taken for transporting the specimen without a transport media.
    e.g:
    i. Stuart's medium – non-nutrient soft agar gel containing a reducing agent
    ii. Buffered glycerol saline – enteric bacilli
- **Differential media**
  - A media which has substances incorporated in it enabling it to distinguish between bacteria.
  - e.g. MacConkey's medium
    i. Peptone
    ii. Lactose
    iii. Agar
    iv. Neutral red
    v. Taurocholate
  - Distinguish between lactose fermenters & non-lactose fermenters.

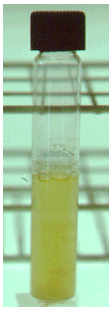
- Lactose fermenters – **Pink** colonies
- Non-lactose fermenters – colorless colonies

III. Based on Oxygen requirement

On the basis of their oxygen requirements bacteria can be classified into following types:

- **Aerobic media**
  - This media is used to grow aerobic bacteria
- **Anaerobic media**
  - This media is used to grow anaerobic organisms.
    e.g. Robertson's cooked meat medium, Thioglycolate medium.

CULTURE METHODS

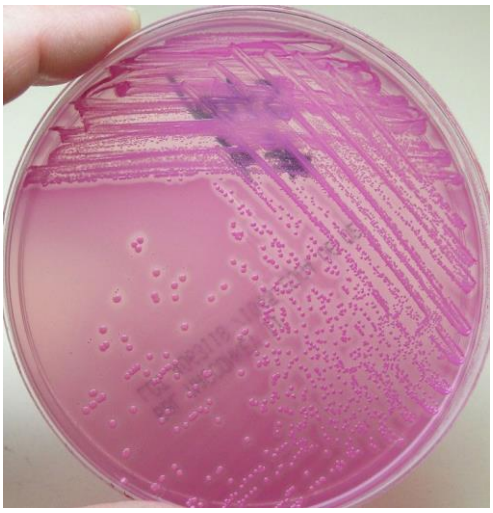
- Culture methods employed depend on the purpose for which they are intended.
- The indications for culture are:
  - To isolate bacteria in pure cultures.
  - To demonstrate their properties.
  - To obtain sufficient growth for the preparation of antigens and for other tests.
  - To determine sensitivity to antibiotics.
  - To estimate viable counts.
  - Maintain stock cultures.

Culture methods include:

- Streak culture
- Lawn culture
- Stroke culture
- Stab culture
- Pour plate method
- Liquid culture
- Anaerobic culture methods

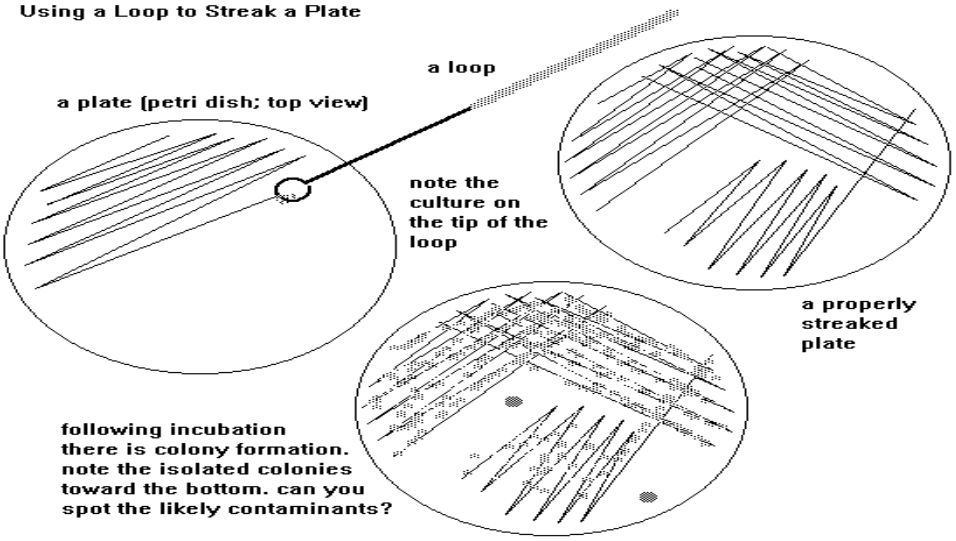
1- STREAK CULTURE

"Streak culture is used for the isolation of bacteria in pure culture from clinical specimens."



In this method platinum wire or nichrome wire is used.

- One loopful of the specimen is transferred onto the surface of a well dried plate and is spread over a small area at the periphery.
- The inoculum is then distributed thinly over the plate by streaking it with a loop in a series of parallel lines in different segments of the plate.
- On incubation, separated colonies are obtained over the last series of streaks.



2- LAWN CULTURE/SPREAD PLATE METHOD

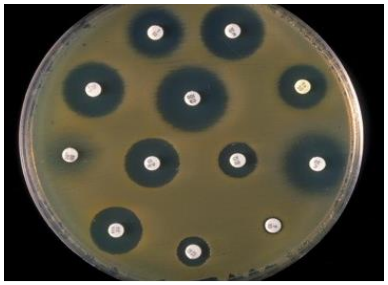
"Lawn Culture/ Spread Plate Method provides a uniform surface growth of the bacterium."

Lawn cultures are prepared by flooding the surface of the plate with a liquid suspension of the bacterium.

Uses

This method is used for:

- Antibiotic sensitivity testing.
- In the preparation of bacterial antigens and vaccines.



Antibiotic Sensitivity Testing

3- POUR PLATE CULTURE

Following procedure is followed for preparing a pour plate culture.

- Agar medium is melted (15 ml) and cooled to 45°C.
- 1 ml of the inoculum is added to the molten agar.
- Mix well and pour to a sterile petri dish.
- Allow it to set.
- Incubate at 37°C, colonies will be distributed throughout the depth of the medium.

Uses

This method is used to

- Give an estimate of the viable bacterial count in a suspension.
- For the quantitative urine cultures.



4- STROKE CULTURE

- Stroke culture is made in tubes containing agar slope / slant.

Uses

- Provide a pure growth of bacterium for slide agglutination and other diagnostic tests.

5- LIQUID CULTURES

Liquid cultures are inoculated by touching with a charged loop or by adding the inoculum with pipettes or syringes.

Uses

- Blood culture
- Sterility tests
- Continuous culture methods

Disadvantage

- It does not provide a pure culture from mixed inocula



6- STAB CULTURE

Stab culture is prepared by puncturing a suitable medium i.e gelatin or glucose agar with a long, straight, charged wire.

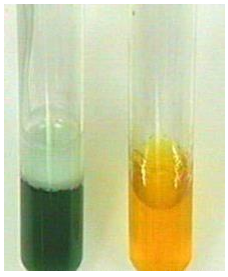
- **Uses**

  Stab culture is used for

  - Demonstration of gelatin liquefaction.
  - Oxygen requirements of the bacterium under study.
  - Maintenance of stoke cultures.



Gelatin Liquefaction



Oxidation-Fermentation Medium

ANAEROBIC CULTURE METHODS

- Anaerobic bacteria differ in their requirement and sensitivity to oxygen.
- Clostridium tetani is a strict anaerobe i.e it grows at an oxygen tension < 2 mm Hg.

Methods

Methods for the preparation of Anaerobic Culture are:

- Production of vacuum
- Displacement of oxygen with other gases
- Chemical method
- Biological method
- Reduction of medium



Production of vacuum

- Incubate the cultures in a vacuum desiccator.

Displacement of oxygen with other gases

- Displacement of oxygen with hydrogen, nitrogen, helium or CO_2. e.g. Candle jar

Chemical method

- Alkaline pyrogallol absorbs oxygen.

McIntosh – Fildes' anaerobic jar

- Consists of a metal jar or glass jar with a metal lid which can be clamped air tight.
- The lid has 2 tubes – gas inlet and gas outlet
- The lid has two terminals – connected to electrical supply.
- Under the lid – small grooved porcelain spool, wrapped with a layer of asbestos.

Working

- Inoculated plates are placed inside the jar and the lid clamped air tight.
- The outlet tube is connected to a vacuum pump and the air inside is evacuated.
- The outlet tap is then closed and the inlet tube is connected to a hydrogen supply.
- After the jar is filled with hydrogen, the electric terminals are connected to a current supply, so that the palladinised asbestos is heated.
- Act as a catalyst for the combination of hydrogen with residual oxygen.

Gaspak

- Commercially available disposable envelope.
- Contains chemicals which generate H_2 and CO_2 on addition of water.
- Cold catalyst – in the envelope
- Indicator is used – reduced methylene blue.
  - Colorless – anaerobically
  - Blue color – on exposure to oxygen



McIntosh – Fildes' anaerobic jar



Gaspak

Biological method

- Absorption of oxygen by incubation with aerobic bacteria, germinating seeds.

Reduction of oxygen

- By using reducing agents – 1% glucose, 0.1% Thioglycolate

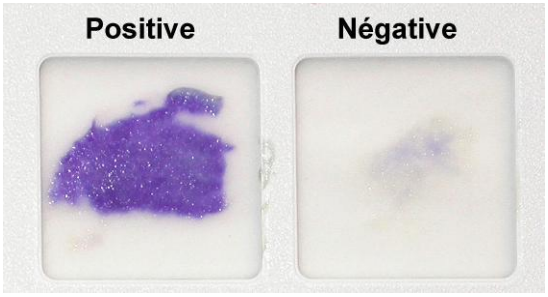
BIOCHEMICAL TEST & REACTIONS

"The biochemical tests and reactions provide additional information for the identification of the bacterium."

The tests include:

- Oxidase test
- Triple sugar iron agar (TSI)
- Indole test
- Citrate utilization
- Urease test

OXIDASE TEST



"Oxidase test detects the presence of an enzyme "oxidase" produced by certain bacteria which will reduce the dye(Kavoc's reagent)."

- Positive test is indicated by the development of a **purple** color.
- Oxidase positive – Pseudomonas, Vibrio, Neisseriae
- Oxidase negative – Salmonella, Shigella

TRIPLE SUGAR IRON AGAR (TSI)

"Triple Sugar Iron Agar (TSI) is a composite media used to study different properties of a bacterium i.e sugar fermentation, gas production and H_2S production."

- In addition to peptone, yeast extract & agar, it contains 3 sugars – Glucose, Lactose, Sucrose.
- The Iron salt – Ferric citrate indicates H_2S production.
- Phenol red is the indicator.
- It is an orange red medium with a slant and a butt. With pH of 7.4

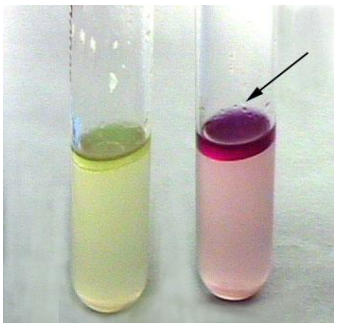
TSI Reactions:



Yellow – Acid
Pink- Alkaline

- Yellow slant / Yellow butt – Lactose fermenters.
- Pink slant / Yellow butt – Non lactose fermenters.
- Pink slant / no color change – Non fermenters
- **Black color** – H_2S production.
- Gas bubbles or crack in the medium – gas production.
- LF – E.coli, Klebsiella
- NLF – Salmonella, Shigella
- H_2S – Proteus

INDOLE TEST



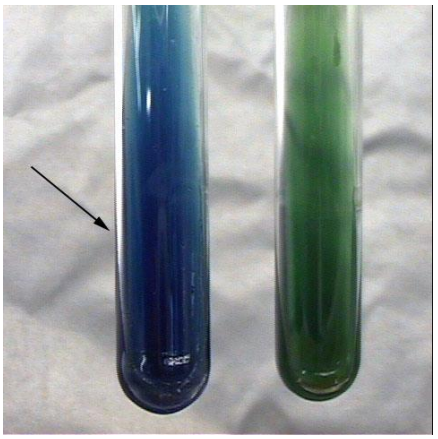
"Indole test is used to detect indole production by the organism."

- They produce indole from tryptophan present in peptone water.
- After overnight incubation, a few drops of indole reagent is added.
- Positive indole test is indicated by a pink ring. **e.g** *E.coli*
- Negative indole test is indicated by yellow ring. **e.g** *Klebsiella, Salmonella*.

CITRATE UTILIZATION TEST

"Citrate Utilization test is used to detect the ability of certain bacteria to utilize citrate as the sole source of carbon."

- This test is carried out in Simmon's Citrate medium which contains Sodium citrate and bromothymol blue as the indicator.
- If citrate is utilized, alkali is produced which turns the medium to blue.
- Citrate positive test gives blue colour.
  e.g *Klebsiella*
- Citrate negative test gives green colour.
  e.g *E.coli*



UREASE TEST

"Urease test is used to detect organisms that produce urease."

- This test is carried out in Christensen's urease medium.
- Urease produced by the organisms split urea into ammonia and CO_2.
- Urease positive test gives pink color.
  e.g *Proteus, Klebsiella*
- Urease negative test gives yellow color
  e.g *E.coli, Salmonella*



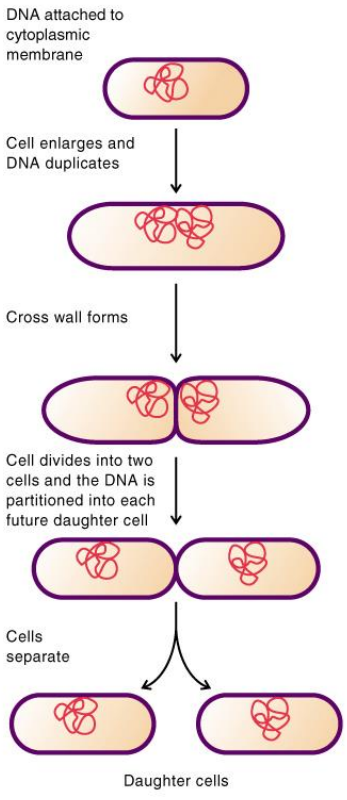
Bacterial Growth

" Growth is defined as an increase in population number and not an increase in cell size"

Growth is carried out by cell division.
Most Bacteria Reproduce by **Binary Fission**

Binary Fission:

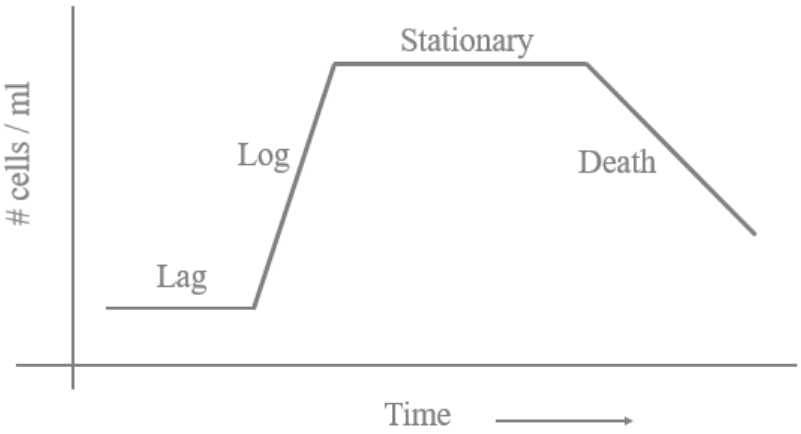
- The cell doubles in size.
- Replicates the chromosome (DNA)
- Forms a septum in the center
- Synthesizes a Cell Wall at the Septum
- Daughter cells separate.



Bacterial Growth Curve

- Two growth curves
  - Normal
  - Synchronous
- All microorganisms undergo similar growth patterns.
- Each growth Curve has 4 Phases

1. **Lag Phase**
   - occurs immediately after inoculation
   - cells do not grow; cells per volume do not increase
2. **Growth Phase**
   - Exponential Phase
   - Log Phase
   - During this phase the microbe are growing at the maximum rate possible.
   - Cells per volume increases dramatically
   - Most research is performed on cells during log phase
3. **Stationary Phase**
   - Growth levels off.
   - Cells per volume does not increase or decrease
   - Growth Rate = Death Rate
   - Due to
     i. Depletion of Nutrients
     ii. Increase in Waste Products
4. **Death Phase**
   - Death Rate exceeds Growth Rate
   - Cells per volume decreases
   - Due to:
     i. Very low concentrations of Nutrients
     ii. Very high concentrations of Waste Products



TERMINOLOGIES

Growth Rate

- Number of doublings (cell divisions) per hour

Doubling Time (Generation Time)

- Length of time required for a cell to divide during log growth
- Doubling Time = 1/Growth Rate

Generation Time of Various Microorganism

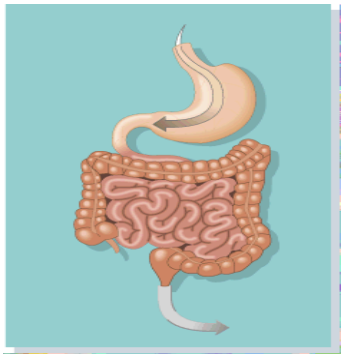
Microorganism	Temperature (°C)	Generation Time (Hours)
Bacteria		
Beneckea natreigens	37	0.16
Escherichia coli	40	0.35
Bacillus subtilis	40	0.43
Staphylococcus aureus	37	0.47
Pseudomonas aeruginosa	37	0.58
Clostridium botulinum	37	0.58
Rhodospirillum rubrum	25	4.6-5.3
Anabaena cylindrica	25	10.6
Mycobacterium tuberculosis	37	12
Treponema palladium	37	33
Algae		
Scendesmus quadricanda	25	5.9
Chlorella pyrenoidosa	25	7.75
Asteriomella formosa	20	9.6
Euglena grucilus	25	10.9
Ceratium tripos	20	82.8
Protozoa		
Tetrahymena geleii	24	2.2-4.2
Leishmania donovani	26	10-12
Paramecium caudatum	26	10.4
Acanthamoeba castellanii	30	11-12
Giardia lambia	37	18

Methods for Culturing Microorganisms

There are two methods for culturing microorganisms:

1. **Batch Culture:**
   - the microbes are grown in a closed vessel
   - typically in a test tube or flask
2. **Continuous Culture:**
   - the microbes are grown in vessel which has medium constantly added and spent medium constantly removed
   - performed in a **chemostat**



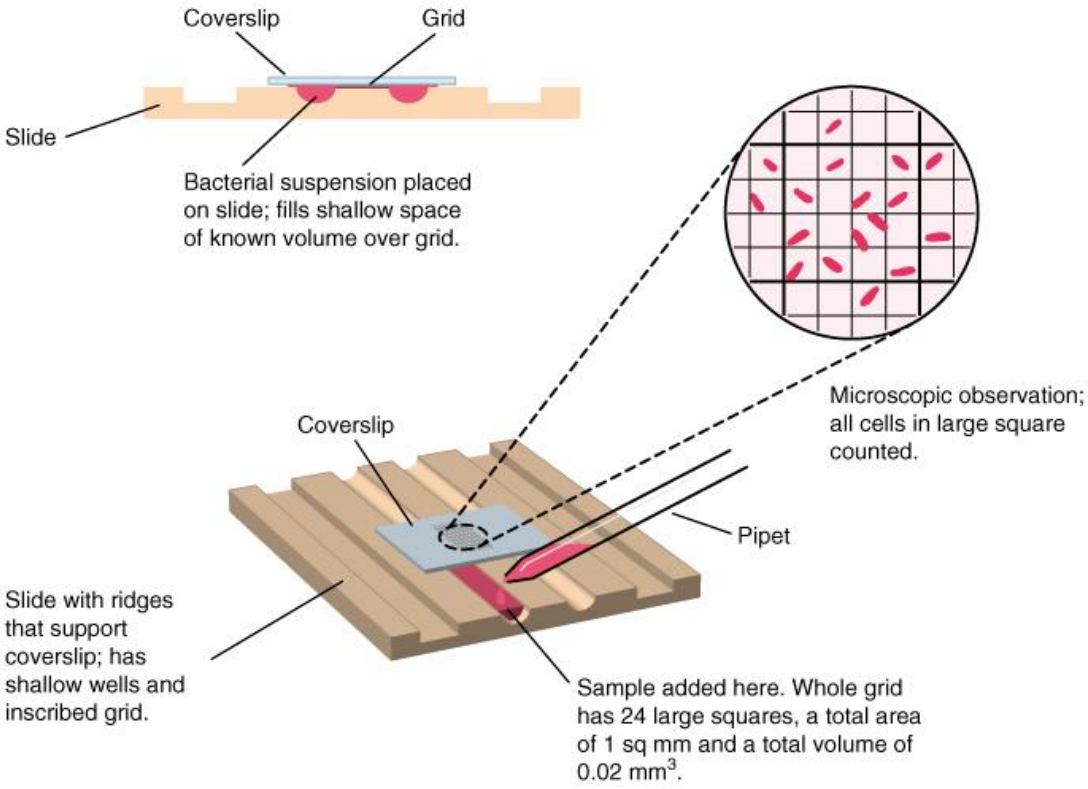


"Extra" Growth Factors

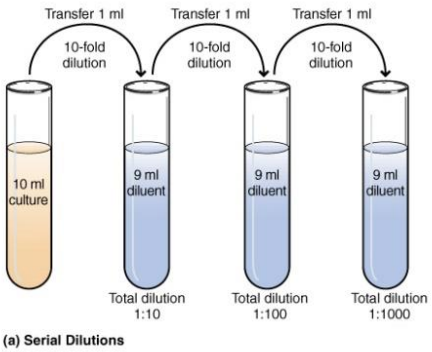
- **Fastidious** Microorganisms
  - microbes which need special organic chemicals added to the medium for growth to occur
  - Ex. Vitamins, amino acids

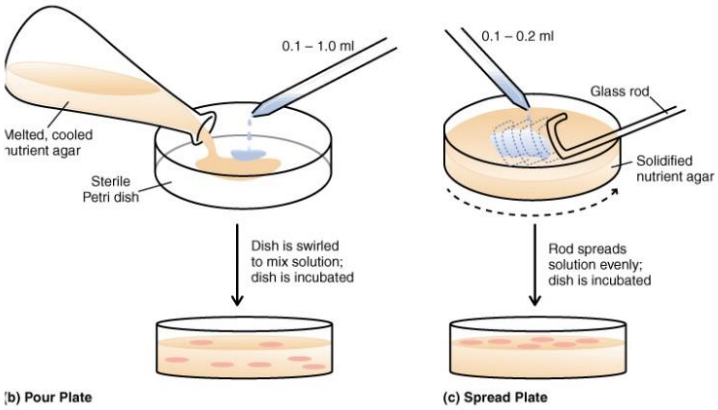
Measuring Microbial Conc.

- **Direct cell counts**
  Using a microscope count the number of cells in a known volume



- **Indirect Methods**
  - Plate Count
    - Dilute the sample and plate a known volume;
    - Incubate
    - Count the number of colonies
      1 colony = 1 cell
    - Calculate cells/ml adjusting for dilution

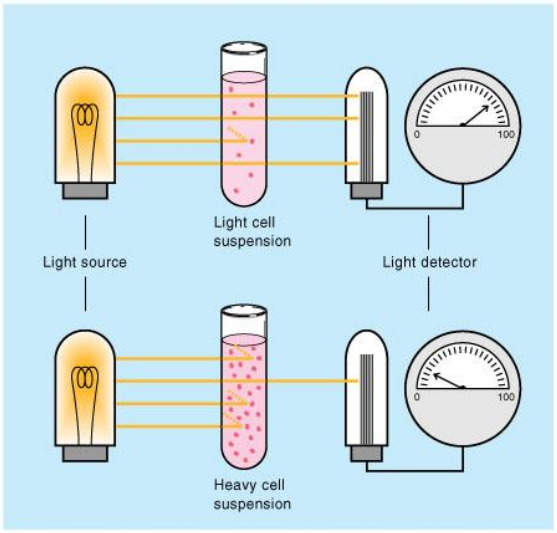




- **Spectrophotometric**
  - Measure the "turbidity"
  - Increase in turbidity = Increase in Cells/ml
  - Use Spectrophotometer
    - Calculates amount of light that passes through the sample and gives Absorbance Units
    - No of cells/ ml are standardized.

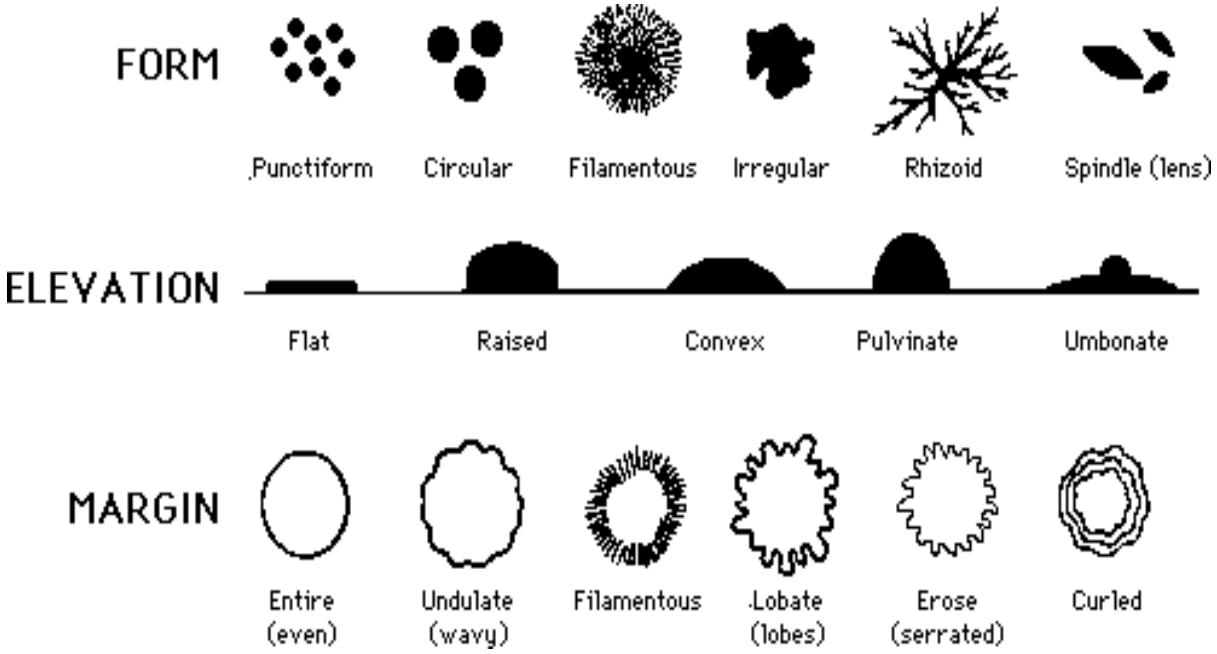
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(a)



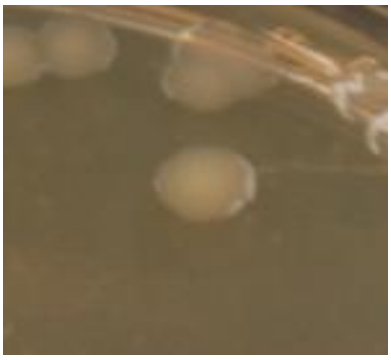
(b)

COLONY CHARACTERISTICS OF BACTERIA



E. coli:

- medium sized colonies
- regular margin
- convex elevation



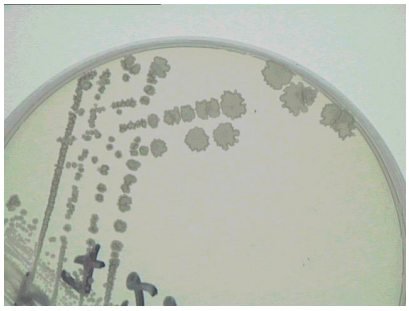
Klebsiella pneumoniae:

- Slightly gummy/wet looking colonies
- Circular
- convex
- entire margin



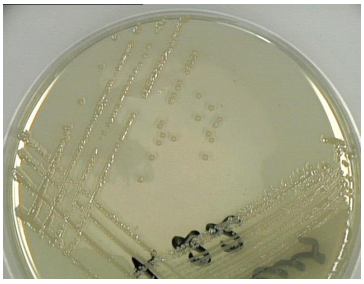
Bacillus subtilis:

- Colonies large
- margin is undulate or lobate
- dry
- flat elevation



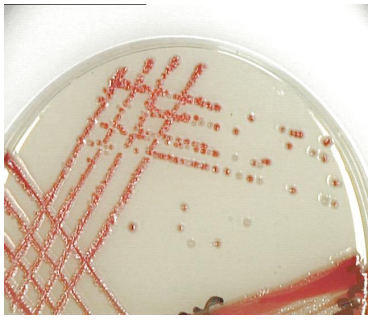
***Staphylococcus aureus*:**

- Circular
- Pinhead
- Convex
- Entire margin



Serratia marcescens:

- Mucoid
- Entire margin
- Umbonate elevation
- Red and white colonies



Growth on Slant



Staphylococcus aureus
Even

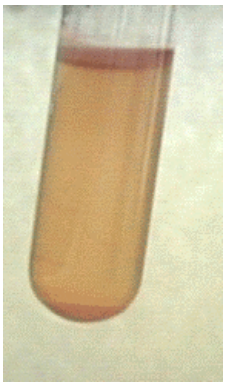


Bacillus subtilis
Spreading

Growth in Broth



Mycobacterium smegmatis



Serratia marcescens
Turbid and sediment

BACTERIAL CULTURE PRESERVATION METHOD

Bacterial Culture is preserved by following ways:

1. Agar Slant Cultures:
2. Agar Slant Culture Covered with Oil (Parafin Method)
3. Saline Suspension
4. Cryo-preservation
5. Preservation at Very Low Temperature
6. Preservation by Drying in Vacuum
7. Preservation by Freeze Drying (Lyophilization)

1. Agar Slant Cultures:

- Agar stabs or slants are prepared after autoclaving the medium. They are inoculated carefully with individual culture and incubated overnight. Then cultures are stored at 5 to -20°C. It must be sub-cultured at an interval of six months.

2. Agar Slant Culture Covered with Oil (Parafin Method):

- After incubation the cultures are incubated to facilitate the growth of organism. Then the cultures are covered with mineral oil to a depth of 1 cm above the slant. They are sub-cultured after the storage of about 1 year.

3. Saline Suspension:

- Bacteria require salts for various metabolic activities. But at high concentration, salts act as bacteriostatic (temporary growth inhibitor). Hence, bacterial culture is preserved in 1% salt concentration in screw-capped tubes to prevent evaporation. The tubes are stored at room temperature.

4. Cryo-preservation:

- The cultures are grown in culture tubes. Then 10-30% of a cryo-protective agent (e.g. glycerol or dimethyl sulfoxide) is added into it so that it could protect the culture from cold sock. The culture is dispensed into ampoules. The ampoules are properly sealed and kept in liquid nitrogen at -176 to -196°C.

5. Preservation at Very Low Temperature:

- Pure cultures can be successfully stored at 0-4°C either in refrigerators or in cold-rooms. This method is applied for short duration (2-3 weeks for bacteria and 3-4 months for fungi) because the metabolic activities of the microorganisms are greatly slowed down but not stopped. Thus their growth continue slowly, nutrients are utilized and waste products released in medium. This results in, finally, the death of the microbes after sometime

6. Preservation by Drying in Vacuum:

- The cultures are dried over $CaCl_2$ in a vacuum, then stored in a refrigerator. At such conditions, the organisms survive for longer period than the air dried cultures.

7. Preservation by Freeze Drying (By lyophilization):

- In this method the microbial suspension is put in small ampoules and frozen through drying under vacuum. Vacuum drying results in sublimation of cell water. Then the ampoules are sealed off in vacuum. The lyophilised cultures are stored at 4°C. The cultures remain viable for about 10 years. By this method a large number of cultures are maintained without variation in their characteristics.

BACTERIAL DISEASES

1. Botulism:

- **Bacteria:** *Clostridium botulinum*
- **Family:** *Clostridiaceae*
- **Genus:** *Clostridium*
- **Shape:** Rod shaped
- **Type:**
  - Gram positive
  - Spore forming
  - Motile
- **Habitat:** *Clostridium botulinum* spores are most commonly found in oxygen-poor environments with low acidity. As a result, they can be found in soil and marine environments all throughout the world.
- **Mode of Transmission:** Though ingestion of inadequately cooked food or processed or refrigerated foods in which toxin has formed, particularly canned and alkaline foods, the spores get into an open wound and are able to reproduce in an anaerobic environment.
- **Vaccination:** A vaccine against botulism exists but it is rarely used as its effectiveness has not been fully evaluated and it has demonstrated negative side effects.
- **Treatment:** Antitoxin, antibiotics, mechanical ventilation

2. Diphtheria:

- **Bacteria:** *Corynebacterium diphtheria*
- **Family:** *Corynebacteriaceae*
- **Genus:** *Corynebacterium*
- **Shape:** Rod shaped
- **Genome:**
  - Gram positive
  - Non-spore forming
  - Non-motile
- **Habitat:** Soil, water, plants, and food products.
- **Mode of Transmission:** Through people by direct contact or through the air.
- **Vaccination:** A diphtheria vaccine is effective for prevention and available in a number of formulations.
- **Treatment:** Erythromycin or Benzylpenicillin

3. Gonorrhea:

- **Bacteria:** *Neisseria gonorrhoeae*
- **Family:** *Neisseriaceae*
- **Genus:** *Neisseria*
- **Shape:** Diplococci
- **Type:** Gram negative
- **Habitat:** Human body
- **Mode of Transmission:** Through vaginal, oral, or anal sex
- **Vaccination:** Not available
- **Treatment:** An injected single dose of ceftriaxone (a third-generation cephalosporin) along with azithromycin administered orally.

4. Legionnaire's Disease:

- **Bacteria:** *Legionella pnemophila*
- **Family:** *Legionellaceae*
- **Genus:** *Legionella*
- **Shape:** Bacillus
- **Type:**
  - Gram negative
  - Non-spore forming
  - Flagellated
- **Habitat:** Fresh water.
- **Mode of Transmission:** By breathing is mist that contains bacteria.
- **Vaccination:** Not available.
- **Treatment:** Antibiotics usually Fluoroquinolones, Azithromycin, or Doxycycline.

5. Leprosy:

- **Bacteria:** *Mycobacterium leprae or Mycobacterium lepromatosis*.
- **Family:** *Mycobacteriaceae*
- **Genus:** *Mycobacterium*
- **Shape:** Rod shaped
- **Type:**
  - Acid Fast
  - Pleomorphic
  - Pathogenic
- **Habitat:** Most common in warm, wet areas in the tropics and subtropics.
- **Mode of Transmission:** Through a cough or contact with fluid from the nose of a person infected by leprosy.
- **Vaccination:** The Bacillus Calmette–Guérin (BCG) vaccine offers a variable amount of protection against leprosy in addition to its target of tuberculosis.
- **Treatment:** Dapsone, Rifampicin, and Clofazimine.

6. Leptospirosis:

- **Bacteria:** *Leptospira*
- **Family:** *Leptospiraceae*
- **Genus:** *Leptospira*
- **Shape:** Spirochete
- **Type:**
  - Gram negative
  - Pathogenic
- **Habitat:** High humidity and neutral (6.9–7.4) pH are necessary for their survival in the environment, with stagnant water reservoirs—bogs, shallow lakes, ponds, puddles, etc.—being the natural habitat for the bacteria.
- **Mode of Transmission:** By both wild and domestic animals. The most common animals that spread the disease are rodents. It is often transmitted by animal urine or by water or soil containing animal urine coming into contact with breaks in the skin, eyes, mouth, or nose.
- **Vaccination:** Vaccines for animals exist for certain type of *Leptospira* which may decrease the risk of spread to humans
- **Treatment:** Antibiotics such as doxycycline, penicillin, or ceftriaxone.

7. Listeriosis:

- **Bacteria:** *Lysteria monocytogenes*
- **Family:** *Lysteriaceae*
- **Genus:** *Lysteria*
- **Shape:** Rod shaped
- **Type:**
  - Gram Positive
  - Non-spore forming
  - Motile
- **Habitat:**
- **Mode of Transmission:** Through the ingestion of contaminated food products. *Listeria* has been isolated from raw meat, dairy products, vegetables, fruit and seafood
- **Vaccination:** Not available.
- **Treatment:** Ampicillin generally is considered antibiotic of choice.

8. Lyme Disease:

- **Bacteria:** *Borrelia burgdorferi*, *Borrelia afzelii*, and *Borrelia garinii*.
- **Family:** *Spirochaetaceae*
- **Genus:** *Borrelia*
- **Shape:** Spirochete
- **Type:** Diderm i.e. double membrane bacteria (not called gram positive or gram negative)
- **Habitat:** In ticks
- **Mode of Transmission:** Spread by bites of infected ticks of genus *lxodes*.
- **Vaccination:** LYMErix
- **Treatment:** Antibiotics usually doxycycline or amoxicillin, cefuroxime axetil, and azithromycin.

9. Mycoplasma pneumonia:

- **Bacteria:** *Mycoplasma pneumoniae*
- **Family:** *Mycoplasmataceae*
- **Genus:** *Mycoplasma*
- **Shape:** Spherical to flamentous.
- **Type:** Lack peptidoglycan cell wall
- **Habitat:** *Mycoplasma pneumoniae* grows exclusively by parasitizing mammals.
- **Mode of Transmission:** Mycoplasma is spread through contact with droplets from the nose and throat of infected people especially when they cough and sneeze.
- **Vaccination:** Not available
- **Treatment:** Antibiotics which target the bacteria by acting on rRNA such as macrolides, tetracycline, ketolides, and fluoroquinolone

10.Pseudomembranous colitis:

- **Bacteria:** *Clostridium difficile*
- **Family:** *Peptostreptococcaceae*
- **Genus:** *Clostridium*
- **Shape:** Rod shaped, pleomorphic
- **Type:**
  - Gram positive
  - Spore forming
- **Habitat:** It inhabits a natural reservoir of soil, faeces of domestic animals and humans, sewage, the human intestinal tract, and retail meat.
- **Mode of Transmission:** Transmitted from person to person by the fecal-oral route.
- **Vaccination:** Not available.
- **Treatment:** Usually by metronidazole.

11.Pulmonary anthrax:

- **Bacteria:** *Bacillus anthracis*
- **Family:** *Bacillaceae*
- **Genus:** *Bacillus*
- **Shape:** Rod shaped
- **Type:**
  - Gram positive
  - Aerobic
  - Spore forming
- **Habitat:** It exist in an inactive (dormant) state called spores. The spores are found in soil, animal carcasses and feces (including sheep, goats, cattle, bison, horses, and deer), and animal products (e.g., hides and wool)
- **Mode of Transmission:** By contact with the bacterium's spores, which often appear in infectious animal products. Contact is by breathing, eating, or through an area of broken skin.
- **Vaccination:** Available.
- **Treatment:** Antibodies or antitoxins.

12.Rocky Mountain Spotted Fever:

- **Bacteria:** *Rickettsia rickettsii*
- **Family:** *Rickettsiaceae*
- **Genus:** *Rickettsia*
- **Shape:** Coccobacillus
- **Type:** Gram negative.
- **Habitat:** Obligate intracellular parasites..
- **Mode of Transmission:** Spread to humans by American dog ticks, Rocky Mountain wood ticks and Brown dog ticks.
- **Vaccination:** Not available
- **Treatment:** Doxycycline.

13. Salmonellosis:

- **Bacteria:** *Salmonella*
- **Family:** *Enterobacteriacae*
- **Genus:** *Salmonella*
- **Shape:** Rod shaped
- **Type:** Gram negative
- **Habitat:** Salmonella bacteria can survive for some time without a host; they are frequently found in polluted water, with contamination from the excrement of carrier animals being particularly important
- **Mode of Transmission:** By eating contaminated meat, eggs, or milk.
- **Vaccination:** Not available
- **Treatment:** Antibiotic like ceftriaxone and azithromycin.

14. Tetanus:

- **Bacteria:** *Clostridium tetani*
- **Family:** *Clostridiaceae*
- **Genus:** *Clostridium*
- **Shape:** Rod shaped
- **Type:**
  - Gram positive
  - Motile
  - Spore forming
- **Habitat:** Commonly found in soil, saliva, dust, and manure.
- **Mode of Transmission:** The bacteria generally enter through a break in the skin such as a cut or puncture wound by a contaminated object.
- **Vaccination:** Available
- **Treatment:** Tetanus immunoglobulin, Metronidazole, diazepam, magnesium sulphate etc.

15. Tuberculosis:

- **Bacteria:** *Mycobacterium tuberculosis*
- **Family:** *Mycobacteriaceae*
- **Genus:** *Mycobacterium*
- **Shape:** Rod shaped
- **Type:**
  - Either stains weak gram positive or does not stain because of high lipid and mycolic acid in cell wall
  - Non-motile
  - Non-spore forming
- **Habitat:** Human pathogen
- **Mode of Transmission:** Tuberculosis is spread through the air when people who have active TB in their lungs cough, spit, speak, or sneeze
- **Vaccination:** Available
- **Treatment:** Treatment of TB uses antibiotics to kill the bacteria. Effective TB treatment is difficult, due to the unusual structure and chemical composition of the mycobacterial cell wall, which hinders the entry of drugs and makes many antibiotics ineffective

16.Tularemia:

- **Bacteria:** *Francisella tularensis*
- **Family:** *Francisellaceae*
- **Genus:** *Francisella*
- **Shape:** Coccobacillus
- **Type:**
  - Gram negative
  - Non-spore former
  - Non-motile
- **Habitat:** Facultative intracellular bacterium
- **Mode of Transmission:** The bacterium is typically spread by ticks, deer flies, or contact with infected animals. It may also be spread by drinking contaminated water or breathing in contaminated dust.
- **Vaccination:** Not available
- **Treatment:** Antibiotic Streptomycin, Gentamicin, doxycycline, or ciprofloxacin.

17.Typhoid Fever:

- **Bacteria:** *Salmonella typhi*
- **Family:** *Enterobacteriaceae*
- **Genus:** *Salmonella*
- **Shape:** Rod shaped
- **Type:**
  - Gram negative
  - Motile (flagellated)
- **Habitat:** Humans
- **Mode of Transmission:** It is spread by eating or drinking food or water contaminated with the feces of an infected person**.**
- **Vaccination:** Available
- **Treatment:** Antibiotics like azithromycin, fluoroquinolones or third generation cephalosporins.

18.Whooping Cough:

- **Bacteria:** *Bordetella pertussis*
- **Family:** *Alcaligenaceae*
- **Genus:** *Bordetella*
- **Shape:** Coccobacillus
- **Type:**
  - Gram negative
  - Encapsulated
- **Habitat:**
- **Mode of Transmission:** It is an airborne disease which spreads easily through the coughs and sneezes of an infected person.
- **Vaccination:** Pertussis vaccine
- **Treatment:** Antibiotics such as erythromycin, azithromycin, clarithromycin, or trimethoprim/sulfamethoxazole.

CONTROLLING MICROORGANISMS

CHAPTER NO. 03

Dr. Qazi Adnan

STERILIZATION

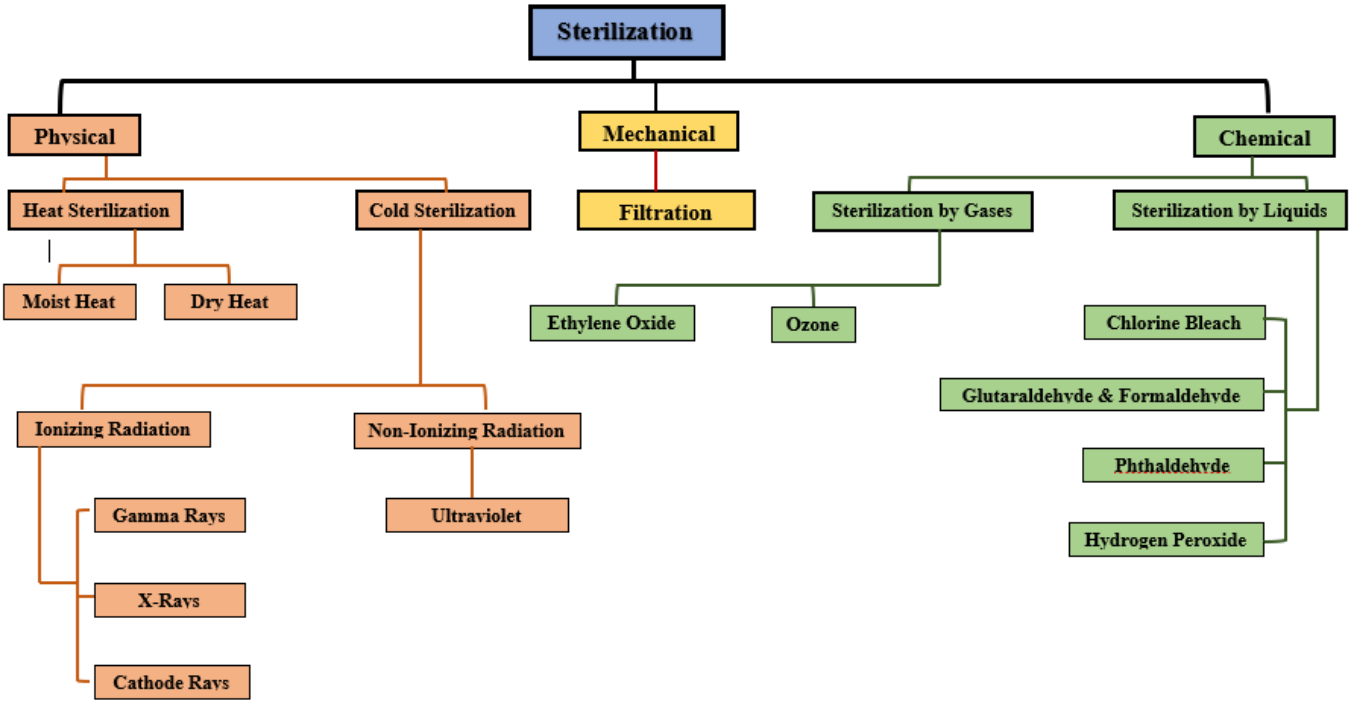
" It is the process by which all living cells, viable spore, viruses and viroids are either killed or removed."

It is a latin word which means unable to produce offsprings. **Sterilization** is a term referring to any process that eliminates (removes) or kills all forms of microbial life, including transmissible agents (such as fungi, bacteria, viruses, spore, etc.) present on a surface, contained in a fluid, in medication, or in a compound such as biological culture media.

Methods for Sterilization

There are three methods of sterilization:

- Physical Method
- Chemical Method
- Mechanical Method



Physical Method of Sterilization

*"Physical methods include processes where no gross chemical or biological changes are carried out and strictly **physical** phenomena are used for sterilization."*

There are two types of physical sterilization

1. Heat Sterilization
2. Cold Sterilization

1. Heat Sterilization

It is a method in which the object is sterilized by using heat

There are further two types of heat Sterilization

i. Moist Heat Sterilization
ii. Dry Heat Sterilization

i. Moist Heat Sterilization:

"Moist heat sterilization describes sterilization techniques that uses hot air that is heavily laden with water vapor and where this moisture plays the most important role in the sterilization."

- In this method steam is used.
- Steam is also further produced by two methods:
  a. Steam under pressure
  b. Tyndallization or Intermittent Sterilization

a. Steam Under Pressure:

Under pressure steam is produced in an autoclave.

Autoclave:

- It was invented by Charles Chamberland
- An autoclave is an instrument used to sterilize equipment and supplies by subjecting them to high pressure saturated steam at 121 °C for around 15–20 minutes depending on the size of the load and the contents.













Temperature, pressure and time

- Normally at sea level the atmospheric pressure is 1 atm.
- By increasing the pressure the boiling point of water is also increased
- So by increasing the pressure by 5 psi, temperature increases at 109 ^{0}C
- And by increasing the pressure at 10 psi the boiling point of water increases to 115 ^{0}C.
- If pressure is kept at 15 psi the temperature will be 121 ^{0}C and the time required for sterilization will be 15 min
- At 20 psi the temp is 125-126 ^{0}C and the time required is 10 min
- At 30 psi the temp is 134-135 ^{0}C and the time is 3 min

Container	Volume	Time
Test Tube	10ml	15 minutes
Flask 125ml	95ml	15 minutes
Flask 2000ml	150ml	30 minutes
Flask 9000ml	6750ml	70 minutes

Mode of Action (MOA) of Autoclave

Mode of action of autoclave is coagulation and denaturation of proteins.

Precaution of Autoclave

- Removal of air
- **Downward displacement** (or gravity type) - As steam enters the chamber, it fills the upper areas first as it is less dense than air. This compresses the air to the bottom, forcing it out through a drain which often contains a temperature-sensing device.

Steam pulsing

- Air dilution by using a series of steam pulses, in which the chamber is alternately pressurized and then depressurized to near atmospheric pressure.

Vacuum pumps

- A vacuum pump sucks air or air/steam mixtures from the chamber.

Superatmospheric cycles

- Achieved with a vacuum pump. It starts with a vacuum followed by a steam pulse followed by a vacuum followed by a steam pulse. The number of pulses depends on the particular autoclave and cycle chosen.

Uses of Autoclave

- Heat resistant materials are sterlized
- Glassware
- Some plastic material
- Clothing e.g. surgical dressings
- Ruber e.g. surgical gloves
- Metalic instruments
- Surgical instruments
- Liquids
- Some paper material
- Culture media

b. Tyndallization or Intermittent Sterilization:

Named after **John Tyndallis** a lengthy process designed to reduce the level of activity of sporulating bacteria. The process involves a chamber of boiling water which provides the steam

Steam is passed for 30-60 min from the materials to be sterilized

- This temperature is not sufficient to kill the spores hence the material is incubated for next 24 hours to germinate the spores and then again steam is passed over the material.
- This process is repeated for three times or days.
- Because the temperature never gets above 100 ^{0}C
- Tyndallization is ineffective against prions.(is an infectious agent composed of protein in a misfolded form)

Applications

- Intermittent sterilization is done for
- Heat sensitive culture media
- Sera
- Egg
- Some heat sensitive carbohydrates

Direct Boiling

- Ineffective method for sterilization

ii. Dry Heat Sterilization:

It uses hot air that is either free from water vapor or has very little of it, where this moisture plays a minimal or no role in the process of sterilization.

It is performed by following methods:

a. Hot air oven
b. Incineration
c. Direct Flame
d. Infra-red Radiations

a. Hot Air Oven:

"Hot air ovens are mostly used for sterilization purposes in which high temperature is kept for a time period to bring about sterilization."

Mode of Action

Denaturation of proteins by dehydration

Temperature and Time

- 121^{0}C for 10 hrs
- 140 ^{0}C for 3 hrs
- 160 ^{0}C for 2 hrs
- 170 ^{0}C for 1 hr
- 180 ^{0}C for 30 min

Materials sterilized

- Heat resistant materials
- Glassware
- Surgical instruments





b. Incineration:

"Incineration is a waste treatment process that involves the combustion of organic substances contained in waste materials."

Incineration and other high temperature waste treatment systems are described as "thermal treatment".

- Temperature of incinerator is kept
- 800^{0}C-6500 ^{0}C
- Incineration of waste materials converts the waste into ash.

Applications

- Large incinerators are regularly employed in hospitals and research labs for complete destruction and disposal of infectious materials such as syringes, needles, cultural materials, dressings, bandages, bedding and pathology samples.






c. Direct Flame:

- Incineration on direct flame or heating coils is also a good method for sterilization of loops.
- The temperature of bunsen burner reaches at 1870 ^{0}C.







d. Infra-Red Sterilization:

- Another sterilization-by-heating technique uses infrared to kill microorganisms in a few seconds using temperatures up to 1500°F/815°C.
- Infrared rays that lie in the waveband range of 0.8–15 µm can be easily generated. With the help of an absorbent surface they are changed into sensible heat, producing high levels of increasing temperature the energy attack on the surface which is changeable into heat depends on the same geometric conditions as those used for UV lights.
- Infrared rays cause sterilization by heat generation. Substances that need to be sterilized are placed in a moving conveyer belt and passed through a tunnel which is previously heated using infrared rays to reach a temperature of 180 degrees. The substances are exposed to that high temperature for a fixed time 7.5 minutes.

Applications:

- The substances become sterile included glassware and metallic instruments.
- It is basically used in central sterile supply department.

Advantages:

- IR emitters a great level in the heating rates and temperatures can be reached by using different wavelengths and densities.
- IR emitters can direct the energy to the required areas that needs treating.
- IR emitters are easy to switch on and off in a few seconds to suit the production requirements.
- IR is an energy source and it isn't affected by the number of the components present in the oven.
- IR is a lot more compact, typically 1/4 to 1/3 the space.
- IR systems normally need low maintenance like changing filters or emitters.
- IR emitters with a high transfer of energy enable surfaces to be heated more rapidly to overcome conduction losses.
- IR emitters can be used to heat components in a vacuum chamber.

2. Cold Sterilization:

Cold sterilization is done by using radiations:

RADIATIONS

"Energy transmitted through space in a variety of forms is generally called radiation."

- For our purposes, the most significant type of radiation is probably electromagnetic radiation.
- There are further two types radiations:
  i. Ionizing Radiations
  ii. Non-Ionizing Radiations

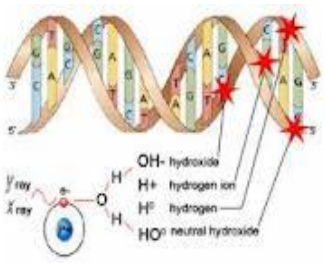
i. Ionizing Radiations:

Ionizing radiation is a type of short wavelength, high intensity radiation that is used to destroy all microorganisms during sterilization.

Mode of Action:

- The target of ionization radiation is DNA, which will undergo mutations on broad scale.
- Secondary lethal function is to bring about changes in organelles and the production of toxic chemical substances.

How Toxicity is produced?



•The interaction between high-energy gamma radiation and matter forms ion pairs by ejecting electrons, leading to free-radical formation. The free radicals are extremely reactive because each has an unpaired electron on one of its outer orbital.

•Free-radical reactions may involve gas liberation, double-bond formation, exchange reactions, electron migration, and cross-linking. In fact, any chemical bond may be broken and any potential chemical reaction may take place.

Dose:

- The dosage of radiation is measured in Grays.
- Depending upon the application, exposure ranges from 5-50 kilograys.

Gray:

The gray is the SI unit of absorbed radiation dose of ionizing radiation (for example, X-rays), and is defined as the absorption of one joule of ionizing radiation by one kilogram of matter (usually human tissue).

A whole-body exposure to 5 or more gray of high-energy radiation at one time usually leads to death within 14 days. This dosage represents 375 joules for a 75 kg adult .Since gray are such large amounts of radiation, medical use of radiation is typically measured in milligray (mGray)

Forms of Ionizing Radiations:

The forms of ionizing radiations used for sterilization are known as:

a. Gamma Irradiation
b. X-rays Irradiation
c. Cathode Rays Irradiation

a. Gamma rays

"Gamma radiations are high energy radiations emitted from certain radioactive isotopes such as ^{60}Co."

- Gamma rays are similar to X-Rays but are of shorter wavelength and higher energy.
- They are capable of great penetration into matter, and they are lethal to all lives, including microorganisms

Applications:

- Foods and medical devices
- Sterilization of single-use medical supplies such as syringes, implants, catheters, IV sets, surgical gloves, gauze
- Elimination of organisms from pharmaceuticals such as ointments and solutions.
- Sterilization of tissue/biological based products
- Microbial reduction on consumer products
- Pharmaceutical products cover a wide range of materials (e.g., containers, closures, excipients)
- Eye drops containers

b. X-Rays:

X-Rays are lethal to microorganisms and higher forms of life.

They have considerable energy and penetration ability. However, they are impractical for purposes of controlling microbial populations because:

- They are very expensive to produce in quantity.
- They are difficult to utilize efficiently, since radiations are given off in all directions from their point of origin.

Applications:

- X-Rays have been widely employed experimentally to produce microbial mutants.

c. Cathode Rays:

- Special types of equipment have been designed which produce electrons of very high intensities, and these electrons are accelerated to extremely high velocities.
- These intense beams of accelerated electrons are microbicidal as well as having other effects on biological and non-biological materials

Applications:

- Cathode rays are used today for sterilization of surgical supplies, drugs, and other materials.
- One of the unique features of the process is that the material can be sterilized after it has been packaged (the radiations penetrate the wrappings) at room temperature.

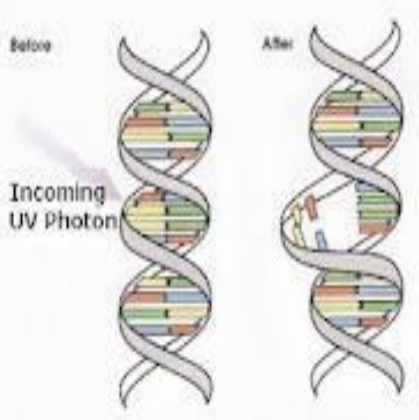
Limitation:

- Electron beam radiation has limited power of penetration; but within its limits of penetration, sterilization is accomplished on very brief exposure.

ii. **Non-Ionizing Radiations:**

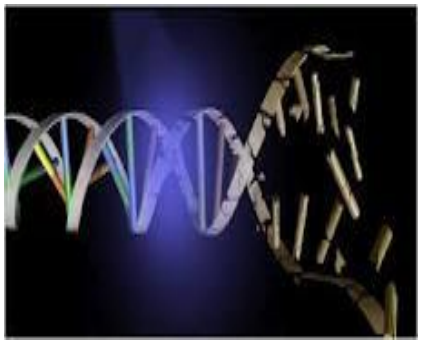
a. Non-ionizing radiations best exemplified by UV.
b. UV ranges in wavelength from approximately 100nm to 400nm. It is most lethal from 240nm to 280nm (with a peak at 260nm)

Mode of Action:



- Excites atoms by raising them to a higher energy state, but it does not ionize them. This atomic excitation, in turn, leads to the formation of abnormal bonds. Within molecules such as DNA and is thus a source of mutations.
- As UV radiation passes through a cell, it is initially absorbed by DNA. Specific molecular damage occurs on the pyrimidine bases (thymine and cytosine), which form abnormal linkages with each other called pyrimidine dimmers. These bonds occur between adjacent bases on the same DNA strand and interfere with normal DNA replication and transcription. The results are inhibition of growth and cellular death.

Applications:



- UV radiation is usually directed at disinfection rather then sterilization. Germicidal lamps can cut down the concentration of air born microbes as much as 99%
- They are used in hospital rooms, operating rooms, schools, food operation areas, and dental offices
- Ultraviolet disinfection of air has proved effective in reducing post-operative infection, preventing the transmission of infections by respiratory droplets, and reducing the growth of microbes in food processing plants and slaughter houses
- UV radiation can be used to treat drinking water and to purify other liquids (milk and fruit juices) as an alternative to heat.
- Ultraviolet treatment has proved effective in freeing vaccines and plasma from contaminants.
- The surfaces of solid, non-porous materials such as walls and floors, as well as meat, nuts, tissues for grafting, and drugs, have been successfully disinfected with UV

Limitations

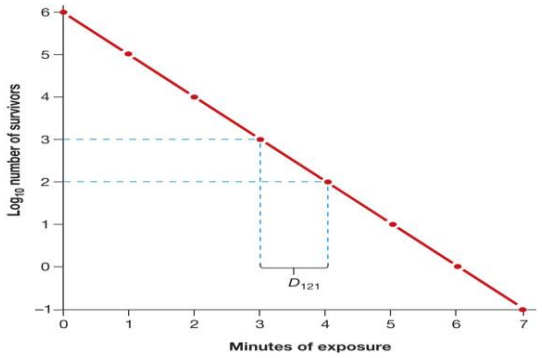
- One major disadvantage of UV is its poor powers of penetration through solid materials such as glass, metals, cloth, plastic, and even paper.
- Another drawback to UV is the damaging effect of over exposure on human tissues, including sun burn, retinal damage, cancer, and skin wrinkling.

THERMAL STERILIZATION

Thermal Death Point:
It is the lowest temperature required to kill all of the microorganisms in a suspension.

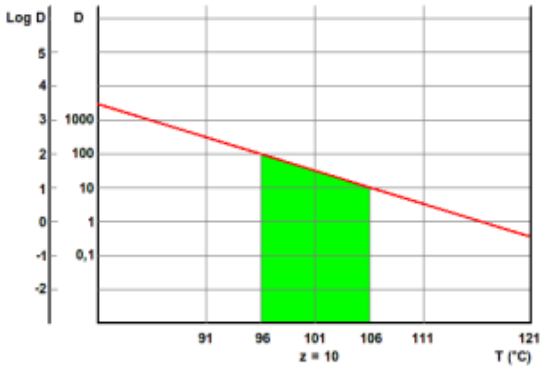
Thermal Death Time:
It is the shortest time required to kill all of the microorganisms in a suspension at a specific temperature.

D-value:
The time required to kill 90% of the microorganisms in a suspension at a specific temperature.



Z-value
The z-value is a measure of the change of the D-value with varying temperature.
It is the increase in temperature needed to reduced D-value to 1/10of original value

$$z = \frac{T_2 - T_1}{logD_1 - logD_2}$$



F-value
It is the time in minutes at a specific temperature (121°C or 250 F) needed to kill the population of cells and spores.

Example

- *Clostridium botulinum*
  - D-value at 121 °C is 0.204 min.
  - F-value at 121 °C is 2.5min
  - Z-value 10°C, it takes 10°C change in the temperature to alter the D-value tenfold. So, at 111°C D-value increase to 2.04 min and F-value at 111°C is 25min

D-value and Z-value for some food-borne pathogens

Organisms	D-value	Z-value
Clostridium botulinum	D_{121} = 0.204	10
Clostridium perfringens	D_{90} = 3-5	6-8
Salmonella	D_{60} = 0.39-0.40	4.9-5.1
Staphylococcus aureus	D_{60} = 5.17-5.37	5.2-5.8

Chemical Method of Sterilization

"Chemical sterilization refers to a technique of sterilization by making use of a chemical agent."

Chemical sterilization is typically used for devices that would be sensitive to the high heat used in steam sterilization, and for devices that may be damaged by irradiation (rubbers and plastics can become more brittle after irradiation.)

Types of Chemical Sterilization:

Chemical Sterilization can be of several types:

1. Sterilization by Gases
2. Sterilization by Liquids

1. Sterilization by Gases:

Sterilizing gases are typically used when exposure to other methods (heat or radiation) could damage the materials or equipment.

The most common gases used for sterilization are:

i. Ethylene Oxide
ii. Ozone

i. Ethylene Oxide:

Ethylene oxide (EO or EtO) gas is commonly used to sterilize objects sensitive to temperatures greater than 60 °C and/or radiation such as plastics, optics and electrics.

Standards for EtO:

- A clean object can be sterilized if treated for 5-8 hours at 38°C or 3-4 hours at 54°C
- Gas concentration between 200 and 800 mg/l
- Relative humidity should be 40-70%
- Because pure EtO is explosive, it is usually supplied in a 10 to 20% concentration mixed with either CO_2 or dichlorodifluoromethane

Precaution

- Extensive aeration of sterilized materials is necessary to remove residual EtO because it is so toxic.

Mode of action

- EtO is a very strong alkylating agent, and it reacts vigorously with functional groups of DNA and proteins. Through these actions, it blocks both DNA replication and enzymatic actions

Sterilization power:

- Ethylene oxide penetrates well, moving through paper, cloth, and some plastic films and is highly effective
- EtO can kill all known viruses, bacteria and fungi, including bacterial spores.

Applications:

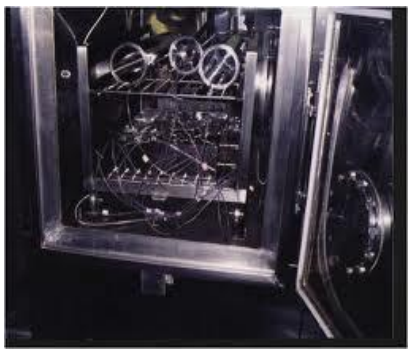
- EtO is an effective way to sterilize and disinfect plastic materials and delicate instruments in hospitals and industries.
- EtO can safely sterilize prepackaged medical devices and surgical supplies, syringes, and disposable petri plates.

Disadvantage:

- One of its disadvantages is its comparatively slow action upon microorganisms
- It is highly flammable.
- It is highly toxic.
- It is carcinogenic.







ii. Ozone:

- **Ozone (O_3)** is an unstable, color-less gas.
- It is a powerful oxidizer and a potent germicide.
- In fact, ozone possesses much higher disinfection capabilities than chlorine and other commonly used disinfectants.
- It is a toxic and unstable gas that must be produced on-site, so it is not practical to use in many settings.
- Ozone offers many advantages as a sterilant gas capable of destroying a wide range of pathogens, including prions without the need for handling hazardous chemicals since the ozone is generated within the sterilizer from medical grade oxygen.

Mode of Action:

- Its mode of action is oxidation.

Concentration used:

- 0.3-5 ppm

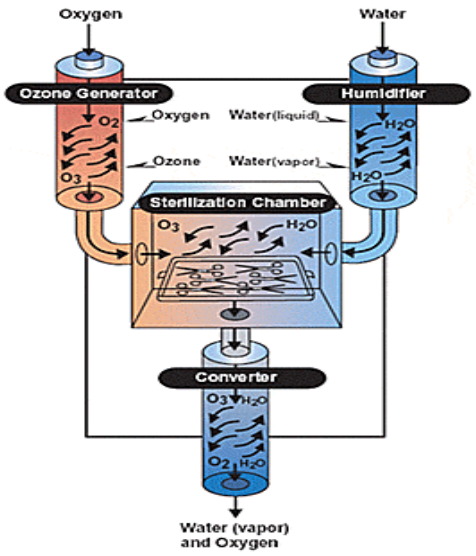
Applications:

- Mostly used for the sterilization of water

Concentration of O_3	(ppm)	Contact Time (min)
Reverse osmosis water	0.3-0.5	4-5
Drinking Water	1.0-2.0	5-10
Pool	0.3-0.7	1
Seafood Wash	0.1-0.15	1-2
Fruit and Vegetable Wash	0.2-0.4	1-5
Cooling Tower	0.2-0.5	2
Pre-Surgical Wash	3-5	3-5

❖ The equipment is mainly used in water treatment project. It can produce great deal of ozone to clean and sterilize. The ozone-sterilization has advantages of quick speed. Harmlessness, and it could not cause pollution again. It is the ideal sterilizer for bottled mineral water and pure







2. Sterilization by Liquids:

"Liquid Chemical Sterilization is used to sterilize heat sensitive devices which can be immersed."

- ❖ The method requires the item to be completely immersed in a germicidal solution for a prescribed period of time to kill all the bacterial micro-organisms.
- ❖ Liquid chemical sterilization involves a two-part process:
  - Devices are treated with a liquid chemical germicide (LCG).
  - The processed devices are rinsed with water to remove the chemical residues.
- ❖ The liquids used for sterilizing are as follows:
  i. Chlorine Bleach
  ii. Glutaraldehyde and Formaldehyde
  iii. Phthaldehyde
  iv. Hydrogen Peroxide

i. Chlorine Bleach:

It is another accepted liquid sterilizing agent.
Household bleach consists of 6.25% sodium hypochlorite.

Concentration Used:

- It is usually diluted to 1/10 immediately before use; however, to kill *Mycobacterium tuberculosis* it should be diluted only 1/5, and 1/2.5 to inactivate prions.
- For water it is used 0.6-1 ppm

Time:

- Bleach will kill many organisms immediately, but for full sterilization it should be allowed to react for 20 minutes.

Mode of action

- Oxidation
- In solution, these compounds combine with water and release hypochlorous acid (HOCl) and then atomic oxygen [O], which oxidizes the sulfhydryl (S-H) group on amino acid cysteine and interferes with disulfide (S-S) bridges on numerous enzymes

Disadvantage:

- Bleach will kill many, but not all spores.
- It is also highly corrosive.
- Bleach decomposes over time when exposed to air, so fresh solutions should be made daily.

ii. Glutaraldehyde and Formaldehyde:

- Solutions are accepted liquid sterilizing agents.
- To kill all spores in a clear liquid can take up to 22 hours with glutaraldehyde and even longer with formaldehyde.
- Glutaraldehyde and formaldehyde are volatile, and toxic by both skin contact and inhalation.

Disadvantage:

- Long duration for sterilization
- Causes prions resistance to itself and also to other chemical sterilants.

iii. **Phthalaldehyde:**

- Ortho-phthalaldehyde (OPA) is a chemical sterilant.
- Typically used in a 0.55% solution, high-level disinfectant at an immersion time of 12 minutes at 20°C and 5 minutes at 25°C.
- OPA shows better myco-bactericidal activity than glutaraldehyde.
- Also effective against glutaraldehyde-resistant spores.
- OPA has superior stability, is less volatile, and does not irritate skin or eyes, and it acts more quickly than glutaraldehyde.

Mode of action:

- Forms covalent cross-links with functional groups of proteins.

Disadvantages:

- It is more expensive and will stain proteins (including skin) gray in color.
- Some side effects from equipment sterilized using this reagent have been reported. For example, two cases of anaphylaxis following cystoscopy with endoscopes sterilized with OPA were reported

iv. **Hydrogen peroxide:**

- Hydrogen peroxide (H_2O_2) is a colourless, caustic liquid that decomposes in the presence of light, metals, or catalase into water and oxygen gas. Early formulations were unstable and inhibited by organic matter, but manufacturing methods now permit synthesis of hydrogen peroxide so stable that even dilute solutions retain activity through several months of storage.

Mode of action:

- Strong oxidizing agent
- The germicidal effects of hydrogen peroxide are due to the direct and indirect actions of oxygen. Oxygen forms hydroxyl free radicals (OH), which, like the superoxide radical, are highly toxic and reactive to cells.

Microbicidal Activity:

- Hydrogen peroxide is bactericidal, virucidal and fungicidal and in higher conc. sporicidal.

Concentration used:

- Mostly 0.2-3%.
- In medical sterilization hydrogen peroxide is used at higher concentrations, ranging from around 35% up to 90%.

Time for sterilization:

- Up to 28-30 min

Materials sterilized:

- Hospital Endoscopes
- Catheters







Mechanical Methods of Sterilization

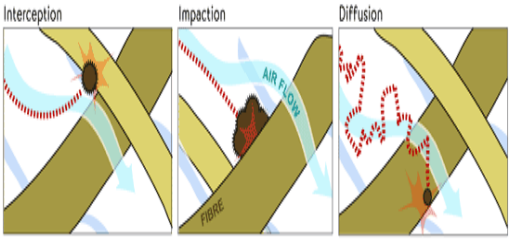
"Mechanical method for sterilization involves filtration."

FILTRATION

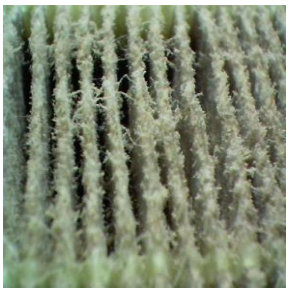
- It is a type of sterilization in which the sterilization is carried out by using filters.
- Air and liquids are sterilized using filtration.

1. Sterilization of Air:

- Air is sterilized by using HEPA Filters.
- **HEPA Filter:**
  - It is the abbreviation of High Efficiency Particulate Air) Filer
  - Its efficiency is 99.97%
  - Its pore size is 0.3 µm.
  - **Working:**
    - **Interception:** When particles come within one radius of a fiber and stick to it.
    - **Impaction:** When larger particles collide with fibers and embed in them.
    - **Diffusion:** When the smallest particles collide with gas molecules. This slows down their velocity and makes them more are likely to be stopped by the other two mechanisms



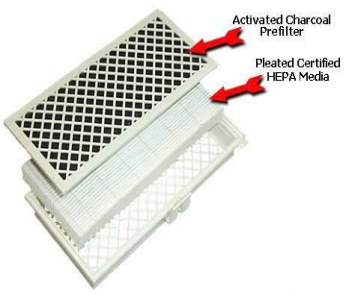
  - **Components:**
    i. HEPA Filter Media
    ii. HEPA Filter Separators
    iii. HEPA Filter Pack
    iv. HEPA Filter Sealant
    v. HEPA Filter Seal



    Typically, a microfine fiberglass media, synthetic fibers, expanded film such as Polytetrafluorethylene (PTFE). Close pleating is necessary to fit all the required media into the desired space, because the paper has a high resistance to airflow and the media velocity is usually in the range of 6 feet per minute



  - **When to Change Your HEPA Filter:**
    HEPA filters can have a lifespan of 1-5 years, depending on the prefiltration that is used
  - **Applications:**
    i. Medical/Biological HEPA Filter Systems
    ii. Commercial/Industrial HEPA Filter System
    iii. Residential HEPA Filter Systems

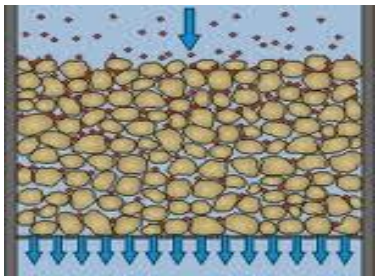


Medical/Biological HEPA Filter Systems:
These can be of following types:

- Biological Safety Cabinet HEPA Filter Systems
- Laminar Flow Bench HEPA Filter Systems
- Operating Room Ceiling Mounted HEPA Filters
- Isolation Room Portable HEPA Filters

2. Sterilization of Liquids:

- Sterilization of liquids is carried out by filtration.
- There are two types of filters used for sterilization of liquids:
  - Membrane filters
  - Depth filters
- **Pore size:**
  - There pore size is 0.22-0.45 µm.
  - Some filters of pore size 0.01 µm are also available which can retain viruses
- **Thickness:** 0.1 mm
- **Materials of Filter:**
  - Filters are made up of cellulose esters and plastic polymers
- **Material Filtered:**
  The materials that are filtered are as follows:
  - Serum
  - Blood products
  - IV fluids
  - Enzymes





DISINFECTION

"The process of cleaning something, especially with a chemical in order to destroy bacteria."

- Disinfection does not necessarily kill all microorganisms, especially resistant bacterial spores; it is less effective than sterilization, which is an extreme physical and/or chemical process that kills all types of life.

Disinfectant:

"Disinfectants are antimicrobial agents that are applied to the surface of non-living objects to destroy microorganisms that are living on the objects."

- Disinfectants are different from other antimicrobial agents such as antibiotics, which destroy microorganisms within the body, and antiseptics, which destroy microorganisms on living tissue

Types of Disinfection

There are two types of disinfection:

- Physical Methods
- Chemical Methods

Physical Methods of Disinfection:

Physically disinfection can be done by following methods:

1. Direct Boiling
2. Pasteurization

1. Direct Boiling:

Objects can be disinfected by boiling it directly. Boiling can kill the microorganisms present on the objects.
e.g. Water is disinfected by boiling.

2. Pasteurization:

"Pasteurization is a process of heating a food, usually a liquid, to a specific temperature for a definite length of time and then cooling it immediately."

- This process slows microbial growth in the food.
- Pasteurization aims to reduce the number of viable pathogens, so they are unlikely to cause disease
- **Methods of Pasteurization:**
  There are three methods of Pasteurization:
  i. **Long Time Method:** In this method the pasteurization is brought about by heating at 63°C for 30 minutes.
  ii. **High Temperature/Short Time Method:** In this method pasteurization is achieved by heating at 71.7^{o}C for 15-20 seconds.
  iii. **Ultra-High Temperature:** In this method pasteurization is done at 140°C to 150°C for 3 to 1 seconds

- **Aims of Pasteurization:**
  - One important aim in pasteurization is to prevent the transmission of milk borne diseases from infected cows or milk handlers
  - The primary targets of pasteurization are non-spore forming pathogens:
    - *Salmonella* species (a common cause of food infection),
    - *Campylobacter jejuni* (acute intestinal infection),
    - *Listeria monocytogenes* (listerioses),
    - *Brucella* species (undulant fever),
    - *Coxiella burnetii* (Q fever),
    - *Mycobacterium bovis*,
    - *M. tuberculosis*
    - Several enteric viruses.



Chemical Methods of Disinfection:

Disinfection can also be brought about by chemical means using chemical reagents which are used to kill microorganisms.

Chemical Disinfectant:

Chemical disinfectant is the chemical used to disinfect objects.
Following disinfectants are usually used for disinfection purposes:

1. Alcohols
2. Dyes
3. Hydrogen Peroxide
4. Aldehydes
5. Alkalis and Acids
6. Heavy Metals
7. Phenol and Phenolic Compounds
8. Detergents
9. Halogens

1. Alcohols:

- Alcohols are protein denaturants, and accounts for their antimicrobial activity.
- Alcohols are also solvents for lipids, and hence they may damage lipid complexes in cell membrane.
- They are also dehydrating agents.

i. Methyl alcohol:

- Less bactericidal than ethyl alcohol; furthermore, it is highly poisonous. Even the fumes of this compound may produce permanent injury to the eyes, and is not generally employed for the destruction of microorganisms

ii. Ethyl Alcohol:

- In conc. between 50 and 90%, is effective against vegetative or non-sporeforming cells. For practical applications 70% of alcohol is generally used

iii. The Higher Alcohols (Propyl, Butyl, Amyl, and others):

- More germicidal than ethyl alcohol. In fact, there is a progressive increase in germicidal power as the molecular weight of alcohols increases (Since alcohols of molecular weight higher than that of propyl alcohol are not miscible in all proportions with water, they are not commonly used as disinfectants). Propyl and isopropyl alcohols in conc. ranging from 40-80% are bactericidal for vegetative cells.

- **Applications of Alcohol:**
  - Alcohol is effective in reducing the microbial flora of skin and for the disinfection of clinical oral thermometers
  - Ethyl alcohol is germicidal, non-irritating, and inexpensive. Solution of 70-95% are routinely used as skin degerming agent because the surfactant action removes skin oil and some microbes sheltered in deeper skin layers (One limitation to its effectiveness is the rate at which it evaporates)
  - Ethyl alcohol is occasionally used to disinfect electrodes, face masks, and thermometers which are first cleaned and then soaked in alcohol for 15-20 minutes. Isopropyl alcohol, sold as rubbing alcohol, is even more microbicidal and less expensive than ethanol

2. Dyes:

- Two classes of dye compounds which have antimicrobial properties are of special interest. These are triphenylmethane and acridine dyes

i. Triphenylmethane dyes:

- Included in this category are malachite green, brilliant green, and crystal violet. Gram +ve organisms are more susceptible to lower concentrations of these compounds than are Gram –ve ones. Crystal violet will inhibit Gram +ve cocci, E.coli, and S.aureus.
- **Mechanism of action:**
  They exert their inhibitory effect by interfering with cellular oxidation processes

- **Application:**
  - Certain media can be made selective by the incorporation of low concentrations of the crystal violet, brilliant green or malachite green.
  - Crystal violet has also been used as a fungicide

3. Hydrogen Peroxide:

- The germicidal effects of hydrogen peroxide are due to the direct and indirect actions of oxygen. Oxygen forms hydroxyl free radicals (OH), which, like the superoxide radical, are highly toxic and reactive to cells
- **Applications:**
  - Hydrogen peroxide is also a versatile disinfectant for soft contact lenses, surgical implants, plastic equipments, utensils, beddings and room interiors

4. Aldehydes:

- Two of the most effective are formaldehyde and glutaraldehyde. Both are highly microbicidal, and both have the ability to kill spores (sporicidal).

i. Formaldehyde:

- It is a gas that is stable only in high conc. and at elevated temperatures. At room temperatures it polymerizes, forming a solid substance. The important polymer is paraformaldehyde, a colourless substance which rapidly yields formaldehyde upon heating. Formaldehyde is also marketed in aqueous solution as formalin, which contains 37-40% formaldehyde.
- **Mechanism of action:**
  - Formaldehyde is an extremely reactive chemical. It combines readily with vital organic nitrogen compounds such as proteins, nucleic acids. It is likely that interaction of formaldehyde with these cellular substances accounts for its antimicrobial actions.
- **Application:**
  - Formaldehyde in gaseous form can be used for disinfection and sterilization of enclosed areas.
  - Formalin and paraformaldehyde are two principle sources of formaldehyde when it is used for gaseous disinfection. Foramalin tincture (8%) has limited use as a disinfectant for surgical instruments

ii. Glutaraldehyde:

- Glutaraldehyde is a yellow acidic liquid with a mild odour
- **Mechanism of action:**
  - The mechanism of activity involves cross linking protein molecules on the cell surface. In this process, amino acids are alkylated, meaning that a hydrogen atom on an amino acid is replaced by glutaraldehyde molecule itself. It can also irreversibly disrupt the activity of enzymes within the cell
- **Application:**
  - Glutaraldehyde is rapid and broad spectrum, and is one of the few chemicals officially accepted as a sterilant and high level disinfectant. It kills spores in 3 hours and fungi and vegetative bacteria in a few minutes. Viruses, including the most resistant forms, appear to be inactivated after relatively short exposure times

5. Alkalis and acids:

- Conditions of very low or high pH can destroy or inhibit microbial cells; but they are limited in applications due to their corrosive, caustic and hazardous nature
- **Application:**
  - Aqueous solutions of ammonium hydroxide remain a common component of detergents, cleansers, and deodorizers.
  - Organic acids are widely used in food preservation because they prevent spore germination and bacterial and fungal growth and because they are generally regarded as safe to eat.
  - Acetic acid is a pickling agent that inhibits bacterial growth.

- Propionic acid is commonly incorporated into breads and cakes to retard mold.
- Benzoic and sorbic acids are added to beverages, syrups and margarine to inhibit yeast.

6. Heavy metals:

- Various forms of metallic elements mercury, silver, gold, copper, arsenic and zinc have been applied in microbial control over several centuries
- **Mechanism of action:**
  - Mercury, silver and most other metals exert microbicidal effects by binding onto functional groups of proteins (mostly sulfhydryl) and inactivating the, rapidly bringing metabolism to a standstill. This mode of action can destroy many types of microbes, including vegetative bacteria, fungal cells and spores, algae, protozoa, and viruses (but not endospores).
- **Applications:**
  - Weak (0.001-0.2%) organic mercury tinctures such as thymerosol and nitromersol are fairly effective antiseptics and infections preventives, but they should never be used on broken skin because they are harmful and can delay healing.
  - The organic mercurials also serve as preservative in cosmetics and ophthalmic solutions.

7. Phenol and phenolic compounds:

- Phenol (carbolic acid) is poisonous compound derived from the distillation of coal tar
- **Mechanism of action:**
  - Depending upon the concentration of the phenolic compound to which microbial cells are exposed causes disruption of cells, precipitation of cell protein, inactivation of enzymes, and leakage of amino acids from the cells
- **Application:**
  - Aqueous solutions form 2-5% can be employed to disinfect such materials as sputum, urine, feaces, contaminated instruments and utensils.
  - One of the widely used phenolic derivatives is o-phenylphenol. Combination of compounds of this class with detergents results in products with good disinfectant as well as detergent properties

i. **Cresols:**
   - These are simple phenolic derivatives that are combined with soap for intermediate or low levels of disinfection in the hospital.

ii. **Hexylresorcinol:**
   - Derivative of phenol; is marketed in a solution of glycerin and water. It is a strong surface-tension reductant, which may account in part for its high bactericidal activity.

iii. **Qualitrol:**
   - It is composed of:
     - ✓ Benzyl Chloro Phenol: 10% w/v
     - ✓ Ortho Phenyl Phenol: 12% w/v
     - ✓ Tertiary Amyl Phenol: 5% w/v

iv. **NAC-PHENE 256:**

- A hospital grade disinfectant cleaner designed for meeting the demands of the health care community. This product is effective against tuberculosis and the AIDS virus
- It is composed of:
  - ✓ o-Phenylphenol
  - ✓ o-Benzyl-p-Chlorophenol
  - ✓ p-Tertiary Amyl Phenol
  - ✓ Sodium Dodecyl diphenyloxide Disulfonate
  - ✓ Isopropanol
- **Directions for Use:**
  - ✓ This is an extremely effective cleaner and disinfectant, recommended for general hospital use on hard, non-porous surfaces such as floors, walls, woodwork, and equipment in all departments. For disinfection: Pre-clean surfaces before disinfection. Apply ½ oz. to 1 gallon of water to hard non-porous surfaces using cloth, sponge, or mop. Thoroughly wet surfaces for 10 minutes. Allow surfaces to air dry. Prepare a fresh solution daily

8. Detergents:

- Detergents are organic molecules that serve as wetting agents and emulsifier because they have both polar hydrophilic and non-polar hydrophobic ends. They are surface tension depressants.
- **Classification:**
  - Anionic detergents e.g. Sodium lauryl sulfate.
  - Cationic detergents e.g. Cetylpyridinium chloride.
  - Nonionic surfactants. These do not possess significant antimicrobial activity.
  - (Cationic detergents are regarded as more germicidal than anionic compounds).
- **Soaps:**
  - They destroy only highly sensitive form such as agents of gonorrhea, meningitis, and syphilis. The common hospital pathogen *Pseudomonas* is so resistant to soap that various species grow abundantly in soap dishes
  - Soap function primarily as cleansing agents and sanitizers in industry and the home. Soaps mechanically remove large amounts of surface oil, greases, and other debris that contain microorganisms.
  - The most popular of these disinfectants are quaternary ammonium compounds characterized by positively charged quaternary nitrogen and a long hydrophobic aliphatic chain.
  - **Mechanism of action:**
    - □ These include Denaturation of proteins, interference with glycolysis and membrane damage.
  - **Applications:**
    - □ Quaternaries are widely used for control of microorganisms on floors, walls, and other surfaces in hospitals, nursing homes and other public places.
    - □ They are used to sanitize food and beverages utensils in restaurant as well as surfaces and certain equipment in food processing plant.

- Other applications are to be found in the dairy, egg and finishing industries to control microbial growth and surfaces of equipment and the environment in general.

9. Halogens:

- The halogens iodine and chlorine are important antimicrobial agents
  - **Iodine:**
    - Mostly used as antiseptic
  - **Chlorine:**
    - Chlorine has been used for disinfection and antisepsis for approximately 200 years. The major forms used in microbial control are liquid and gaseous chlorine (Cl_2), hypochlorite (OCl), and chloramines (NH_2Cl).
    - **Mechanism of action:**
      - In solution, these compounds combine with water and release hypochlorous acid (HOCl) and then atomic oxygen [O], which oxidizes the sulfhydryl (S-H) group on amino acid cysteine and interferes with disulfide (S-S) bridges on numerous enzymes. The resulting Denaturation of enzyme is permanent and suspends metabolic reactions.
    - **Antimicrobial spectrum:**
      - Chlorine kills not only bacteria and endospores but also fungi and viruses
    - **Applications:**
      - Gaseous and liquid chlorine are used almost exclusively for large-scale disinfection of drinking water, sewage, and wastewater from such sources as agriculture and industry.
      - Chlorination to conc. of 0.6 to 1.0 parts of chlorine per million parts if water is safe to drink.

a. **Hypochlorites:**
   - Common house hold bleach contains sodium hypochlorite
   - The scope of applications is too broad, including sanitization and disinfection of food equipment in dairies, restaurants, canneries and treatment of swimming pools

b. **Chloramines (dichloramine, halazone):**
   - Are being employed more frequently as an alternate to pure chlorine in treating water supplies

ANTISEPTICS:

"Antiseptics are antimicrobial substances that are applied to living tissue/skin to reduce the possibility of infection, sepsis, or putrefaction"

- The major antiseptics used in daily life are as follows:

1. Alcohols:

- Mostly 70% alcohols are used, ethanol and isoprolyl alcohol are commonly used

2. Hydrogen peroxide (H_2O_2):

- Used as Antiseptic in washing wounds and also used in ear washing.

3. Heavy metals:

- A silver compound with several applications is silver nitrate solution. Solutions of silver nitrate (1-2%) can also be used for topical germicides on mouth ulcers.
- Silver sulfadiazine ointment, when added to dressings, effectively prevents infection in burn patients, and pure silver is now incorporated into catheters to prevent urinary tract infections in the hospital.
- A 1% solution of silver nitrate is added to the eyes of infants to prevent ophthalmic gonorrhea.





4. Detergents:

- Hand wash and soaps are used as antiseptics

5. Phenols:

- Used as a "scrub" for pre-operative hand cleansing. Used in the form of a powder as an antiseptic baby powder. Also used in mouthwashes and throat lozenges, where it has a painkilling effect as well as an antiseptic one. Other phenolic antiseptics include historically important, but today rarely used (sometimes in dental surgery) thymol, hexachlorophene are used.

6. Halogens:

i. **Iodine:**

- **MOA:** iodine kills through the inactivation of protein synthesis and disruption of protein and nucleic acid structure in a microbe
- All classes of microorganisms are killed by iodine if proper concentrations and exposure times are used. (Iodine activity is not as adversely affected by organic matter and pH as chlorine is)

- **Tincture of iodine:**
  - The element iodine is used as germicidal agent in a form referred to as tincture of iodine. There are several available preparations, such as 2% iodine + 2% sodium iodide diluted in alcohol, 7% iodine + 5% KI in 83% alcohol and 5% iodine in 10% KI in aqueous solution.





- **Iodophores:**
  - Iodophores are complexes of iodine and a neutral polymer such as polyvinyl alcohol. Iodophores are water soluble and non-staining. This formulation allows slow release of free iodine and increases its degree of penetration. These compounds have largely replaced free iodine solution in medical antisepsis because they are less prone to staining or irritating tissues.





7. Mouthwash:

- Mouthwash or mouth rinse is a product used to enhance oral hygiene
- **Active ingredients:**
  - Active ingredients in commercial brands of mouthwash can include thymol, eucalyptol, hexetidine, methyl salicylate, menthol, chlorhexidine gluconate, benzalkonium chloride,cetylpyridinium chloride,methylparaben, hydrogen peroxide, bromide and sometimes fluoride,enzymes, and calcium. Ingredients also include water, sweeteners such as sorbitol, sucralose, sodium saccharin.
- **Brands:**
  - Astring-O-Sol
  - Colgate
  - Meridol
  - Listerine
  - Oral-B
  - Scope (mouthwash)

8. Dettol:

- The active ingredient in Dettol that confers its antiseptic property is chloroxylenol (C_8H_9ClO), an aromatic chemical compound. Chloroxylenol comprises 4.8% of Dettol's total mixture,with the rest composed of pine oil, isopropanol, castor oil soap, caramel, and water.

IMMUNOLOGY

CHAPTER NO. 04

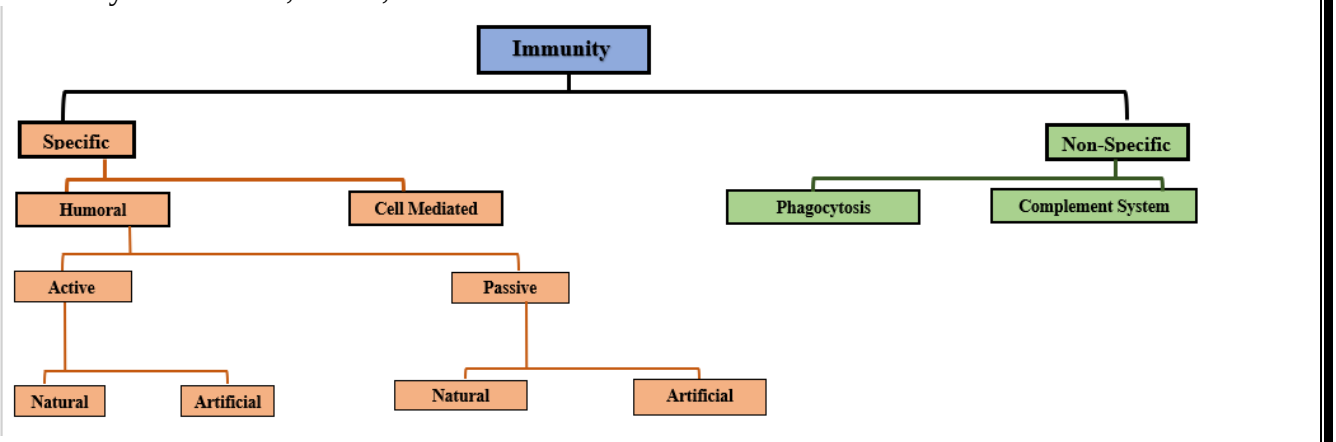
Dr. Qazi Adnan

IMMUNOLOGY

"The branch of medicine and biology concerned with immunity."

Introduction:

- Immunology charts, measures, and contextualizes the physiological functioning of the immune system in states of both health and diseases; malfunctions of the immune system in immunological disorders (such as autoimmune diseases, hypersensitivities, immune deficiency, and transplant rejection); and the physical, chemical, and physiological characteristics of the components of the immune system *in vitro*, *in situ*, and *in vivo*.



Host Defenses:

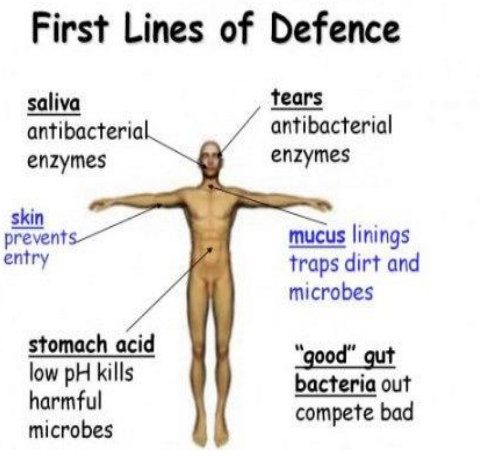
Immunology is the study of the specific defenses of a host against diseases. In addition to these specific immune responses, host defenses also include non-specific resistance. Acting together, the components of the immune system work to protect the host against infections.

1. Non-Specific Defenses:

- The human body has many non-specific defenses to prevent disease.
- These include:
  - i. First Line Defense
  - ii. Phagocytosis
  - iii. Inflammation
  - iv. Complement System

i. First Line Defense:

- Several components make up the first line of defense.



- The skin and mucous membranes serve as physical barriers to prevent microorganisms and harmful substances from entering the human body.
- The very high acidity of gastric juices can destroy many pathogens.
- Tears prevent the microorganism from growing on the surfaces of eye-ball.
- Tears, perspiration and saliva contains lysozyme enzymes which breakdown the cell-wall of bacteria.
- First Line Defenses also include proteins such as the iron binding trans-ferans which inhibit bacterial growth by reducing the amount of available Iron in blood.

ii. Phagocytosis:

"*Phagocytosis is the ingestion of a solid such as a microorganism or cellular debris by a eukaryotic cell."*

- Phagocytosis is used by some protozoans for obtaining nutrients, but this process is also used by certain cells of immune system to fight.
- Certain immune system cells such as macrophages and neutrophils use phagocytosis to ingest and destroy the invading microorganisms.
- These phagocytes attack and destroy invaders using chemotaxis, adherence, ingestion, digestion and elimination

a) Chemotaxis:

"Chemotaxis is the movement of the cell toward or away from a chemical stimulus."

- Phagocytes use pseudopods to move towards microorganism or damaged cells at the site of infection. Often arriving only minutes after the infection. Chemotactic chemicals attract the phagocytes to the site of infection and aids in the adherence of phagocytes to the side of microorganisms.
- Chemotactic chemicals that attract phagocytes include microbial products, components of damaged cells, chemicals released by other white blood cells and peptides derived from complement system. The complement system is another form of host defense.

b) Adherence:

"The plasma membrane of phagocytes attaches to the surface of microorganisms or other foreign material, this process is called adherence."

- Phagocytes can adhere to certain components of microorganisms often interacting with glycoproteins.
- This can be made easier through a process called opsonization in which the microorganisms are coded with serum proteins called opsonins. Opsonins function as handles for phagocytosis and include certain components of the compliment system as well as components of the humoral immune system called antibodies.

c) Ingestion:

- During ingestion the macrophage expands pseudopods to surround and engulf the microorganism.
- The pseudopods fuse and forming a sac called phagosome.

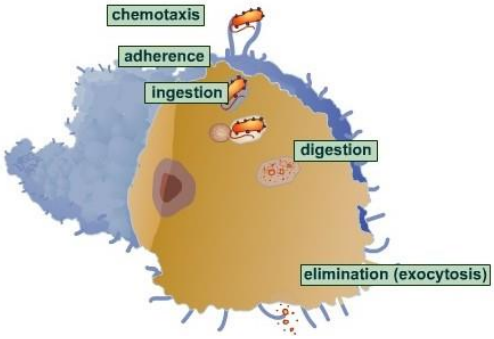
d) Digestion:

- The phagosome enters the cytoplasm where it fuses with lysosomes which contain digestive enzymes and other antimicrobial substances.
- The fused structure is called a phagolysosome.
- Digestion of most bacteria is complete within 10 to 30 minutes.

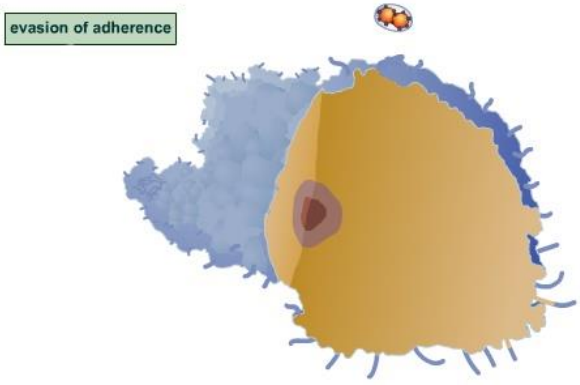
e) Elimination:

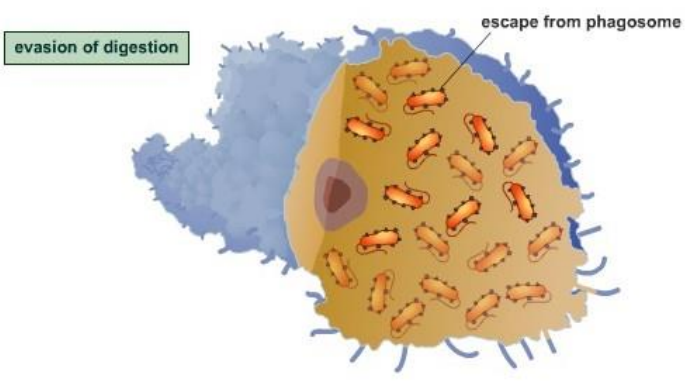
"After phagocytosis the remaining debris is eliminated from the cell by exocytosis."

- In which the phagolysosome fuses with the plasma membrane and expels its contents.
- Some of the contents of phagolysosome may also be presented on the cell membrane through a process called antigen processing and presentation.



- **Evasion of Adherence:**
  - Some bacteria have structures that help them avoid adherence. For example, the protein of Streptococcus pyogenes and capsules of microorganisms such as *Streptococcus pneumoniae* and *Hemophilus influenzae* type B inhibit the attachment of phagocyte to the surfaces of bacteria, making the bacteria slippery. These microorganisms can only be phagocytized if the phagocyte grabs the bacteria against the rough surface such as walls of the blood vessel, a connective tissue fiber or a blood clot.
- **Evasion of Digestion:**
  - Some organisms are ingested but not killed.
  - Some attack the phagosome membrane allowing to escape from the phagosome to reproduce inside the phagocyte. They then escape from the phagocyte by using the same process on the cytoplasmic membrane.
  - Others prevents the phagosome from fusing with the lysosome. While others cause the release of phagocyte's lysosomal enzyme into the cytoplasm killing the phagocyte.
  - Some microorganism such as *Coxiela brunetii*, the causative agent of Q-fever even require the low pH inside the phagolysosome to reproduce.





iii. Inflammation:

"Damage to the body tissues through heat, chemicals, sun burns, cuts or microbial infections triggers a general non-specific defensive response called inflammation."

- Signs and symptoms of inflammation include redness, pain, heat, swelling and sometimes loss of function.
- Inflammation has three main functions:
  - To destroy the agent causing injury.
  - To limit the effect of the agents on rest of the body.
  - To repair or replace the damaged tissue.
- Inflammation consists of the following stages:

a) Vasodilation:

- Damaged cells release several chemicals such as histamines, prostaglandins and leukotrienes. Certain components of the complement system can also stimulate the release of histamines.
- One function of these chemicals is to cause **vasodilation**.
- Vasodilation results in more blood and therefore more phagocytes, oxygen and nutrients being delivered to the site of injury. This increased blood flow results in redness, heat, swelling and some of the pain associated with inflammation.

b) Phagocyte Migration:

- The increased blood flow delivers phagocytes such as monocytes and neutrophils to the scene. The phagocytes stick to walls of blood vessels in the process called margination then squeeze through the gaps between the cells in a process called emigration or diapedesis. Within about hour of tissue damage the phagocytes begin to destroy the invading microorganisms by phagocytosis. After engulfing a large number of microorganisms and damaged tissue, the phagocytes die forming puss.

c) Tissue Repair:

- The final stage of inflammation is tissue repair.
- The increased delivery of nutrients and oxygen from vasodilation aids in this process. Other components of the blood lead to the formation of blood clot which seals off the site of injury.

d) Destruction of the Injury-causing Agent:

- The migration of phagocytes and other factors to the site of injury usually results in the destruction of microorganism. Phagocytes directly attack the microorganisms but other components of the body's defense systems such as the compliment system are involved.

e) Repairing and Replacing Damaged Tissue:

- Inflammation leads to repair and replacement of damaged tissue when possible. Different tissues have different capacities to regenerate themselves. While the outer layer of skin is completely able to repair a small cut leaving no scar. Cardiac muscle tissue does not regenerate at all.

iv. Complement System:

- The compliment system is composed of several proteins that circulate in the blood serum.
- Activation of the compliment system initiates a cascade of events that aids in the destruction of invading microorganisms.
- The compliment system helps inducing the inflammatory response.
- It produces opsonin and chemotactic factors that aid in phagocytosis.
- Components of complement system also directly attack the cell membrane of microorganisms causing cytolysis.

2. Specific Defenses:

- The human body has two main divisions of specific defense:
  i. Humoral Immunity
  ii. Cell-Mediated Immunity
- Antigen presenting cells aid in the activation of these defenses.
- Antigen processing and presentation is used by the host to provide a window into the cell for components of the immune system that are responsible for tracking down infected or cancerous cells in the body. Most nucleated cells present a sampling of the proteins produced inside them and specialized antigen presenting cells present peptides from the proteins of the engulfed invaders.

i. Humoral Immunity:

- Humoral Immune response consists of the activity of B-cells and the antibodies they produce.
- The process of B-cell activation is called clonal selection.
- **Process:**
  - During this process a B-cell with a specific antigen receptor recognizes an antigen. And this cell proliferates to produce a multitude of cells also called a clone of cells with the same specificity. Which differentiates into memory B-cells and plasma cells.
  - The antibodies produced by the plasma cells bind to the antigens to produce the humoral response.

a) B-Cell Activation:

- In the simplest form of B-cell activation an antigen called a T-independent-antigen enters the lymphatic tissue and binds to an antigen receptor also called a B-Cell Receptor (BCR) on the surface of a B-cell.
- These receptors are membrane bound versions of the antibodies that can be produced by that cell.
- Each of the B-cell receptor on a B-cell is identical and binds specifically to one antigen. However, the body produces billions of different B-cells each of which produces a different receptor making it likely that atleast one B-cell will recognize the antigen.

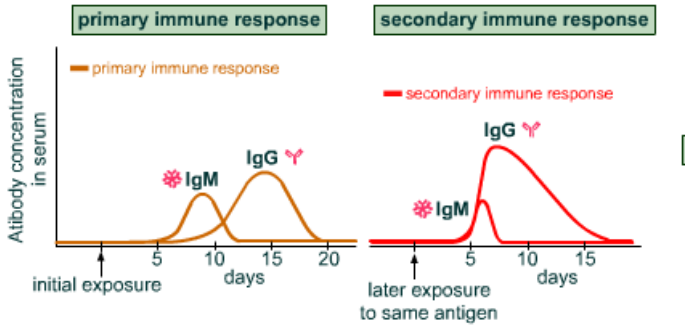
b) B-Cell Proliferation:

- Once a B-cell becomes activated, it proliferates or divides forming a clone of cells specific to the same antigen. While some of these B-cells differentiate into long lived memory B-cells, most differentiate into plasma cells which secrete antibodies against the antigen that activated them.
- The antibodies can enter the bloodstream and travel throughout the body.
- Plasma cells only live for a few days but each one of them can produce two thousand antibodies every second resulting in an enormous quantity of circulating antibodies.

c) Antibody Functions:

- The antibodies produced by plasma cells have several functions.
- Since each antibody has atleast two antigen binding sites.
- Antibodies can bind to more than one microbial cell resulting in agglutination or clumping.
- This agglutination helps remove the microorganisms from the body.
- Antibodies can also neutralize toxins by resisting bacteria by binding to critical portions of the toxins or blocking proteins necessary for attachment to cells.
- Antibodies also act as opsonins for phagocytosis and activate the compliment system.

- **Primary Immune Response:**
  - The humoral immune system experiences periods of ramping up and gradual diminishing in response to the foreign invader. After the first exposure to an antigen there is a lag period when the exposed person's serum does not contain any detectable antibodies. After several days the concentration of antibodies slowly rises followed by a gradual decline. This response is called a primary immune response.
  - **Secondary Immune Response:**
    - During the primary immune response, some activated B-cells differentiate into long lived memory cells. If these cells become stimulated by the original antigen even years later, they rapidly produce antibodies secreting plasma cells. This faster stronger response is called a secondary immune response or a memory response.



Types of Humoral Immunity:

- It is of further of two types;
  a) Active Immunity
  b) Passive Immunity

a) Active Immunity:

- Immunity to disease based on previous exposure to antigen is called as active immunity.
- It can be of further two types i.e. Natural or Artificial
  - **Natural:** When a person is exposed to an infection (antigen) and becomes ill and in most cases survives then this immunity, developed against that disease is called naturally induced immunity or auto immune response.
  - **Artificial:** The use of vaccines, which stimulate the production of antibodies in the body and making person immune against the disease of infection

b) Passive Immunity:

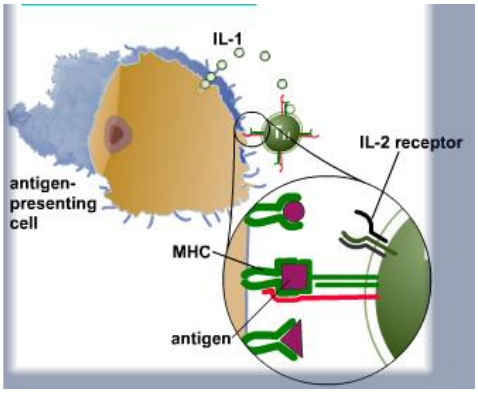
- In contrast to active immunity in which case the antigens are introduced to stimulate the production of antibodies by artificial or natural method; antibodies are inserted in the form of antisera, to make a person immune against a disease is called as passive immunity.
- It is also of further two types i.e. Natural and Artificial
  - **Natural:** It is the passive immunity produced naturally.
    E.g. As in the case of transfer of antibodies from mother to fetus across the placenta.
  - **Artificial.** It is the insertion of antibodies by artificially.
    E.g. As in the case of active infections of hepatitis, rabies, snake bite venom passive immunity is introduced artificially.

ii. Cell-Mediated Immunity:

- Cell-mediated immunity is a host defense used to fight intracellular pathogens and abnormal body cells such as cancer cells
- The protagonists of cell-mediated immunity are T-lymphocytes which can be divided into two groups:

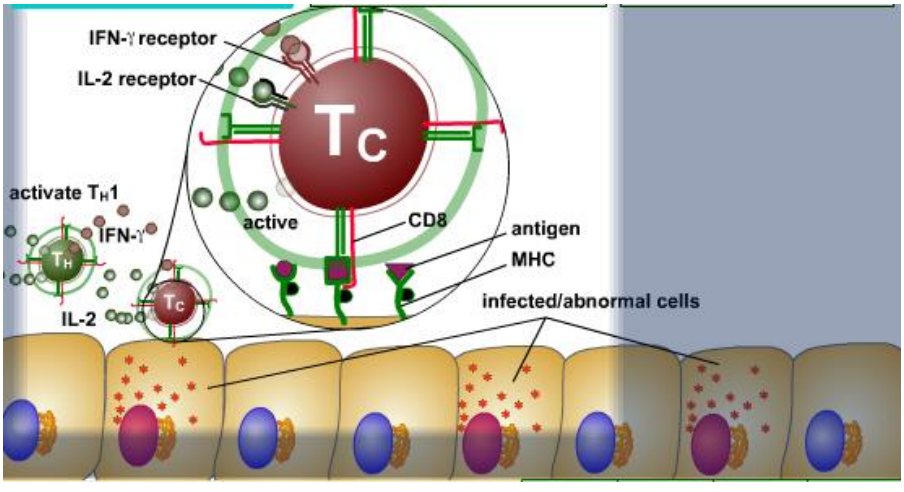
a) Helper T-cells (T_H cells):

- Helper T-cells secrete cytokines to activate other cells of the immune system.
- **Helper T-Cells Activation:**
  - ✓ Each helper T-cell produces a unique T-cell receptor (TCR) which corresponds to a specific antigen.
  - ✓ Each Helper T-cell also displays a receptor called CD4 on its surface which recognizes the MHC molecule of antigen presenting cells.
  - ✓ When helper T-cell encounters an antigen presenting cell with that antigen displayed in on MHC molecule it signals the antigen presenting cell to release Interleukin-1 (IL-1).
  - ✓ The IL-1 stimulates the helper T-cell to produce IL-2 as well as IL-2 receptor. The helper T-cell can then auto stimulate itself keeping it active as it proliferates.
- **Helper T-cells Proliferation:**
  - ✓ Once a helper T-cell becomes activated it proliferates forming a clone of cells specific to the same antigen.
  - ✓ Some of the helper T-cells differentiate into long lived memory T-cells.
  - ✓ But most mature to produce cytokines to help with the immune response.
- **Helper T-cells functions:**
  - ✓ Helper T-cells differentiate into two sub-populations called TH1 cells and TH2 cells
    - T_H1 cells produce cytokines that primarily activate the cell-mediated immunity such as macrophages, cytotoxic t-cells and natural killer cells.
    - T_H2 cells produce cytokines that activates the b-cells involved in humoral immunity.



b) Cytotoxic T-cells (T_C cells):

- Cytotoxic T-cells attack infected cells directly.
- **Cytotoxic T-Cells activation;**
  - ✓ Each cytotoxic T-cell produces a unique T-cell receptor which corresponds to a specific antigen.
  - ✓ Each cytotoxic T-cell also produces a receptor called CD8 which recognizes the MHC molecule of infected cells.
  - ✓ When a cytotoxic T-cell encounters an infected cell with the specific antigen bound in an MHC molecule, the cytotoxic T-cell produces receptors for IL-2 and other cytokines such as gamma interferon or IFN-gamma.
  - ✓ The cytotoxic T-cell can then be activated by nearby cytokine secreting active helper T-cells.
- **Cytotoxic T-Cells Proliferation:**
  - ✓ Once a cytotoxic T-cell becomes activated it proliferates forming a clone of cells specific to the same antigen.
  - ✓ Some of these cytotoxic T-cells differentiate into long lived memory T-cells. But most mature to attack and destroy the infected cells.
- **Cytotoxic T-cells functions:**
  - ✓ Cytotoxic T-cells can destroy infected or abnormal cells through multiple pathways.
  - ✓ The best understood of this pathway is the perforin granzyme cytotoxic pathway.
  - ✓ When the cytotoxic T-cells encounter an infected cell with the target antigen bound in an MHC molecule. The T-Cells secrete two types of protein molecules i.e. **perforin and granzyme.**
    - The perforin forms a pore in the membrane of the target cell allowing granzyme to enter the cell.
    - The granzyme activates the enzymes inside the cell leading to cell death.



HYPERSENSITIVITY

A set of undesirable reactions produced by the normal immune system including allergies and autoimmunology.

Types of hypersensitivity:

There are four types of hypersensitivity:

1. Type-1 Hypersensitivity (Anaphylactic Type)
2. Type-2 Hypersensitivity (Cytotoxic Type)
3. Type-3 Hypersensitivity (Immune Complex Mediated)
4. Type-4 Hypersensitivity (Cell-Mediated or Delayed Hypersensitivity)

1. Type-I hypersensitivity

(Anaphylactic type)

Immediate hypersensitivity reaction, resulting from release of pharmacologically active mediators such as histamines, prostaglandins and leukotrienes.

Diseases

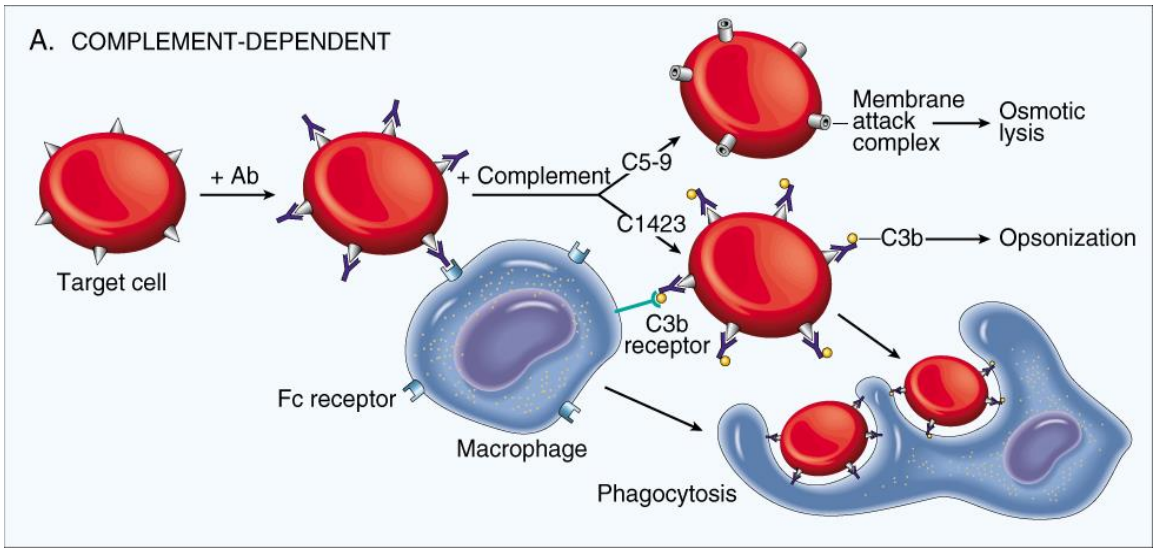
- Urticaria
- Asthma
- Hay fever
- Insect allergy: serious or fatal anaphylaxis may follow. Edema of larynx, with airway obstruction may occur.
- Penicillin induced shock

2. Type-II Hypersensitivity

Cytolytic or cytotoxic reactions

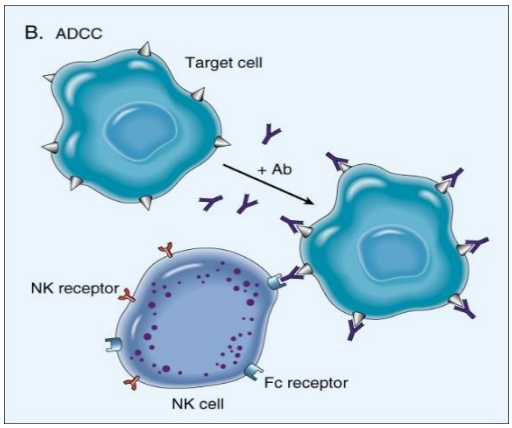
Mechanism:

i. Complement-dependent reactions
   - Transfusion reactions
   - Erythroblastosis fetal
   - Autoimmune hemolytic anemia
   - Certain Drug reactions



Schematic illustration of three different mechanisms of antibody-mediated injury in type II hypersensitivity. A, Complement-dependent reactions that lead to lysis of cells or render them susceptible to phagocytosis. (From Robbins Basic Pathology ,2003)

ii. Antibody Dependent Cell-Mediated Cytotoxicity (ADCC



Antibody-dependent cell-mediated cytotoxicity (ADCC). IgG-coated target cells are killed by cells that bear Fc receptors for IgG (e.g., NK cells, macrophages).
(From Robbins Basic Pathology ,2003）

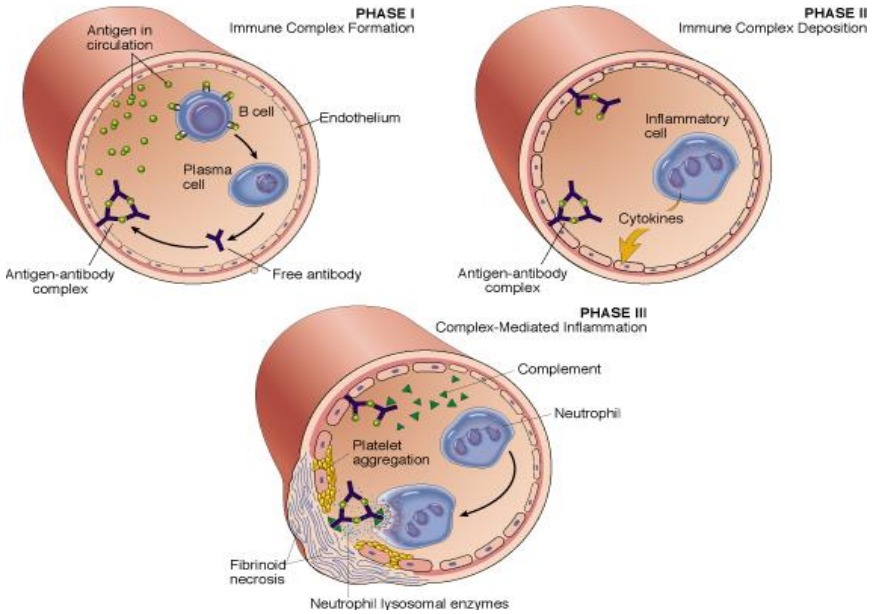
3. Type-III Hypersensitivity (Immune complex-mediated)

Reaction types

- Arthus reaction
- serum sickness
- Collagen diseases

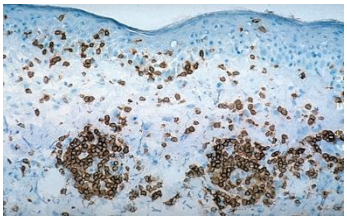
Toxic complex diseases:

- Acute glomerulonephritis
- Rheumatoid arthritis
- Progressive systemic sclerosis
- Dermatomyositis etc.



Schematic illustration of the three sequential phases in the induction of systemic type III (immune complex) hypersensitivity.
(From Robbins Basic Pathology, 2003）

4. Type-IV Hypersensitivity (Cell-Mediated)



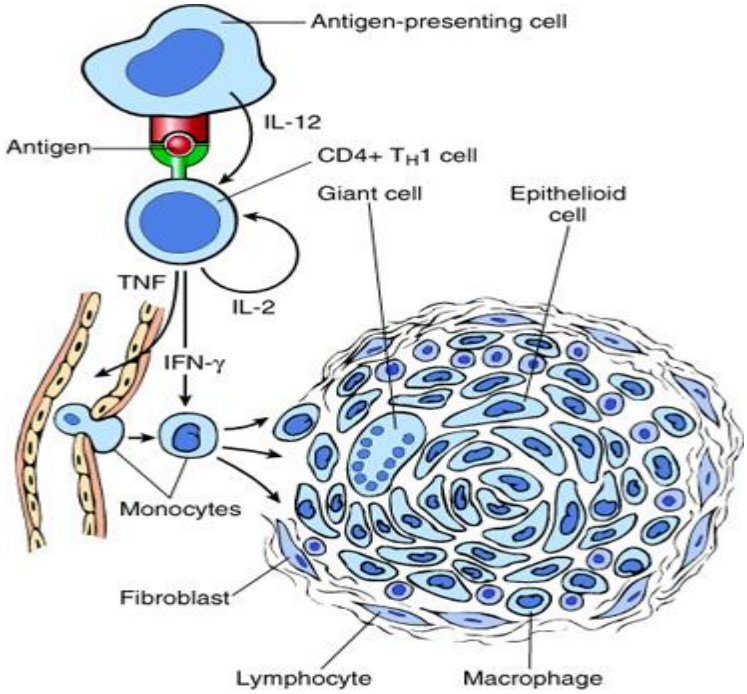
Delayed hypersensitivity reaction

Tissue reaction:

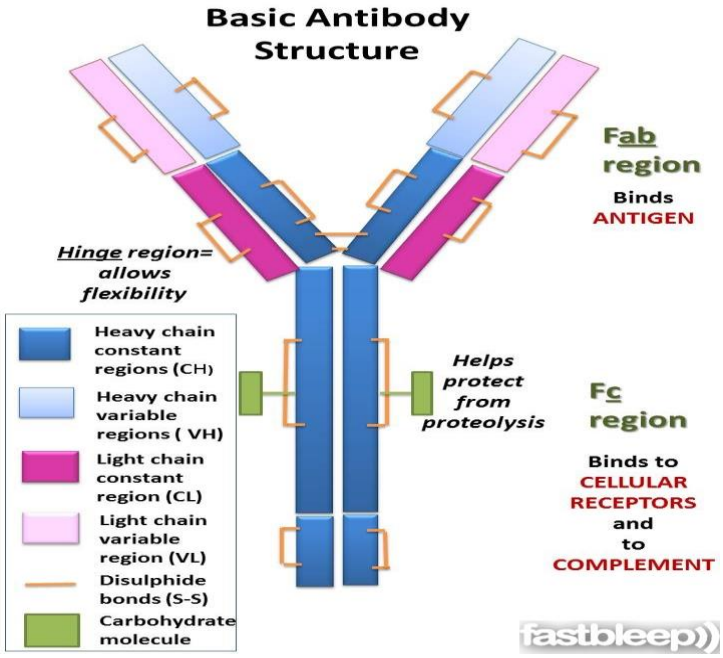
Consist of parenchymal destruction associated with perivascular lymphocytic and macrophage reaction.

Diseases:

- Chronic active hepatitisss
- some Viral infections
- Contact dermatitis
- Graft rejection
- Inflammatory bowel disease.

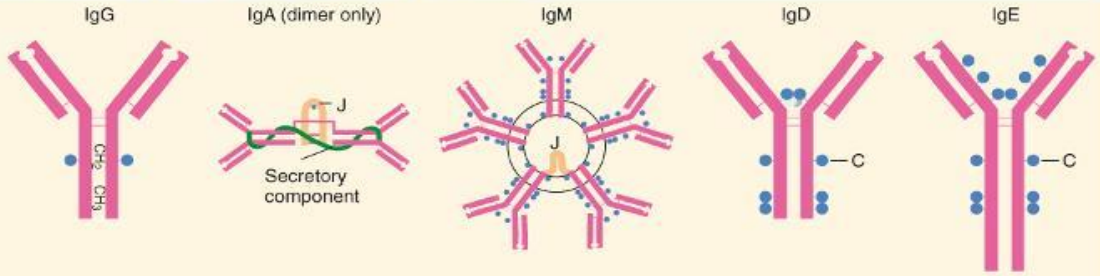


Structure and Functions of Antibodies



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TABLE 15.2 Characteristics of the Immunoglobulin (Ig) Classes



	Monomer	Dimer, Monomer	Pentamer	Monomer	Monomer
Number of Antigen Binding Sites	2	4 2	10	2	2
Molecular Weight	150,000	170,000–385,000	900,000	180,000	200,000
Percentage of Total Antibody in Serum	80%	13%	6%	1%	0.002%
Average Half-Life in Serum (Days)	23	6	5	3	2.5
Crosses Placenta?	Yes	No	No	No	No
Fixes Complement?	Yes	No	Yes	No	No
Fc Binds To	Phagocytes				Mast cells and basophils
Biological Function	Long-term immunity; memory antibodies; neutralizes toxins, opsonizes, fixes complement	Secretory antibody; on mucous membranes	Produced at first response to antigen; can serve as B-cell receptor	Receptor on B cells	Antibody of allergy; worm infections

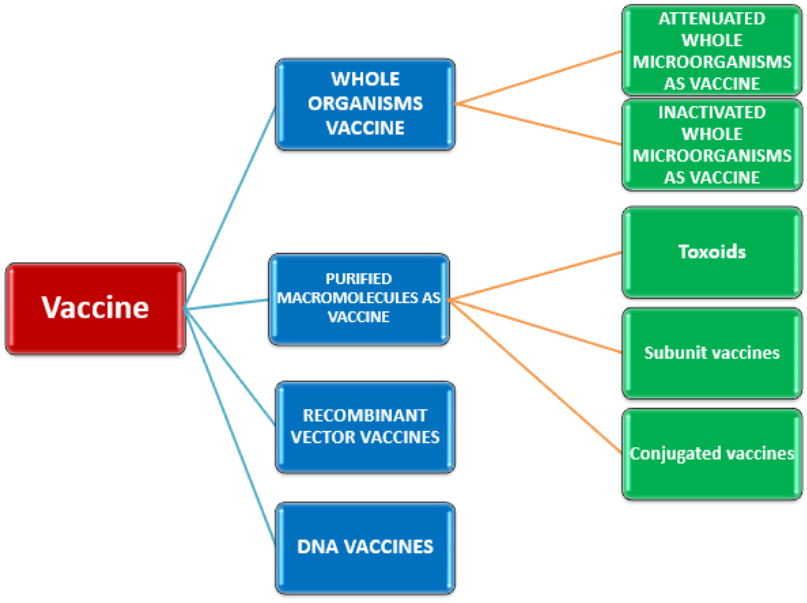
C = carbohydrate. J = J chain.

VACCINE

Definition:

"A vaccine is a biological preparation that improves immunity to a particular disease. A vaccine typically contains an agent that resembles a disease-causing microorganism."

Types of Vaccines



1. Whole Microorganisms Vaccine:

Many of the current vaccines in use for humans that are effective against viral and bacterial diseases consist of whole organisms that are either inactivated (killed) or attenuated (live but nonvirulent). These are termed whole organism's vaccines.

Types:

- Attenuated whole microorganisms as vaccine
- Inactivated whole microorganisms as vaccine

i. Attenuated Whole Microorganisms Vaccine:

Such vaccines use living but attenuated (weakened) microbes. Life-long immunity, especially with virus is often achieved without booster immunization. Its effectiveness is about 95%. Long-term effectiveness probably occur because the attenuated virus replicate in the body, increasing the dose and acting as a series of secondary metabolites.

Examples

- Measles, mumps and rubella injected S.C
- Sabin polio vaccine given Orally





- Chicken and small pox vaccines injected S.C
- Typhoid vaccine given Orally
- Tuberculosis bacillus injected ID





ii. Inactivated Whole Microorganisms Vaccine:

Such vaccines use killed microbes. They are killed by treating with formalin or phenol. Its effectiveness and long-term immunity is compromised.

Examples

- Influenza vaccine injected S.C
- Rabies vaccine injected IM
- Salk polio vaccine injected IM
- Hepatitis A vaccine injected S.C
- Cholera vaccine injected S.C
- Pneumococcal pneumonia injected S.C
- Whooping cough vaccine
- Typhoid vaccine injected S.C





2. Purified Macromolecules as Vaccine:

Such vaccines use only specific, purified macromolecules derived from pathogenic microorganism.

Types

i. Toxoids
ii. Subunit Vaccines
iii. Conjugated Vaccines

i. Toxoids:

A substance that has been treated to destroy its toxic properties but retains the capacity to stimulate production of antitoxins.

Such vaccines contain inactivated exotoxins.

It requires a series of injection for full immunity followed by the booster every 10 years. Booster is for children not for adults.

Example

- Tetanus (purified tetanus toxoids) injected IM
- Diphtheria (purified diphtheria toxoids) injected IM

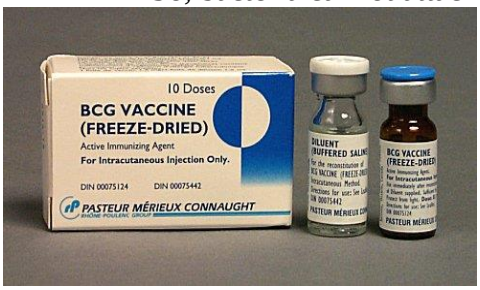
ii. Subunit Vaccines:

Such vaccines use those antigenic fragments of a microorganism that best stimulate an immune response.

Example

Pili of bacteria extracted and purified for use in vaccines to produce anti-pili antibodies.

So, bacteria cannot attach to the tissues and phagocytized.



Tuberculosis Vaccines



Influenza Virus Vaccine

iii. Conjugated Vaccines:

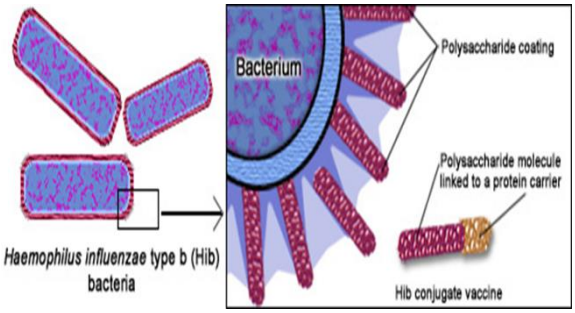
Such vaccines based on capsular polysaccharides and used for poor immune response children. Polysaccharides are T-independent antigens. So, children immune system does not respond well to these antigens until age of 15-24 months. In this vaccine polysaccharides are combined with proteins like diphtheria toxoids to produce conjugated vaccine.

Example

Hemophilus influenzae type b





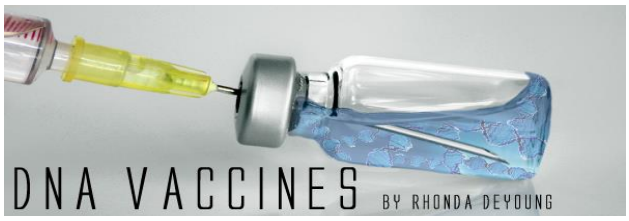


3. Recombinant Vector Vaccines:

- This form of vaccines applies the principle of isolating genes that encode major antigen from the pathogen and insert them into non-virulent viruses or bacteria. These vaccines are delivered by needle injected or by a device called a gene gun. The vector replicating within the host and expressing the gene product of the pathogen encoded antigenic proteins.
- These are experimental vaccines and work is going on measles, HIV.

4. DNA Vaccines:

- A DNA vaccine elicits protective immunity against a microbial pathogen by activating both humoral and cellular immunity. Long lasting memory cells are also generated.
- These are also in trail and influenza and herpes has been tried on humans.



Gene Gun

- Gene guns have also been used to deliver DNA vaccines.





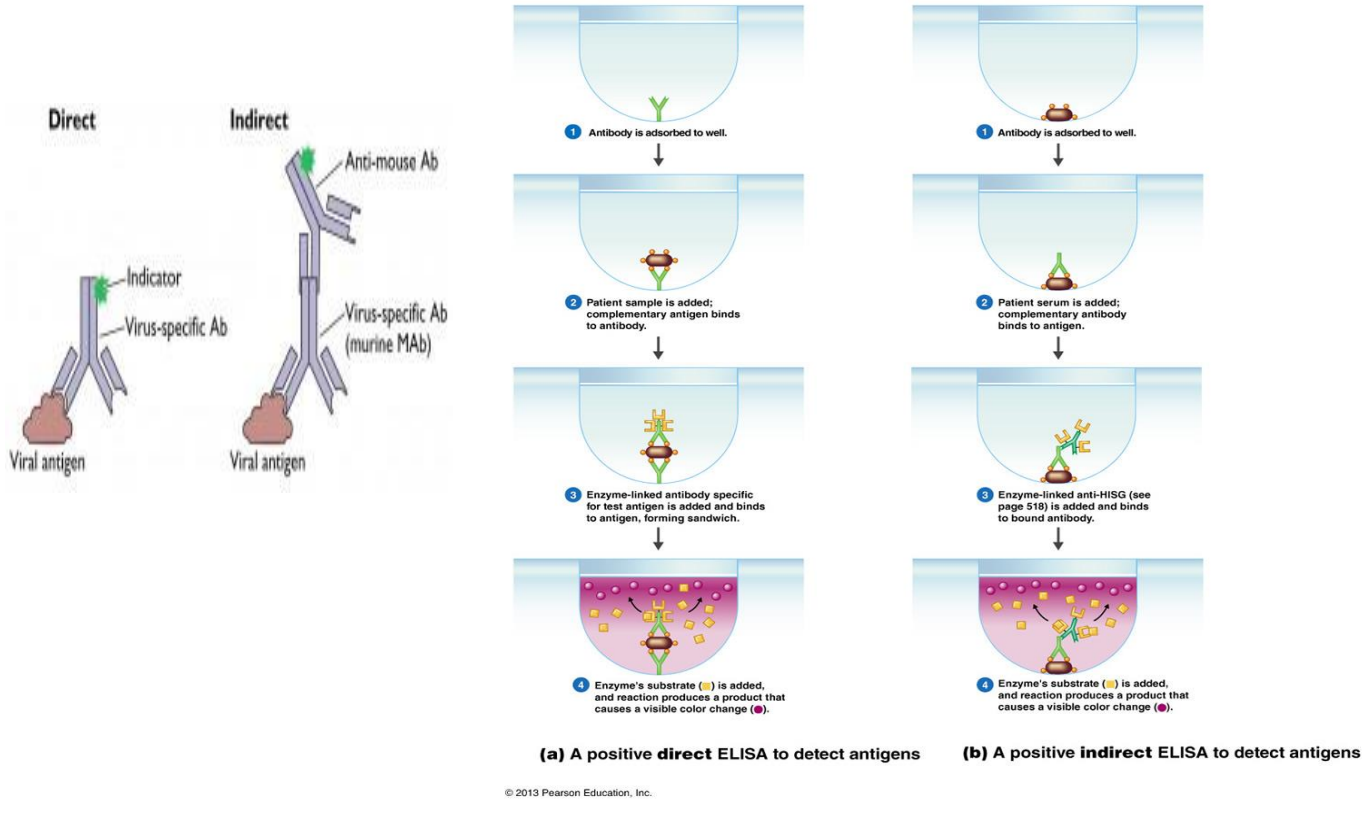
IMMUNOLOGICAL TESTS

- A test based on the principles of antigen-antibody reactions.



1. Enzyme Linked Immuno Sorbent Assay (ELISA):

- ELISA (enzyme-linked immunosorbent assay) is a plate-based assay technique designed for detecting and quantifying substances such as peptides, proteins, antibodies and hormones.
- The assay uses a solid-phase enzyme immunoassay (EIA) to detect the presence of a ligand (commonly a protein) in a liquid sample using antibodies directed against the protein to be measured.
- The ELISA has been used as a diagnostic tool in medicine, plant pathology, and biotechnology, as well as a quality control check in various industries.
- **Types of ELISA:**
  i. Direct ELISA
  ii. Indirect ELISA



(a) A positive direct ELISA to detect antigens **(b) A positive indirect ELISA to detect antigens**

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2. Complement Fixation Test:

- Complement is a protein (globulin) present in normal serum.
- Whole complement system is made up of nine components: C1 to C9
- Complement proteins are heat labile and are destroyed by heating at 56°C for 20 – 30 minutes.
- Complement binds to Ag-Ab complex
- When the Ag is an RBC it causes lysis of RBC's.

Positive Test:

- **Step 1:**

Antigen + Antibody (from serum) + Complement $\xrightarrow[1\text{ Hour}]{\text{At }37°C}$ Complement gets fixed

- **Step 2:**

Fixed Complement complex + Haemolytic system $\xrightarrow[1\text{ Hour}]{\text{At }37°C}$ No Haemolysis (Test Positive)

Negative Test

- **Step 1:**

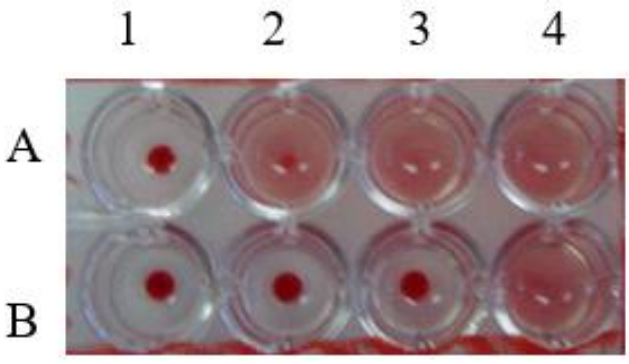
Antigen + Antibody absent + Complement $\xrightarrow[1\text{ Hour}]{\text{At }37°C}$ Complement not fixed

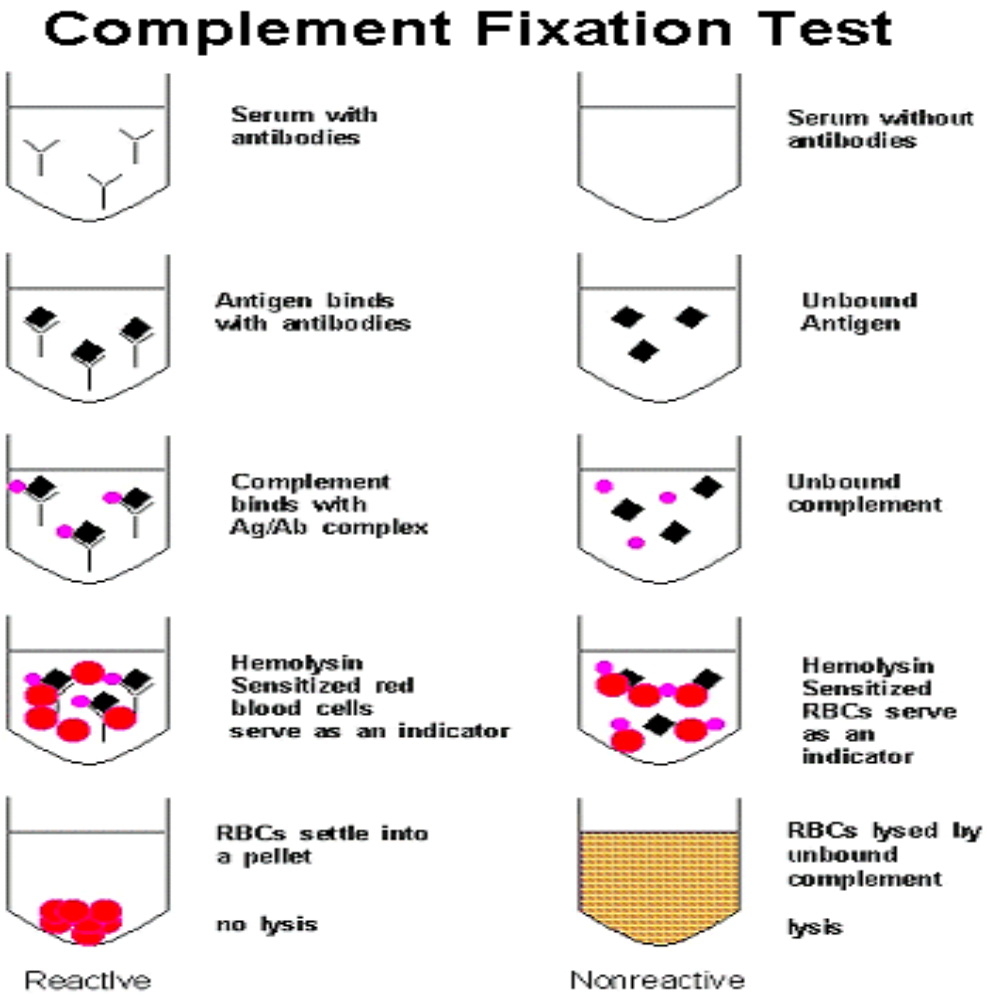
- **Step 2:**

Free Complement + Haemolytic system $\xrightarrow[1\text{ Hour}]{\text{At }37°C}$ Haemolysis (Test Negative)

Results and Interpretations:

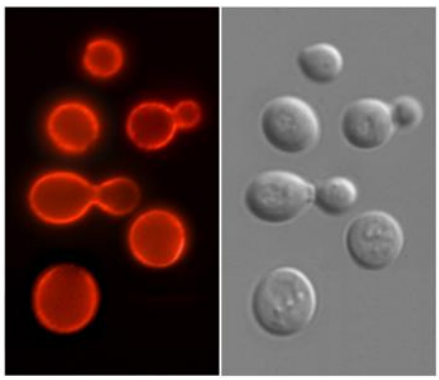
- No haemolysis is considered as a **positive test**.
- Haemolysis of erythrocytes indicative of a **negative test**.
- Microtiter plate showing Haemolysis (Well A3, A4 and B4) and No Haemolysis (Well A1,B1,B2,B3)



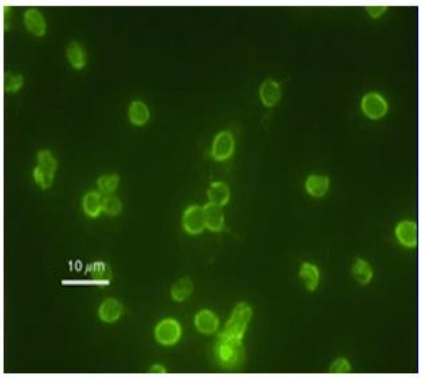


3. Immunofluorescence:

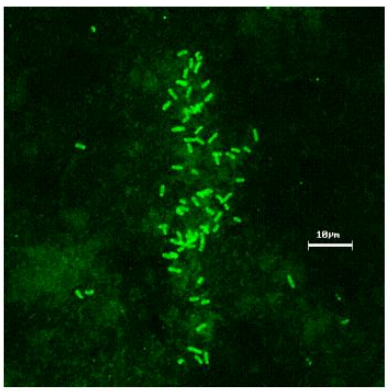
- Immunofluorescence (IF) is a common laboratory technique, which is based on the use of specific antibodies which have been chemically conjugated to fluorescent dyes.
- These labeled antibodies bind directly or indirectly to cellular antigens



Indirect immunofluorescence of S. cerevisiae



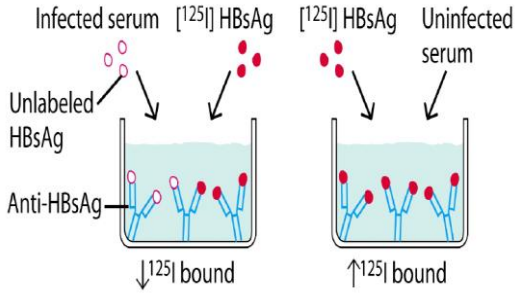
Immunofluorescence image of *Cryptosporidium parvum* oocysts



Concentrated groundwater methanotrophic bacteria on 0.2 µ m filter labeled with fluorescent antibodies.

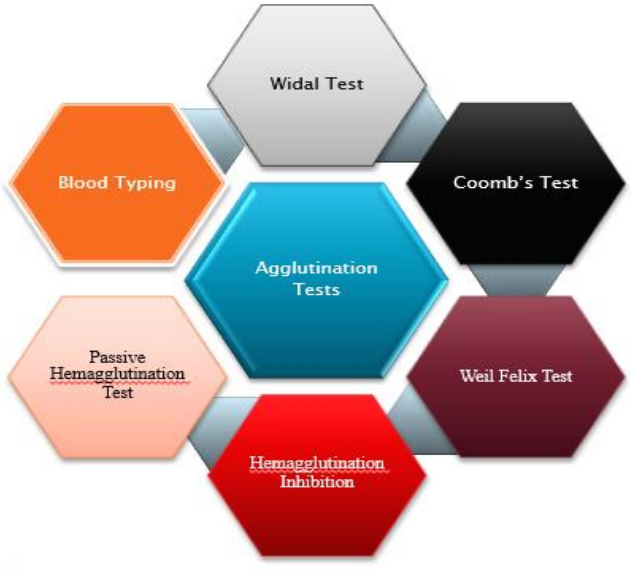
4. Radio Immunoassay:

- A technique for determining antibody levels by introducing an antigen labelled with a radioisotope and measuring the subsequent radioactivity of the antibody component.
- **Method:**
  - To perform a radioimmunoassay, a known quantity of an antigen is made radioactive, frequently by labeling it with gamma-radioactive isotopes of iodine, such as 125-I, attached to tyrosine.
  - This radiolabeled antigen is then mixed with a known amount of antibody for that antigen, and as a result, the two specifically bind to one another.
  - Then, a sample of serum from a patient containing an unknown quantity of that same antigen is added. This causes the unlabeled (or "cold") antigen from the serum to compete with the radiolabeled antigen ("hot") for antibody binding sites.
  - As the concentration of "cold" antigen is increased, more of it binds to the antibody, displacing the radiolabeled variant, and reducing the ratio of antibody-bound radiolabeled antigen to free radiolabeled antigen.
  - The bound antigens are then separated from the unbound ones, and the radioactivity of the free(unbound) antigen remaining in the [supernatant] is measured using a gamma counter.



5. Agglutination Tests:

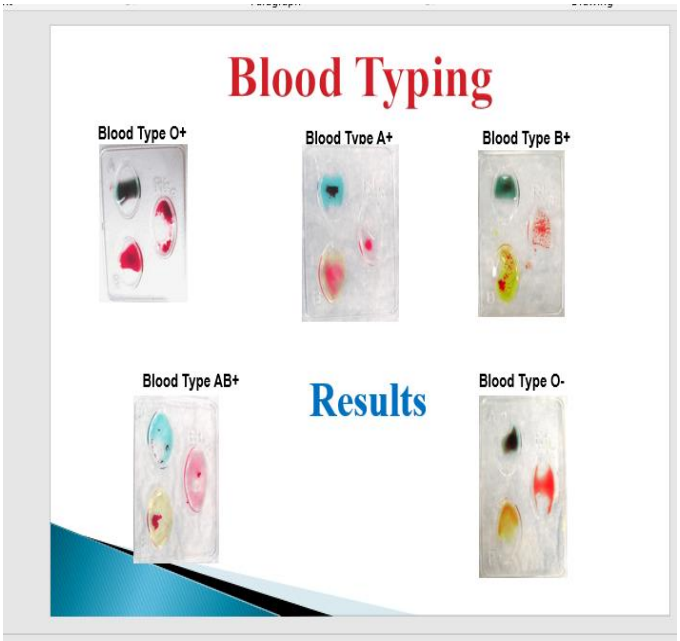
- A blood test used to identify unknown antigens; blood with the unknown antigen is mixed with a known antibody and whether or not agglutination occurs helps to identify the antigen; used in tissue matching and blood grouping and diagnosis of infections.

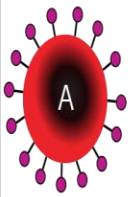
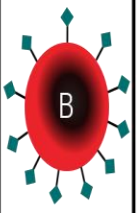
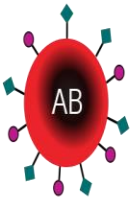
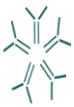


i. Blood Typing:



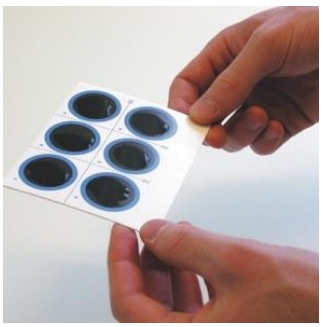
- Blood typing is a method to tell what type of blood you have.
- Our blood type is based on whether or not certain proteins are on your red blood cells.
- Human blood has three types of antigens, antigen A, antigen B and another Rhesus factor known as Rh factor.
- For blood typing three type of antibody reagents are needed namely antibody A, antibody B and antibody D.
- If the blood form agglutinates on adding antiserum A then the blood has antigen A. i.e Blood Group A
- If the blood form agglutinates on adding antiserum B then the blood has antigen B. i.e. Blood Group B.
- If the blood forms agglutinates on adding both antiserum A and antiserum B then the blood has both antigens. i.e Blood Group AB.
- If the blood does not form agglutinates on adding both the antisera, then it is said to have no antigen. i.e. Blood Group O.
- Now by antiserum D if the blood forms agglutinates then it is said to have Rh factor. i.e. + blood type
- If the blood does not form agglutinates on adding antiserum D then it does not have Rh factor. i.e. - blood type.



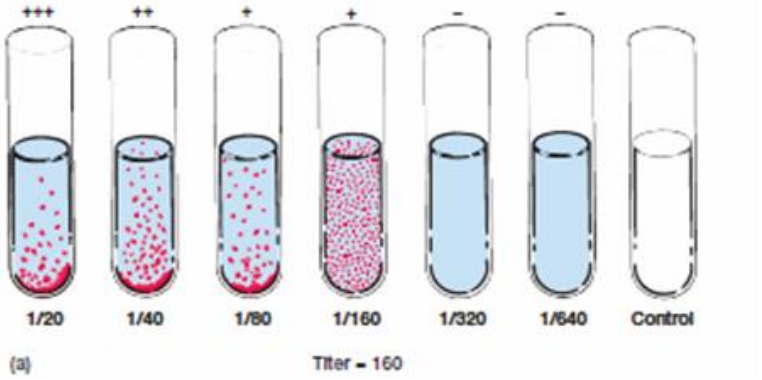
	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies in plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in red blood cell	A antigen	B antigen	A and B antigens	None

ii. Widal Test:

- A test for detecting typhoid fever and other salmonella infections.
- The Widal test is the reaction involving the agglutination of typhoid bacilli when they are present in serum that contain typhoid antibody from an individual who has typhoid fever.





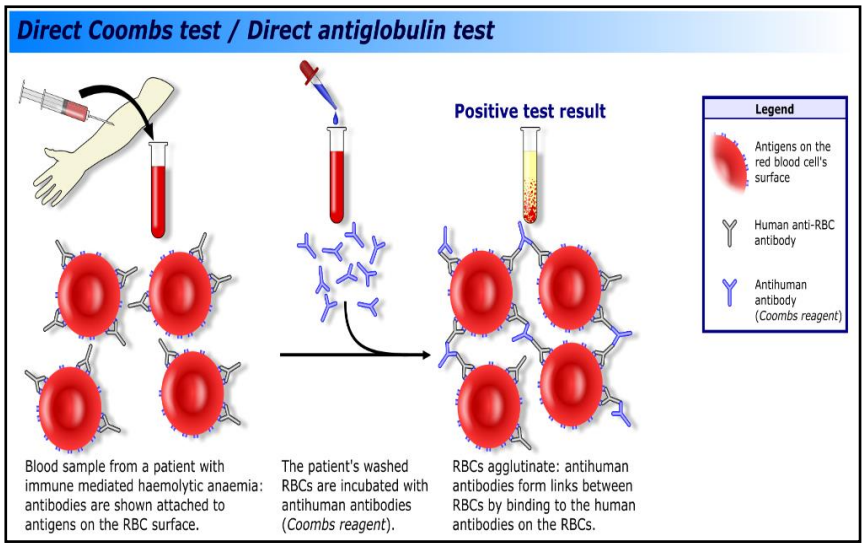


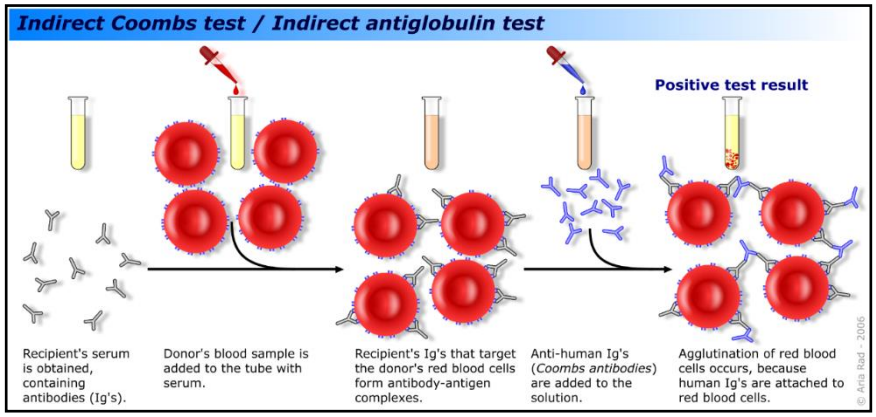
iii. Weil-felix Test:

- The **Weil-Felix test** is an agglutination test for the diagnosis of rickettsial infections.
- Weil-felix test, detects the antirickettsial antibodies in a patient serum by agglutinations of the proteus organisms.

iv. Coomb's Test:

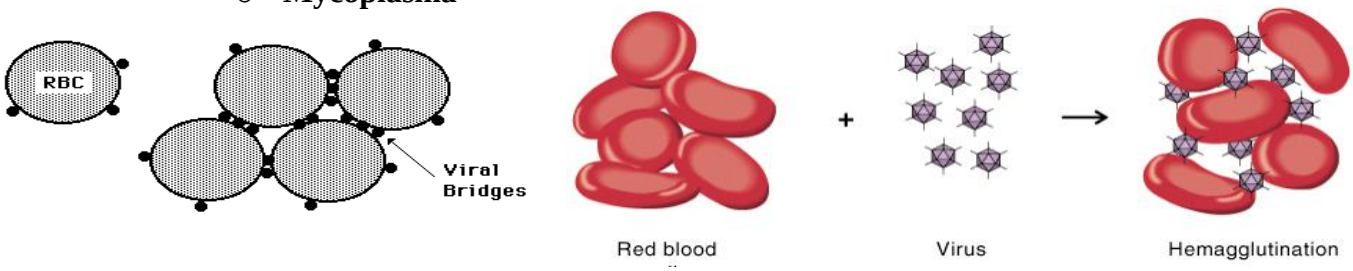
- This test is used for diagnosis of all kinds of Anemias.
- The phenomenon of agglutination of resulting clumping of RBCs can be visualized; when clumping is seen the test is positive and when clumping is not seen the test is negative.
- **Indirect Coombs test (Indirect Antiglobulin test):**
    - This test is performed to detect presence of Rh-antibodies or other antibodies in patients serum in case of the following:
        a) To check whether an Rh-negative women (married to Rh-positive men) has developed Anti Rh-antibodies
        b) Anti D may be produced in the blood of any Rh-negative person by exposure to D antigen by-
        c) Transfusion of Rh positive blood
        d) Pregnancy, if infant is Rh positive (if father is Rh-positive).

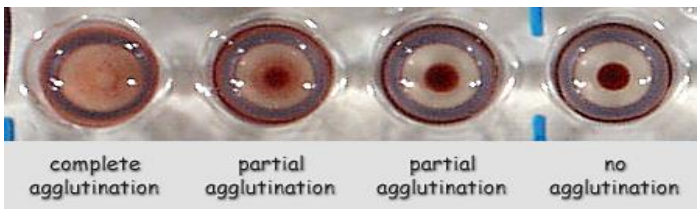




v. Haemagglutination:

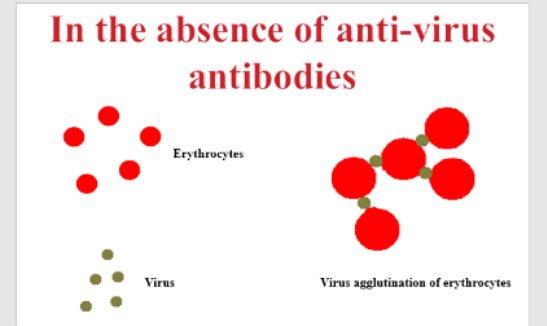
- **Passive Hemagglutination Test:**
  - A passive hemagglutination (PHA) test is used for measles.The PHA test determines exclusively the level of antibody directed to the hemagglutinin protein of measles virus.
- **Viral Haemagglutination:**
  - Some viruses and microbes contain proteins which bind to erythrocytes (red blood cells) causing them to clump together
  - **Adenovirus**
  - **Mycoplasma**





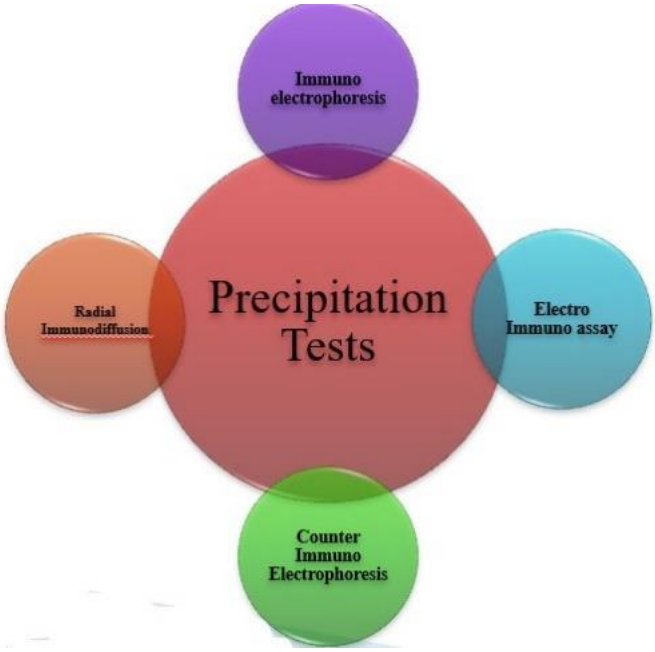
vi. **Hemagglutination/Inhibition:**

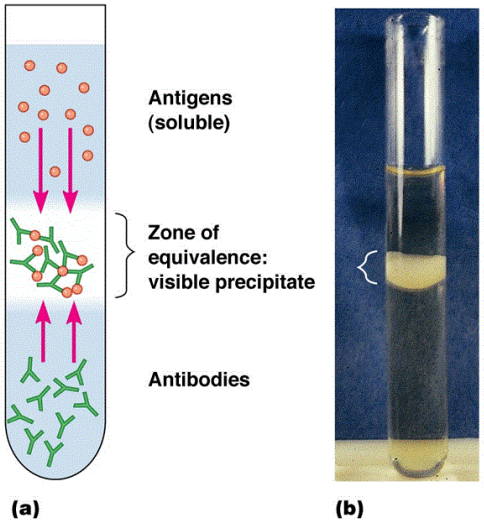
- Influenza virus particles have an envelope protein called the hemagglutinin, or HA, which binds to sialic acid receptors on cells. The virus will also bind to erythrocytes (red blood cells), causing the formation of a lattice. This property is called hemagglutination, and is the basis of a rapid assay to determine levels of influenza virus present in a sample.

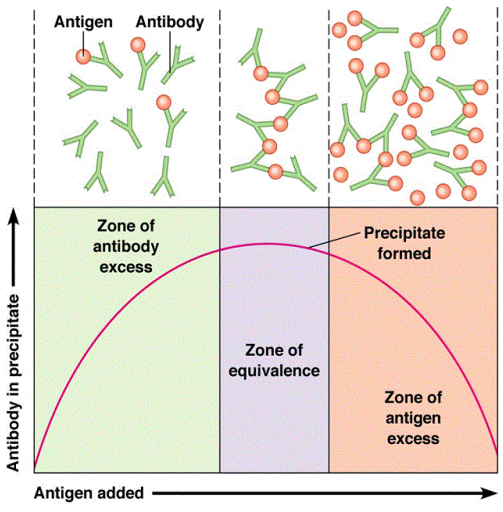


6. Precipitation:

- **Principle**
  - Soluble antigen combines with its specific antibody
  - Antigen-antibody complex is too large to stay in solution and precipitates
  - The interaction of soluble antigens with IgG or IgM antibodies leads to precipitation reactions.

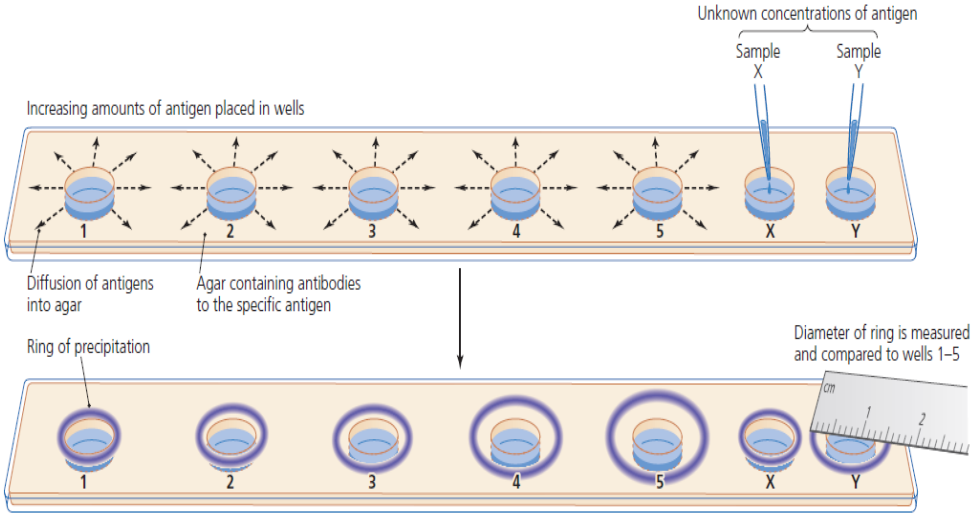


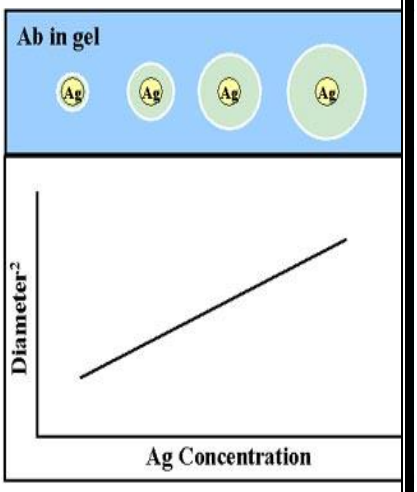




i. Radial immunodiffusion method:

- Radial immunodiffusion (RID) or Mancini method, Mancini immunodiffusion or single radial immunodiffusion assay, is an immunodiffusion technique used in immunology to determine the quantity or concentration of an antigen in a sample.
- **Preparation:**
  - A solution containing antibody is added to a heated medium such as agar or agarose dissolved in buffered normal saline. The molten medium is then poured onto a microscope slide or into an open container, such as a Petri dish, and allowed to cool and form a gel. A solution containing the antigen is then placed in a well that is punched into the gel. The slide or container is then covered or closed to prevent evaporation.
  - The antigen diffuses radially into the medium, forming a circle of precipitin that marks the boundary between the antibody and the antigen. The diameter of the circle increases with time as the antigen diffuses into the medium, reacts with the antibody, and forms insoluble precipitin complexes. The antigen is quantitated by measuring the diameter of the precipitin circle and comparing it with the diameters of precipitin circles formed by known quantities or concentrations of the antigen.
  - Antigen-antibody complexes are small and soluble when in antigen excess. Therefore, precipitation near the center of the circle is usually less dense than it is near the circle's outer edge, where antigen is less concentrated.
  - Expansion of the circle reaches an end point and stops when free antigen is depleted and when antigen and antibody reach equivalence. However, the clarity and density of the circle's outer edge may continue to increase after the circle stops expanding.

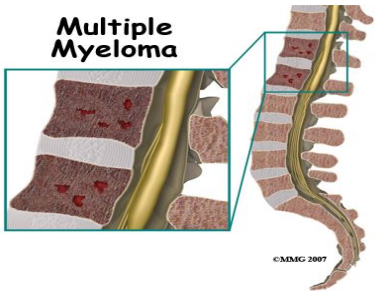


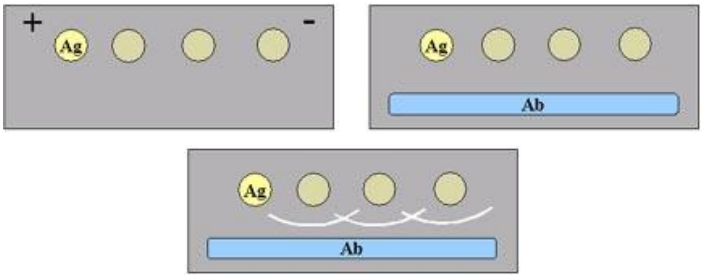


The diameters of the circles increase with time as the antigen diffuses into the medium, reacts with the antibody, and forms insoluble precipitin complexes.

ii. Immunoelectrophoresis Method:

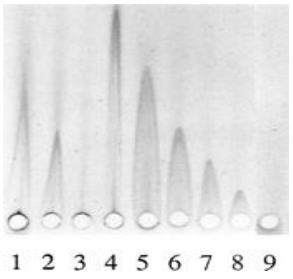
- Immunoelectrophoresis is a general name for a number of biochemical methods for separation and characterization of proteins based on electrophoresis and reaction with antibodies.
- All variants of immunoelectrophoresis require immunoglobulins, also known as antibodies, reacting with the proteins to be separated or characterized.
- The methods were developed and used extensively during the second half of the 20th century. In somewhat chronological order:
- Immunoelectrophoretic analysis (one-dimensional immunoelectrophoresis)
  - ❖ Crossed immunoelectrophoresis (two-dimensional quantitative immunoelectrophoresis)
  - ❖ Rocket-immunoelectrophoresis (one-dimensional quantitative immunoelectrophoresis)
  - ❖ Fused rocket immunoelectrophoresis
  - ❖ Affinity immunoelectrophoresis.
- E.g. Multiple myeloma is a blood cancer in which antibody-producing plasma cells grow in an uncontrolled and invasive (malignant) manner.





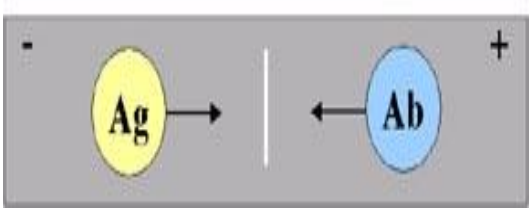
iii. **Electro Immunoassay:**

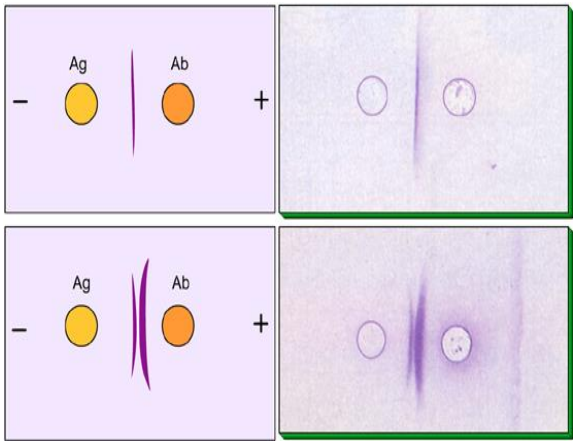
- A form of immunoassay that uses electrophoresis.



iv. Counter Immunoelectrophoresis:

- Used for the rapid diagnosis of some bacterial, mycoplasma and viral infections.
- A laboratory technique used to evaluate the binding of an antibody to its antigen.





VIRUSES

CHAPTER NO. 05

Dr. Qazi Adnan

VIRUSES

"An infective agent that typically consists of a nucleic acid molecule in a protein coat, is too small to be seen by light microscopy, and is able to multiply only within the living cells of a host is called as virus"

- Most viruses that have been studied have a diameter between 20 and 300 nanometers.

Structure:

- A complete virus particle, known as a virion, consists of nucleic acid surrounded by a protective coat of protein called a capsid. These are formed from identical protein subunits called capsomeres.
- Viruses can have a lipid "envelope" derived from the host cell membrane. The capsid is made from proteins encoded by the viral genome and its shape serves as the basis for morphological distinction.

Capsid:

- The protein shell or coat that encloses the nucleic acid genome is called as capsid.

Capsomeres:

- Morphological units seen in the electron microscope on the surface of icosahedral virus particles.
- Capsomeres represent clusters of polypeptides.

Envelope:

- A lipid containing membrane that surrounds some virus particles. It is acquired during viral maturation by a cellular membrane.
- Virus encoded glycoproteins are exposed on the surface of the envelope. These projections are called peplomers.

Nucleocapsid:

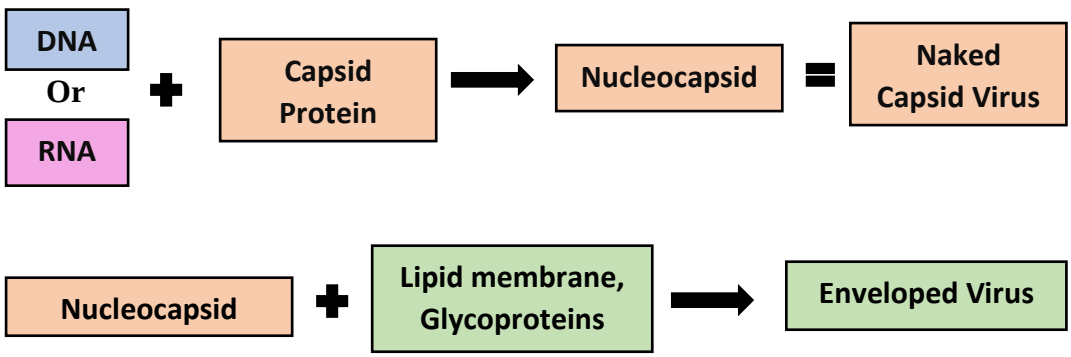
- The protein nucleic acid complex representing the packaged form of the viral genome.
- The term is commonly used in cases where the nucleocapsid is a substructure of a more complex virus particle.

Structural subunit:

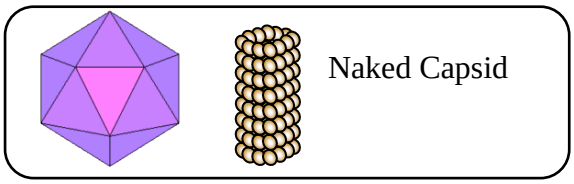
- The basic protein building blocks of the coat. They are usually a collection of more than one nonidentical protein subunit.
- The structural unit is often referred to as a protomer.

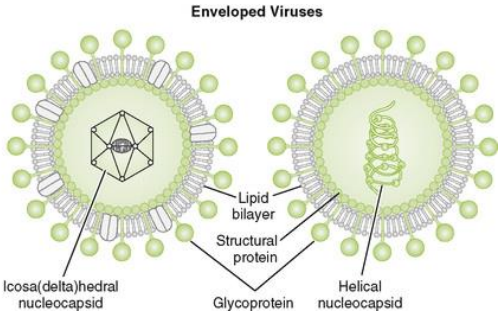
Basic virus structure:

The basic viral structure can be described in the form of flow sheet as follows:



Capsid symmetry:

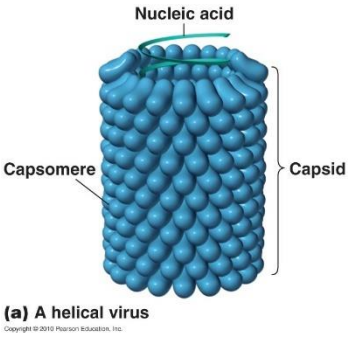




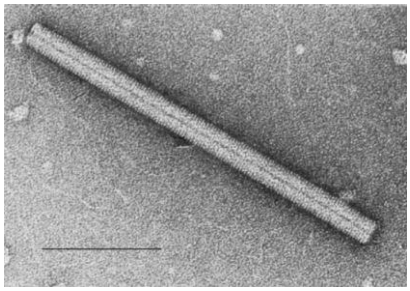
Types of viruses according to Shape:

- ❖ Helical
- ❖ Icosahedral
- ❖ Complex

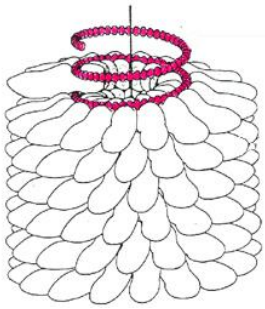
- ❖ **Helical:**
  - These viruses are composed of a single type of capsomer stacked around a central axis to form a helical structure, which may have a central cavity, or hollow tube. This arrangement results in rod-shaped or filamentous virions: these can be short and highly rigid, or long and very flexible.



Helical Naked Capsid Viruses

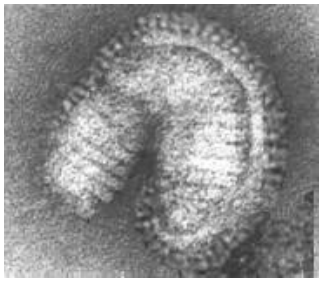


Tobacco Mosaic Virus
Electron Micrograph

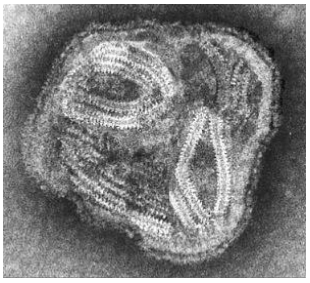


Tobacco Mosaic Virus
Model

Helical Enveloped Viruses



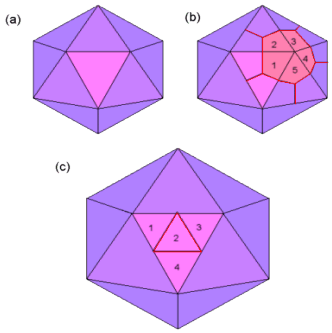
Influenza A Virus
Electron Micrograph



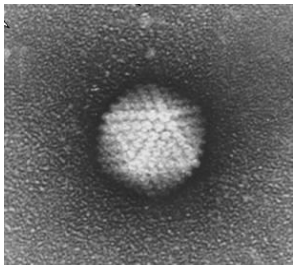
Paramyxovirus
Electron Micrograph

❖ **Icosahedral:**

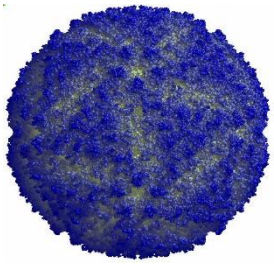
- Most animal viruses are icosahedral or near-spherical with icosahedral symmetry.
- A regular icosahedron is the optimum way of forming a closed shell from identical sub-units.
- The minimum number of identical capsomers required is twelve, each composed of five identical sub-units



Icosahedral Naked Viruses

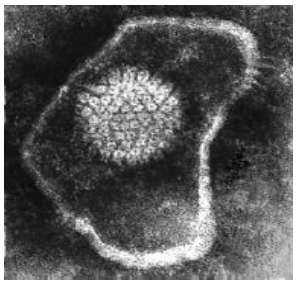


Adenovirus
Electron Micrograph

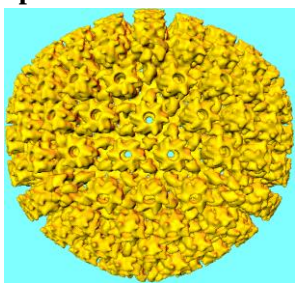


Foot and Mouth Disease virus
Crystallographic Model

Icosahedral Enveloped Viruses



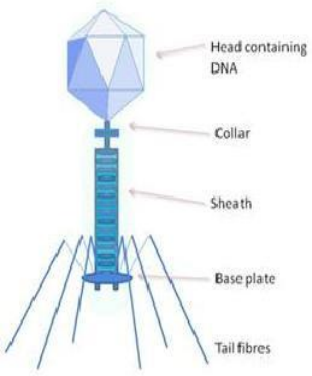
Herpes Simplex Virus
Electron Micrograph



Herpes Simplex Virus
Nucleocapsid cryoEM Model

❖ **Complex:**

- These viruses possess a capsid that is neither purely helical, nor purely icosahedral, and that may possess extra structures such as protein tails or a complex outer wall.
- Some bacteriophages, such as Enterobacteria phage T4 have a complex structure consisting of an icosahedral head bound to a helical tail.



Properties of enveloped viruses

- Envelope is sensitive to
  - Drying
  - Heat
  - Detergents
  - Acid
- **Consequences**
  - Must stay wet during transmission
  - Transmission in large droplets and secretions
  - Cannot survive in the gastrointestinal tract
  - Do not need to kill cells in order to spread
  - May require both a humoral and a cellular immune response

Properties of naked capsid viruses

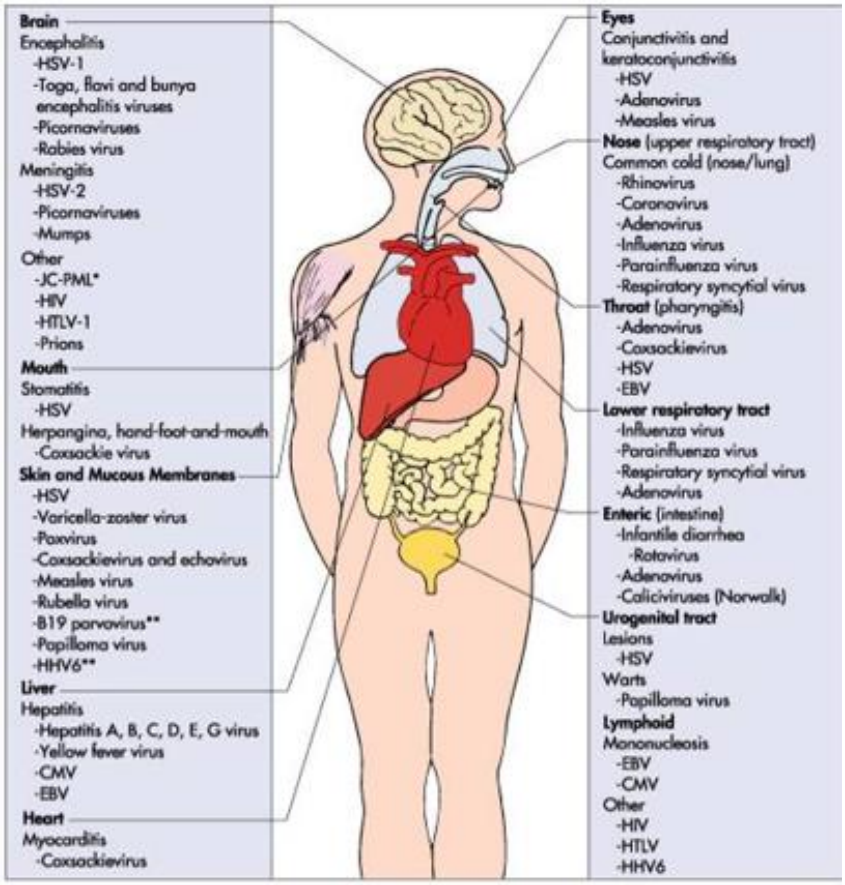
- Capsid is resistant to
  - Drying
  - Heat
  - Detergents
  - Acids
  - Proteases
- **Consequences**
  - Can survive in the gastrointestinal tract
  - Retain infectivity on drying
  - Survive well on environmental surfaces
  - Spread easily via fomites
  - Must kill host cells for release of mature virus particles
  - Humoral antibody response may be sufficient to neutralize infection

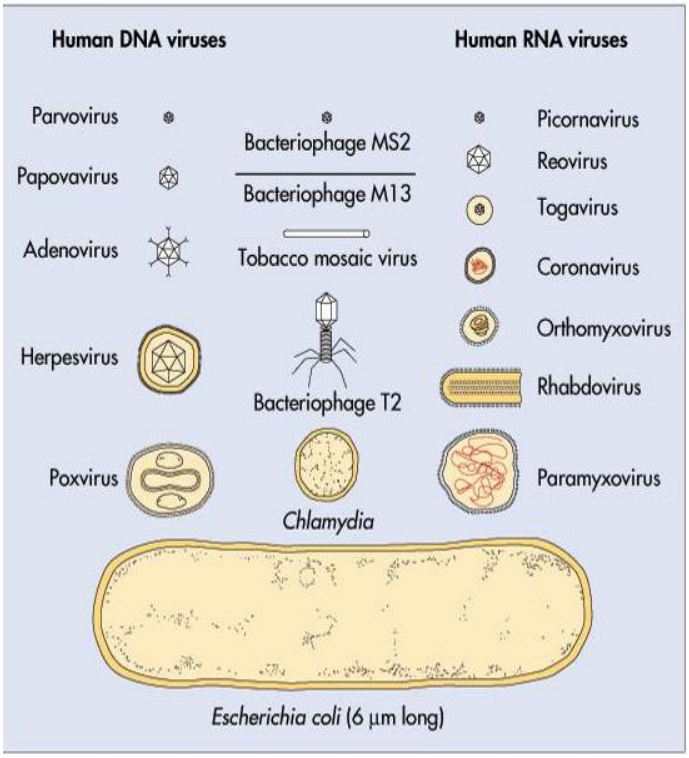
Classification of Human Viruses

"Group"	Family	Genome	Genome size (kb)	Capsid	Envelope
dsDNA					
	Poxviridae	dsDNA, linear	130 to 375	Ovoid	Yes
	Herpesviridae	dsDNA, linear	125 to 240	Icosahedral	Yes
	Adenoviridae	dsDNA, linear	26 to 45	Icosahedral	No
	Polyomaviridae	dsDNA, circular	5	Icosahedral	No
	Papillomaviridae	dsDNA, circular	7 to 8	Icosahedral	No
ssDNA					
	Anellovirus	ssDNA circular	3 to 4	Isometric	No
	Parvoviradae	ssDNA, linear, (- or +/-)	5	Icosahedral	No
Retro					
	Hepadnaviridae	dsDNA (partial), circular	3 to 4	Icosahedral	Yes
	Retroviridae	ssRNA (+), diploid	7 to 13	Spherical, rod or cone shaped	Yes
dsRNA					
	Reoviridae	dsRNA, segmented	19 to 32	Icosahedral	No
ssRNA (-)					
	Rhabdoviridae	ssRNA (-)	11 to 15	Helical	Yes
	Filoviridae	ssRNA (-)	19	Helical	Yes
	Paramyxoviridae	ssRNA (-)	10 to 15	Helical	Yes
	Orthomyxoviridae	ssRNA (-), segmented	10 to 13.6	Helical	Yes
	Bunyaviridae	ssRNA (-, ambi), segmented	11 to 19	Helical	Yes
	Arenaviridae	ssRNA (-, ambi), segmented	11	Circular, nucleosomal	Yes
	Deltavirus	ssRNA (-) circular	2	Spherical	Yes
ssRNA (+)					
	Picornaviridae	ssRNA (+)	7 to 9	Icosahedral	No
	Calciviridae	ssRNA (+)	7 to 8	Icosahedral	No
	Hepevirus	ssRNA (+)	7	Icosahedral	No
	Astroviridae	ssRNA (+)	6 to 7	Isometric	No
	Coronaviridae	ssRNA (+)	28 to 31	Helical	Yes
	Flaviviridae	ssRNA (+)	10 to 12	Spherical	Yes
	Togaviridae	ssRNA (+)	11 to 12	Icosahedral	Yes

Major diseases caused by human viruses

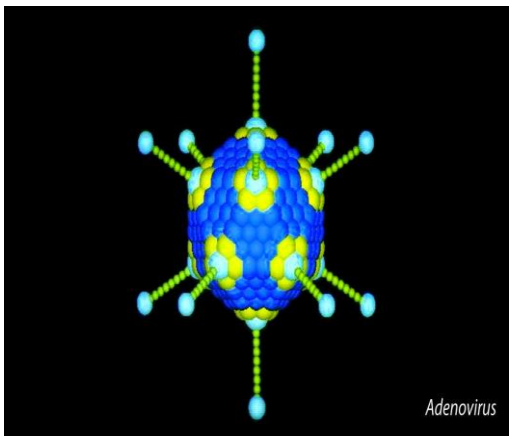
"Group"	Family	Human pathogens (disease)
dsDNA		
	Poxviridae	Variola (smallpox); Orf (pustular dermatitis); Molluscum contagiosum (pustular dermatitis)
	Herpesviridae	Herpes simplex 1,2 (oral, genital herpes); Varicella-zoster (chickenpox); Epstein-Barr (mononucleosis); Cytomegalovirus (neonatal abnormalities); HHV6 (roseola); HHV8 (Kaposi's sarcoma)
	Adenoviridae	Adenovirus (respiratory infection, conjunctivitis)
	Polyomaviridae	Polyomavirus (benign kidney infection, respiratory disease, leukoencephalopathy)
	Papillomaviridae	Papillomavirus (warts, genital carcinoma)
ssDNA		
	Anellovirus	Unknown
	Parvoviradae	B-19 (fifth disease, fetal death)
Retro		
	Hepadnaviridae	Hepatitis B ("serum" hepatitis)
	Retroviridae	HIV (aids); HTLV (leukemia)
dsRNA		
	Reoviridae	Rotavirus (infantile gastroenteritis)
ssRNA (-)		
	Rhabdoviridae	Rabies virus (rabies)
	Filoviridae	Ebola virus (ebola)
	Paramyxoviridae	Parainfluenza virus (respiratory infection); Mumps virus (mumps); Respiratory syncytial virus (respiratory infection); Measles virus (measles)
	Orthomyxoviridae	Influenza virus (influenza)
	Bunyaviridae	Hantaan virus (hemorrhagic fever with renal syndrome)
	Arenaviridae	Lassa fever virus (hemorrhagic fever)
	Deltavirus	Hepatitis D (fulminant acute hepatitis)
ssRNA (+)		
	Picornaviridae	Poliovirus (polio), rhinovirus (URI), Hepatitis A ("infectious" hepatitis)
	Calciviridae	Norwalk (gastroenteritis)
	Hepevirus	Hepatitis E (acute hepatitis)
	Astroviridae	Astrovirus (gastroenteritis)
	Coronaviridae	Coronavirus (respiratory infection)
	Flaviviridae	Yellow fever virus (yellow fever); Hepatitis C (hepatitis)
	Togaviridae	Eastern Equine encephalitis virus (encephalitis); Rubella virus (rubella)

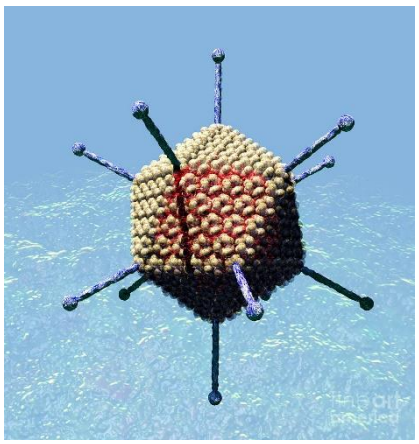




Adenoviridae

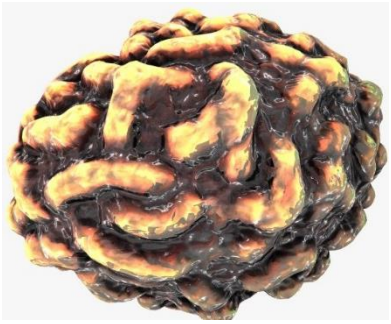
- Medium Sized (90-100 nm)
- Non-Enveloped Icosahedral Virus containing a double stranded DNA genome





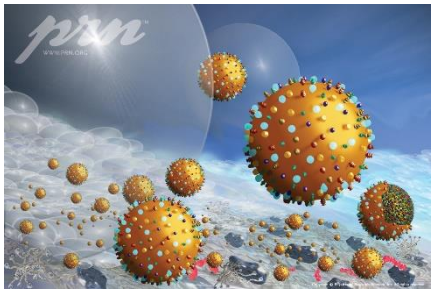
Poxviridae

- Small Pox Viruses
- Enveloped DNA Virus



Herpesviridae

- Icosahedral Enveloped DNA Virus



Polyomaviridae

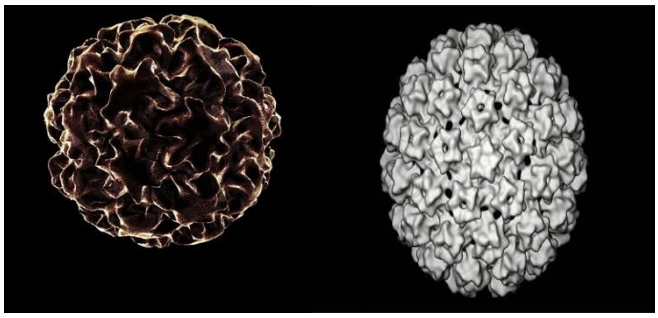
- Icosahedral unenveloped DNA Virus
- Kidney and respiratory infections





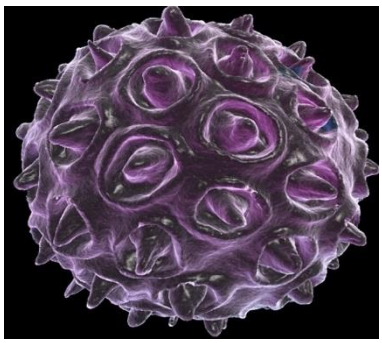
Papillomaviridae

- No Enveloped DNA Viruses
- Tumor and warts



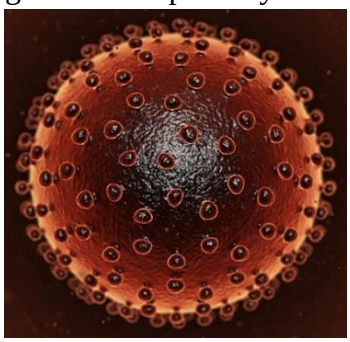
Parvoviridae

- Non-enveloped single stranded DNA viruses
- Gatroenteritis



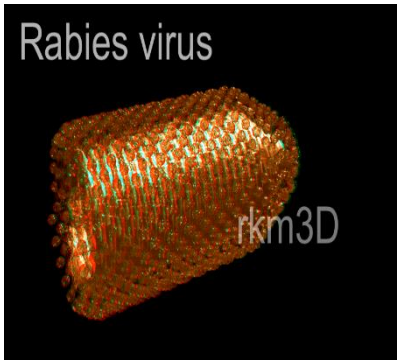
Hepadnaviridae

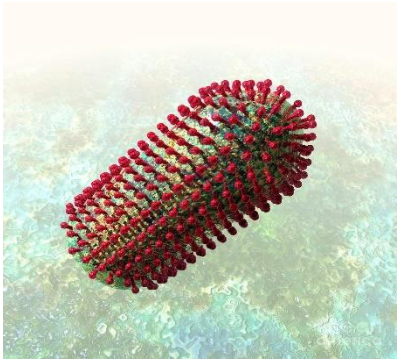
- Enveloped DNA Virus (partially single stranded partially Double Stranded)
- Hepatitis B virus



Rhabdoviridae

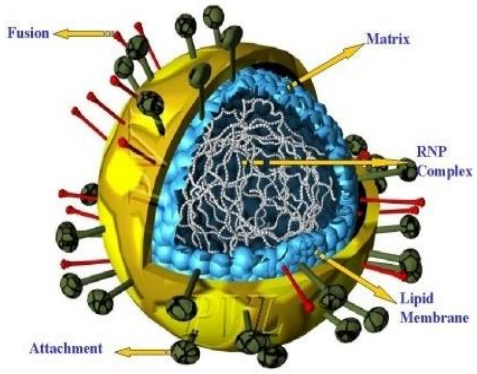
- RNA Virus
- Rabies Virus





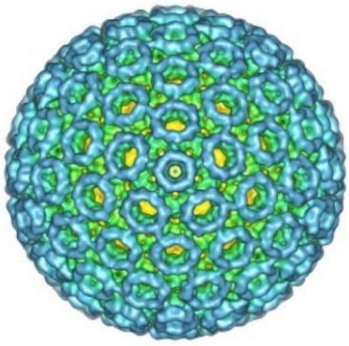
Paramyaxoviridae

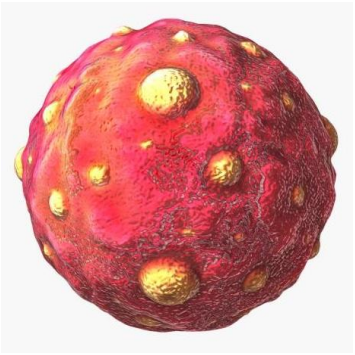
- RNA Virus
- Common cold
- Pneumonia



Bunyaviridae

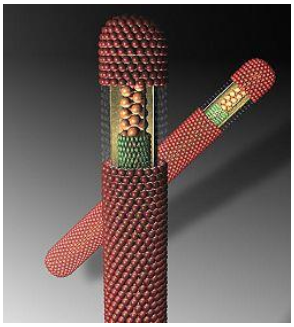
- RNA Virus
- Bunyavirus
- Hantaan virus
- Hemorrhagic Fever
- Congo virus





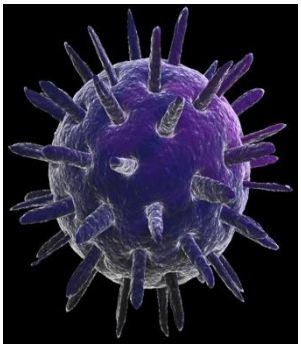
Filoviridae

- RNA Virus
- Hemorrhagic fever



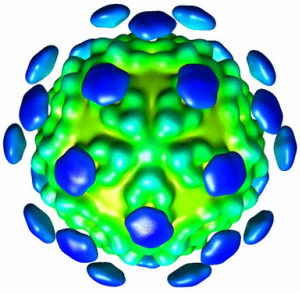
Calciviridae

- RNA Virus
- Norwalk virus
- Causes gastroenteritis



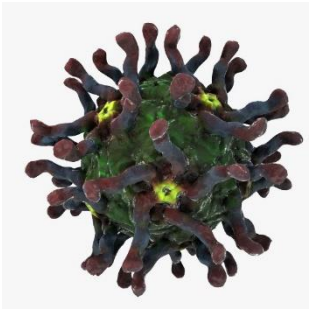
Astroviridae

- RNA Virus
- Gastroenteritis



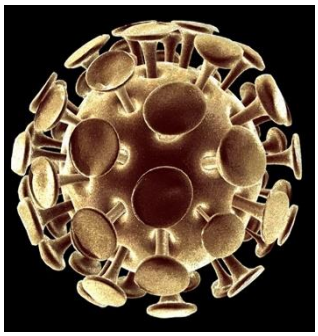
Picornaviridae

- RNA Virus
- Polio virus



Coronaviridae

- RNA Virus
- Respiratory infections
- SARS
  (Severe Acute Respiratory Syndrome)



Togaviridae

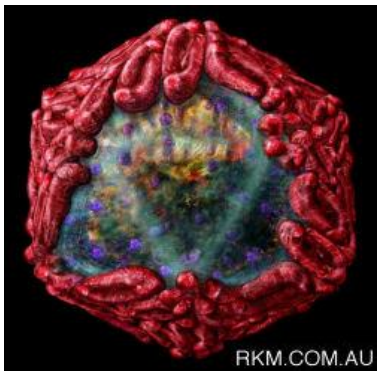
- RNA Virus
- Rubella virus

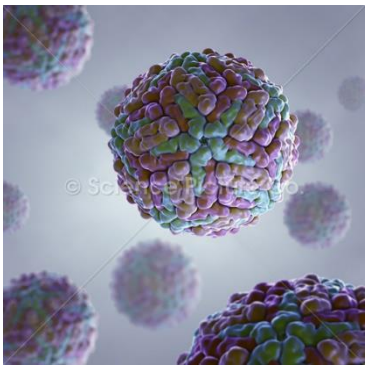




Flavidiridae

- RNA Virus
- Hepatitis C virus & dengue virus





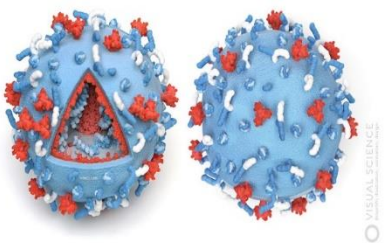
Reoviridae

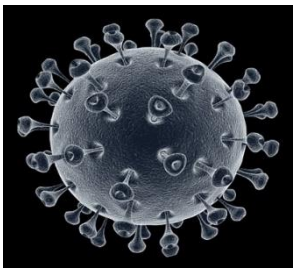
- RNA Virus
- Rota virus



Retroviridae

- RNA Virus
- HIV

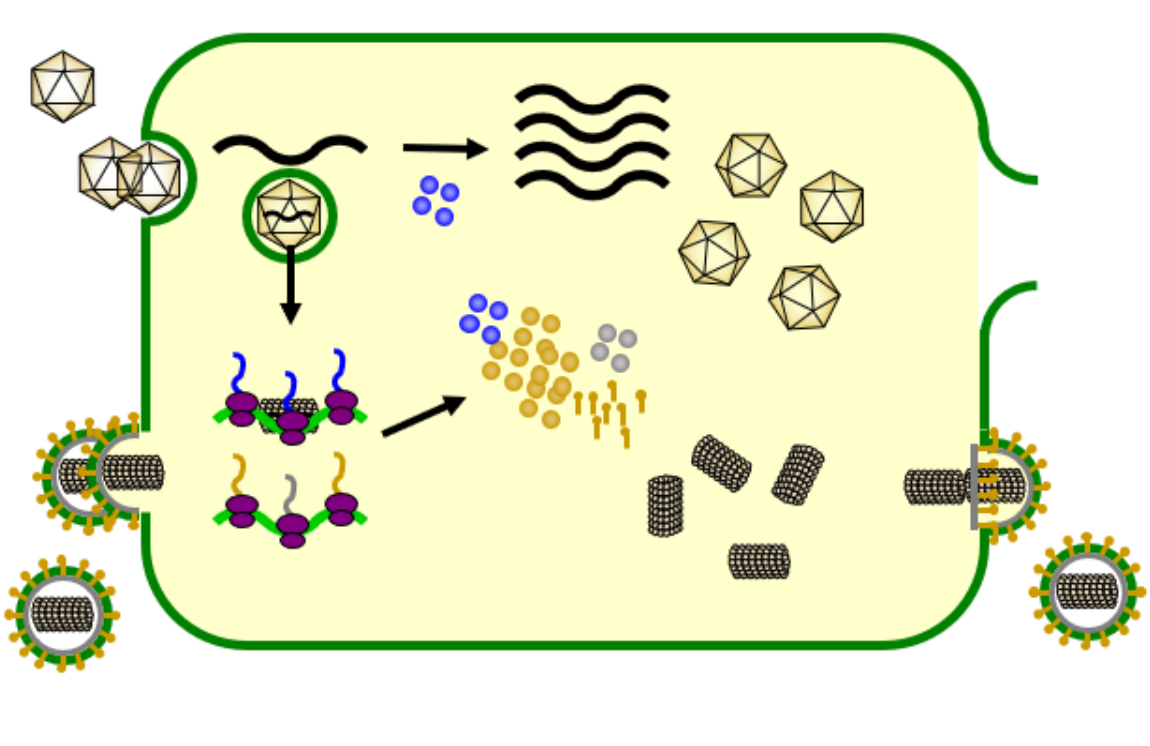




Replication of Virus

"Viral replication is the formation of biological viruses during the infection process in the target host cells."

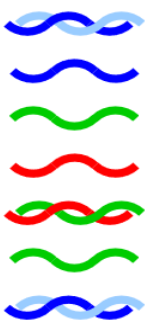
Viruses must first get into the cell before viral replication can occur. Through the generation of abundant copies of its genome and packaging these copies, the virus continues infecting new hosts.



Virus Replication Variance of the themes:

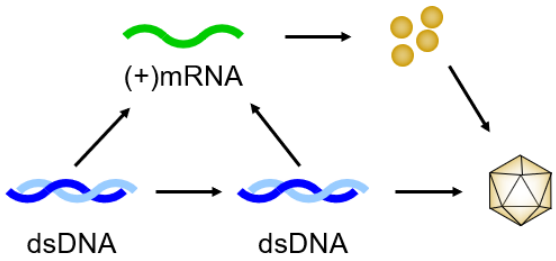
Replication between viruses is greatly varied and depends on the type of genes involved in them.

- dsDNA
- ssDNA
- (+)ssRNA
- (-)ssRNA
- dsRNA
- RNA retro
- DNA retro

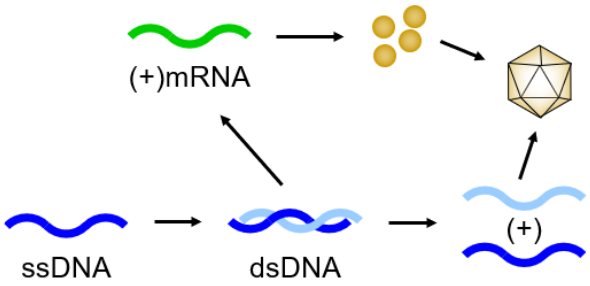


ds = Double Stranded
ss = Single Stranded

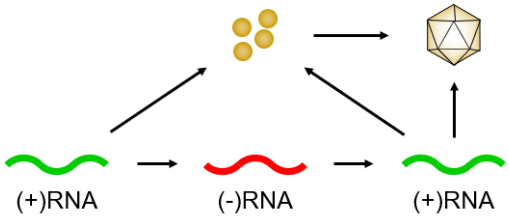
1. **dsDNA Viral Replication:**
   - Once the nucleocapsid of this type of virus enters the cell, it proceeds to the nucleus where it mimics the genome of the host cell.
   - Usually, the viral genome is replicated using the host cell DNA polymerase, and the viral genome is transcribed by the host cell RNA polymerase.
   - The resulting transcripts carrying information encoding viral proteins then transported to the cytoplasm and seen as a template by the host cell ribosomes.



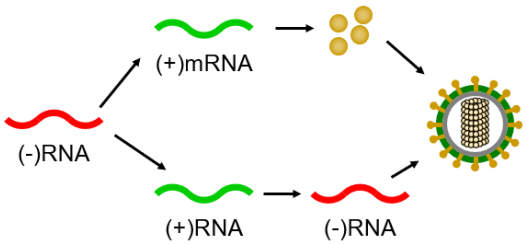
2. **ssDNA Viral Replication:**
   - The virus with a ssDNA genome also faces a serious replication problem in the host cell.
   - When introduced into cells, these genomes can not be used to make viral proteins because the only template for transcription is double-stranded DNA.
   - For this reason, the first step after infection is the conversion of the viral ssDNA into dsDNA using host cell DNA polymerase.



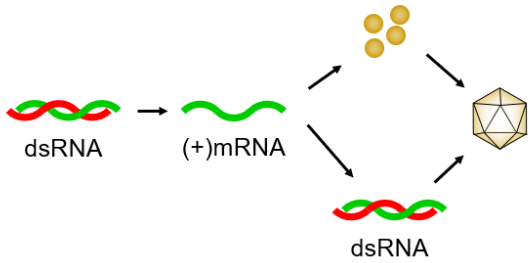
3. **(+) ssRNA Viral Replication:**
   - Positive-sense ssRNA viruses have genetic material that can function both as a genome and as messenger RNA
   - It can be directly translated into protein in the host cell by host ribosomes.
   - The first proteins to be expressed after infection serve genome replication functions.



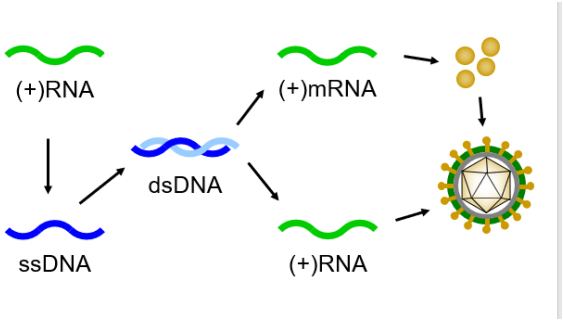
4. **(-) ssRNA Viral Replication:**
   - Negative sense ssRNA viruses need RNA polymerase to form a positive sense RNA.
   - The positive-sense RNA acts as a viral mRNA, which is translated into proteins for the production of new virion materials.
   - With the newly formed virions, more negative sense RNA molecules are produced.



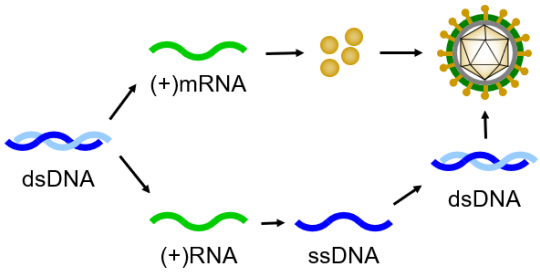
5. **dsRNA Viral Replication:**
   - A small group of viruses carries its genetic information around in the form of double-stranded RNA (dsRNA) molecules.
   - The virions carry an RNA polymerase that transcribes the dsRNA into (+) ssRNA.
   - These transcripts can serve as mRNA that is then translated into the necessary viral proteins or they can act as a template for (-) strand synthesis and be converted back into a dsRNA genome for packaging.



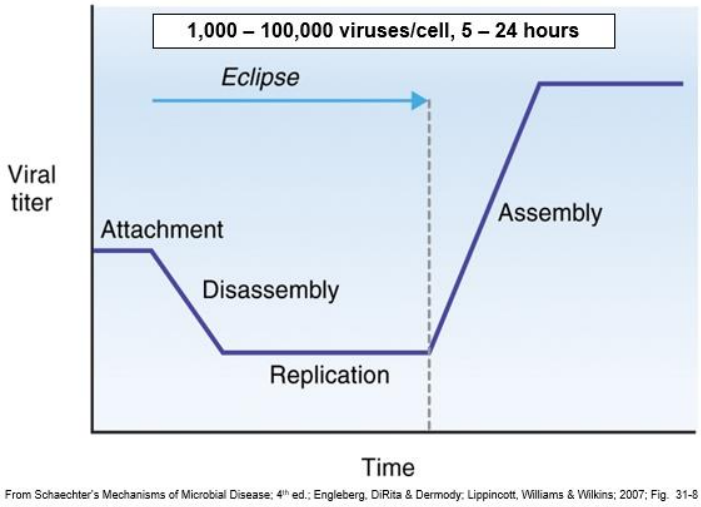
6. **(+) ssRNA Retrovirus Replication:**
   - The virus uses its own reverse transcriptase enzyme to produce DNA from its RNA genome, the reverse of the usual pattern, thus *retro* (backwards).
   - The new DNA is then incorporated into the host cell genome by an integrase enzyme, at which point the retroviral DNA is referred to as a provirus.
   - The host cell then treats the viral DNA as part of its own genome, transcribing and translating the viral genes along with the cell's own genes, producing the proteins required to assemble new copies of the virus.



7. **dsDNA Retrovirus Replication:**
   - Retroviruses are single-stranded RNA animal viruses that employ a double-stranded DNA intermediate for replication.
   - The RNA is copied into DNA by the enzyme reverse transcriptase.



Virus Growth



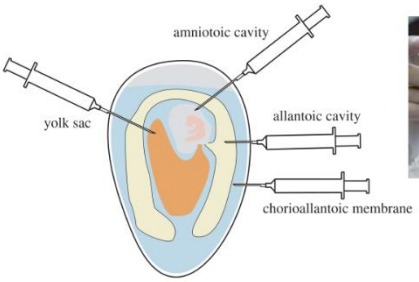
Cultivation of Virus

- Since the viruses are obligate intracellular parasites, they cannot be grown on any inanimate culture medium.
- Viruses can be cultivated within suitable hosts, such as a living cell.
- There is no universal cell that will support all viruses.
- Viruses tend to be host specific; therefore:
    - Human viruses grow best in cells of human origin.
    - Bovine viruses in bovine cells.
    - Canine viruses in canine cells.
    - While some viruses will not grow in-vitro at all
- Therefore, in the laboratory the suspected virus must be grown in a culture method known to support its growth.

Methods for Cultivation of Virus:

- Generally three methods are employed for the virus cultivation
    1. Inoculation of virus into animals
    2. Inoculation of virus into embryonated eggs
    3. Tissue Culture









1. **Inoculation of Virus in Animals:**
   - Laboratory animals play an essential role in studies of viral pathogenesis.
   - Live animals such as monkeys, mice, rabbits, guinea pigs, ferrets are widely used for cultivating virus.
   - Mice are the most widely employed animals in virology.







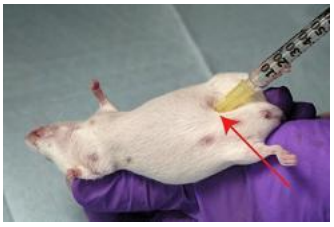


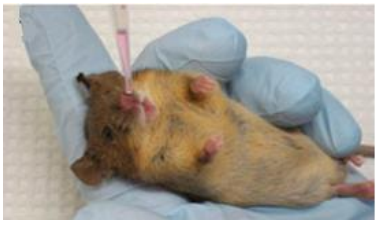


- The different routes of inoculation in mice are:
  - Intracerebral.
  - Subcutaneous.
  - Intraperitoneal.
  - Intranasal.
- After the animal is inoculated with the virus suspension, the animal is:
  - Observed for signs of disease.
  - Observed for visible lesions.
  - Or is killed so that infected tissues can be examined for virus.

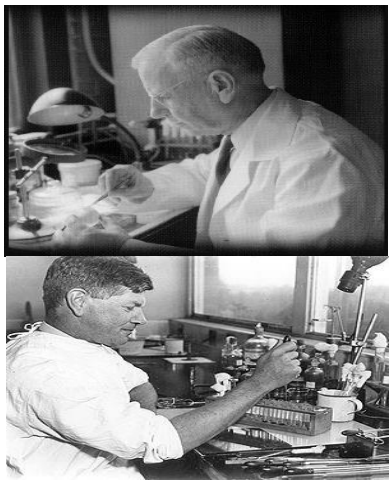








2. **Inoculation of Virus in Embryonated Eggs:**



- Goodpasture and Burnet in 1931 first used the embryonated hen's egg for the cultivation of virus.
- The process of cultivation of viruses in embryonated eggs depends on the type of egg being used.
- Eggs provide a suitable means for:
  - The primary isolation and identification of viruses.
  - The maintenance of stock cultures.
  - The production of vaccines.

Embryonated Egg:

- Referring to an advanced stage of development and not merely after fertilization.
- Terms most often refer to eggs:
  - Embryonated: having an embryo
  - Unembryonated: not having an embryo
  - De-embryonated: having lost an embryo

Inoculation of Virus

- Chicken, duck, and turkey eggs are the most common choices for inoculation
- The egg used for cultivation must be sterile and the shell should be intact and healthy
- Rigorous sterile techniques must be used to prevent contamination by bacteria and fungi from the air and the outer surface of the shell
- The egg must be injected through the shell, usually by drilling a hole or making a small window
- The viral suspension or suspected virus- containing fluid is injected into the fluid of the egg
- The exact tissue that is inoculated is guided by the type of virus being cultivated and the goals of the experiment





Detection of Viral Growth:

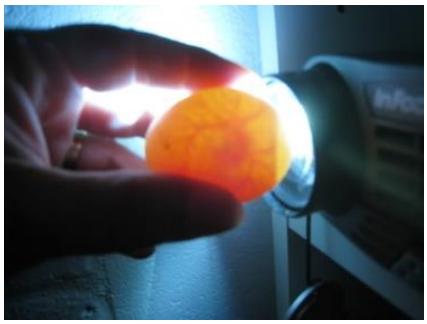


- Viruses multiplying in embryos may or may not cause effects visible to the naked eye
- The signs of viral growth include:
  - Death of the embryo.
  - Defects in embryonic development.
  - Localized areas of damage in the membranes, resulting in discrete opaque spots called pocks.
- If a virus does not produce obvious changes in the developing embryonic tissue, virologists have other methods of detection
- Embryonic fluids and tissues can be prepared for direct examination with an electron microscope
- Certain viruses can also be detected by:
  - their ability to agglutinate red blood cells
  - or by their reaction with an antibody of known specificity

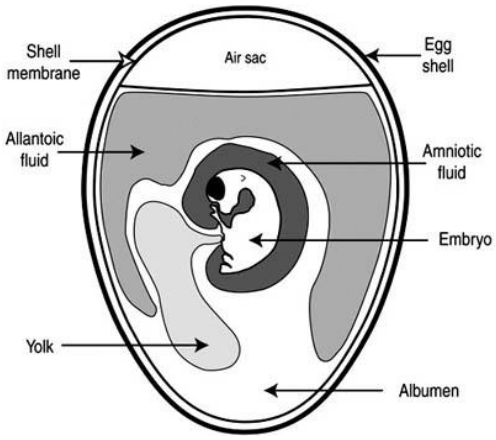
Candling of Egg:

- Candling is the process of holding a strong light above or below the egg to observe the embryo
- A candling lamp consists of a strong electric bulb covered by a plastic or aluminum container that has a handle and an aperture



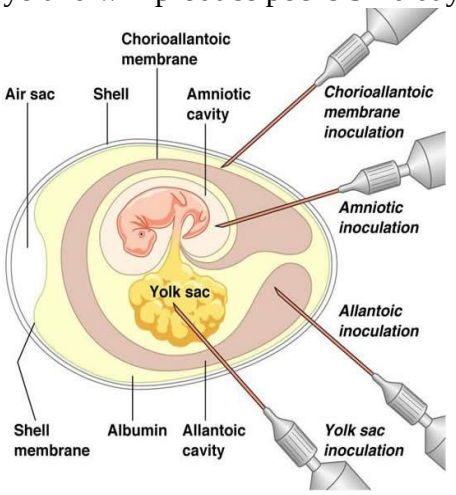


Parts of Embryonated Egg



Routes of Viral Inoculation:

- An embryonated egg offers various sites for the cultivation of viruses
- The different sites of viral inoculation in embryonated eggs are:
  - Chorioallantoic membrane(CAM)
  - Amniotic Cavity
  - Allantoic Cavity
  - Yolk sac
- The chosen route of inoculation and age of the embryo are determined by the given virus selectivity for a certain membrane or developmental stage of the embryo
  - For example Infectious bronchitis virus is propagated in the yolk sac of a 5-6 day old embryo
  - Whereas sarcoma virus is inoculated on the chorioallantoic membrane of a 9-11 day old embryo and will produce pocks 5-10 days post-infection



Chorioallantoic Membrane (CAM):

- This method has been widely used in veterinary virology.
- Many viruses grow readily or can be adapted to grow on the CAM.
- Viruses produce visible foci or 'pocks', inclusion bodies, edema or other abnormalities.
- Each infectious virus particle forms one pock.
- Viruses which can be grown include:
  - Herpes viruses
  - Poxviruses

Amniotic Cavity Inoculation:

- The virus is introduced directly into the amniotic fluid that bathes the developing embryo.
- The volume of fluid in the infected amniotic sac is small (1-2 ml).
- The amniotic route is recommended for the primary isolation of human viruses:
  - mumps virus
  - and influenza A, B and C viruses
- has little application in veterinary virology
- Newly isolated influenza viruses may require several passages before they adapt to growth by other routes, such as allantois.

VIRAL DISEASES

1. Hepatitis A:

- **Virus:** Hepatitis A Virus (HAV)
- **Family:** Picornaviridae
- **Genus:** Hepatovirus
- **Shape:** Icosahedral
- **Genome:**
  - Non-enveloped virus.
  - (+) ssRNA Virus
- **Habitat:** Human and Non-Human primates.
- **Mode of Transmission:** Fecal or oral route.
- **Site of Infection:** Liver
- **Vaccination:** Available
  - **Brand:** Havrix, Vaqta
  - **Duration:** Hepatitis A vaccine is given as two shots, six months apart. Hepatitis A vaccine also comes in combination form containing both Hepatitis A and B vaccine. These can be given to 18 years or older. This form is given as three shots over one month and a booster shot at 12 months.
  - **Site of Delivery:** Intramuscular Injection
- **Treatment:** Available

2. Hepatitis B:

- **Virus:** Hepatitis B Virus (HBV)
- **Family:** Hepadnaviridae
- **Genus:** Orthohepadnavirus
- **Shape:** Icosahedral
- **Genome:**
  - Enveloped Virus
  - Double Stranded DNA Virus
- **Habitat:** Human body.
- **Mode of Transmission:** Contact with blood and body fluids
- **Site of Infection:** Liver
- **Vaccination:** Available.
  - **Brand:** Engerix-B (IM)
  - **Duration:** 1st dose is given after 24 hour of birth with either 2 or 3 more doses are given after that.
- **Treatment:** Oral Antimicrobial agents. Interferon injection.

3. Hepatitis C:

- **Virus:** Hepatitis C Virus (HCV)
- **Family:** Flaviviridae
- **Genus:** Hepacivirus
- **Shape:** Spherical particles surrounded by an icosahedral protective shell of protein.
- **Genome:**
  - Enveloped Virus
  - (+) ssRNA Virus

- **Habitat:** Blood, hepatocytes and other body cells
- **Mode of Transmission:**
  - The hepatitis c virus is a blood-borne virus and the most common modes of infection are through unsafe injection practices, inadequate sterilization of medical equipment, and the transfusion of unscreened blood and blood products.
  - It can also be transmitted by contact with bodily fluids including saliva or semen of an infected person, but this is *rare*
- **Site of Infection:** HCV is carried in blood stream to liver and primarily infects the liver and also affect the digestive system, the immune system, and the brain.
- **Treatment:** Treatment is available including ifn,peg-ifn,ribavirin or any HCV direct-acting antiviral (daa) agent.

4. Hepatitis D:

- **Virus:** Hepatitis D Virus (HDV)
- **Shape:** Small spherical shape
- **Genome:**
  - Circularr (-) ssRNA virus
  - Enveloped
- **Habitat:** Lives waterborne and poor sanitation
- **Mode of Transmission:** Contact with blood and body fluids
- **Site of Infection:** Liver
- **Vaccination:** The vaccine for hepatitis B protects against hepatitis D virus because of the latter's dependence on the presence of hepatitis B virus for it to replicate
- **Treatment:**

5. Hepatitis E:

- **Virus:** Hepatitis E Virus (HEV)
- **Family:** Hepeviridae
- **Genus:** Orthohepevirus
- **Shape:** Icosahedral
- **Genome:**
  - (+) ssRNA Virus
  - Non-enveloped virus
- **Habitat:** Lives in waterborne and poor sanitation
- **Mode of Transmission:** Mostly through oral route and by fecal matter contact
- **Site of Infection:** Liver
- **Vaccination:** Not available
- **Treatment:** Not Available

6. Infentile Gastroentritis:

- **Virus:** Rotavirus A
- **Family:**
- **Genus:**
- **Shape:** Distinct wheel like shape, virus consists of two icosahedral capsid.
- **Genome:**
  - dsRNA Virus
  - Non-enveloped
- **Habitat:** Survive in small intestine near the tips of the villi
- **Mode of Transmission:** Transmitted by fecal or oral route, via contaminated hands, surfaces and objects and possibly by the respiratory route.
- **Site of Infection:** It infects and damages the cells that line the small intestine.
- **Vaccination:**
  - Vaccines against rotavirus A infection are effective in children
    i. Rotarix by GSK
    ii. RotaTaq by Merck
  - Both taken orally
- **Treatment:** No medicine available. But Oral Rehydration Therapy is the primary aim of treatment.

7. Polio Virus:

- **Virus:** Enterovirus C
- **Family:** Picornaviridae
- **Genus:** Enterovirus
- **Shape:** Spherical but closer look to icosahedral
- **Genome:**
  - Non-enveloped
  - (+)ssRNA Virus
- **Mode of Transmission:** GIT, Upper respiratory tract.
- **Site of Infection:** Spinal Cord
- **Vaccination:**
  - Inactivated Polio Vaccine (IPV) parenterally and Oral Polio Vaccine orally
  - Time Span: IPV:14 weeks, OPV: 4 weeks
- **Treatment:** Its treatment includes bed rest, pain killer, anti-spasmodic drugs, antibiotics, physical therapy.

8. Dengue Fever:

- **Virus:** Dengue virus
- **Family:** Flaviviridae
- **Genus:** Flavivirus
- **Shape:** Spherical
- **Genome:** (+) ssRNA virus
- **Habitat:** In mosquitoes and body fluids.
- **Mode of Transmission:** Transmitted by bites of Aedes mosquito.
- **Vaccination:** Not available
- **Treatment:** No treatment available.

9. Chicken Pox:

- **Virus:** Varicella zoster
- **Family:** Herpesviridae
- **Genus:** Varicellovirus
- **Shape:** Icosahedral
- **Genome:**
  - Enveloped
  - Double Stranded DNA
- **Habitat:** Located in neurons in human trigeminal ganglia
- **Mode of Transmission:** Through the droplets released by sneezing or coughing of patient and by contact with blisters.
- **Site of Infection:** Oropharynx, Trigeminal, Cervical, Thoracic and Lumbosaccral dermatome.
- **Vaccination:** Available and delivered by shots.
- **Treatment:** Available as Calamine lotion, Paracetamol and Aciclovir.

10. Congo or Crimea Congo Haemorrhage Fever (CCHF)

- **Virus:** Nairovirus (Tick-borne Virus)
- **Family:** Nairoviridae
- **Genus:** Nairovirus
- **Shape:** Spherical
- **Genome:**
  - (-) ssRNA virus.
  - Enveloped virus.
- **Habitat:** Remain in blood stream
- **Mode of Transmission:** Transmitted by tick bites or through contact with infected animal blood or tissue. Improper sterilization of medical equipments, reuse of needles and contamination of medical supplies.
- **Site of Infection:** Virus causes illness in which fever and haemorrhage.
- **Vaccination:** Not available
- **Treatment:** The antiviral drug ribavirin has been used to treat CCHF infection with apparent benefit. Both oral and intravenous formulation seen to be effective.

11. Warts:

- **Virus:** Papillomavirus
- **Family:** Papillomaviridae
- **Shape:** Icosahedral
- **Genome:**
  - Non-enveloped
  - Double Stranded DNA virus.
- **Habitat:**
- **Mode of Transmission:** Use of public showers, working with meat, eczema, low immune system.
- **Site of Infection:** Skin
- **Vaccination:** available as Gardasil 6 (IM Injection)
  - **Effectiveness:** Resolved in months to years
- **Treatment:** Salicylic acid applied to the skin, cryotherapy.

12. AIDS:

- **Virus:** Human Immunodeficiency Virus (HIV)
- **Family:** Retroviridae
- **Genus:** Lentivirus
- **Shape:** Roughly spherical
- **Genome:**
  - Enveloped virus
  - ssRNA virus
  - RT (RNA Dependent DNA Polymerase)
- **Habitat:** Survives in vital organs of immune system
- **Mode of Transmission:** Bodily fluids such as blood, semen, pre-ejaculate, vaginal secretions and breast milk.
- **Site of Infection:** CD-4 and T-Cells.
- **Vaccination:** Not available
- **Treatment:** It consists of highly active anti—retroviral therapy (HAART) which slows the progression of disease.

13. Influenza:

- **Virus:** Parainfluenza type A, B and C viruses
- **Family:** Paramyxoviridae
- **Shape:** Roughly spherical and somewhat polymorphic.
- **Genome:**
  - Enveloped
  - (-) ssRNA Virus
- **Habitat:** Survive in throat region at actin microfilaments.
- **Mode of Transmission:** Transmitted through direct physical contact and respiratory droplets.
- **Site of Infection:** Superficial epithelial of the upper and lower respiratory tract are primary site of attack. The virus may also agglutinate the RBCs of many animal species.
- **Vaccination:** Vaccines are available namely live attenuated vaccine (Fluzone). Influenza virus undergoes antigenic variations from time to time. It is difficult to prepare sufficient quantity of vaccine.
- **Treatment:** Available including broad spectrum antiviral Ribivirin.

14. Measles:

- **Virus:** Measles morbillivirus
- **Family:** Paramyxoviridae
- **Genus:** Morbillivirus
- **Genome:**
  - Enveloped virus
  - (-) ssRNA virus
- **Habitat:** Survives in air
- **Mode of Transmission:** Through coughing and sneezing.
- **Site of Infection:** Respiratory system.
- **Vaccination:** Available vaccines include universal Mumps, Measles and Rubella Vaccine
- **Treatment:** Oral rehydration solution and supplements

15.Mumps:

- **Virus:** Paramyxovirus
- **Family:** Paramyxoviridae
- **Shape:** Spherical
- **Genome:**
  - Enveloped virus
  - (-) ssRNA virus
- **Habitat:**
- **Mode of Transmission:** Through respiratory droplets and direct contact with infected person.
- **Site of Infection:** Multiple tissues and glands like parotid glands.
- **Vaccination:** Available as live virus
  - **Effectiveness:** 88%
  - **Duration:** Life long
  - **Schedule:** Atleast one dose should be administered with MMR. Single antigen vaccine is not available
- **Treatment:** No specific treatment, pain-killers are given

16.Small Pox:

- **Virus:** Variolo virus
- **Family:** Poxyviridae
- **Genus:** Orthpov-virus
- **Shape:** Large Brick shaped
- **Genome:** Double Stranded DNA virus.
- **Habitat:** Commonly in air.
- **Mode of Transmission:** Spread through air droplets from and infected person to healthy person's nasal, pharyngeal and oral mucosal membranes.
- **Site of Infection:** The virus spreads to the face, forearm, chest, abdomen and back.
- **Vaccination:** Available as Vacinia (Acam 2000) percutaneously. Booster dose may be given after approx. 14 dayss.
- **Treatment:** Only antibiotics are given to counter 2nd bacterial infections.

17.Ebola Haemorrhagic Fever:

- **Virus:** Ebola virus
- **Family:** Filoviridae
- **Genus:** Ebolavirus
- **Shape:** Filamentous, branched and thread like in electron microscope.
- **Genome:**
  - Enveloped Virus
  - (-) ssRNA virus
- **Habitat:** Survives in fruit bats of pteripodidae.
- **Mode of Transmission:** Through direct physical contact and interaction between body fluids with animals like fruit bats and monkeys etc.
- **Site of Infection:** Whole body of animal.
- **Vaccination:** Not available.
- **Treatment:** Only supportive treatment.

18.Rabies:

- **Virus:** Lyssaviruses including Rabies virus and Australian Bat lyssaviruses
- **Family:** Rhabdovirus
- **Genus:** Lyssavirus
- **Shape**: Helical symmetry
- **Genome:**
  - Enveloped virus
  - ssRNA genome.
- **Habitat:** Symptomatic rabid animal's saliva, nerves, cells of cold blooded vertebrates. Sperms, vaginal secretions.
- **Mode of Transmission:** Through transplant surgery, when infected animal scratches or bites or when saliva comes in contact with eyes mouth or nose and through sex.
- **Site of Infection:** Cytoplasm
- **Vaccination:** Vaccines include
  - Human Diploid Rabies Vaccines (HDRV)
  - Purified Chicken Embryo cell Vaccines.
  - Purified vero cell rabies vaccine.
- **Treatment:** Thoroughly washing the wound with soap and water for 5 minutes, povidone-iodine or alcohol also reduces virus, one dose of HRIG and four doses of Rabies vaccine for 14 days.

19.Herpes Virus:

- **Virus:** Herpes simplex virus 1, Herpes Simplex Virus 2.
- **Family:** Herpesviridae
- **Genus:** Simplexvirus
- **Shape:** Icosahedral
- **Genome:**
  - Double Stranded DNA virus
  - Enveloped virus
- **Habitat:**
- **Mode of Transmission:**
  - HSV-1 mainly through oral route
  - HSV-2 mainly through genital route.
- **Site of Infection:** Infections are categorized based on the part of body infected. Oral herpes involves face or mouth.
- **Vaccination:** Not available
- **Treatment:** There is no cure for Herpes Simplex, Once a person has the virus, it remains in the body. The virus lies inactive in the nerve cells until something trigger it to become active again.

NORMAL FLORA

CHAPTER NO. 06

Dr. Qazi Adnan

NORMAL FLORA

MICROBIOLOGY OF AIR:

- The air contains gases, dust particles, dried vapor droplets, in addition to these, air also contains more number of microorganisms.
- The air has vegetative cells and spores of bacteria, fungi and algae and protozoan cyst.
- In the atmosphere, air mainly acts as dispersal or transport medium for microorganisms. When compared to soil or water, air contains less numbers of microorganisms. The microbiology of air can be studied under two headings such as outdoor and indoor microflora.

  Outdoor microflora:
  - The air in the atmosphere, which is found outside the buildings, is referred to as **outside air**. The dominant microflora of outside air are fungi.
  - The two common genera of fungi are
    i. *Cladosporium*
    ii. *Sporobolomyces*
  - Besides these two genera, other genera found in air are *Aspergillus*, *Alternaria*. The outdoor air also contains basidiospores, ascospores of yeast, fragments of mycelium and conidia of molds.
  - Among the bacterial genera *Bacillus* and *Clostridium*, *Sarcina*, *Micrococcus*, *Corynebacterium* and *Achromobacter* are widely found in the outside air.

  Indoor microflora:
  - The air found inside the building is referred to as Indoor air. The commonest genera of fungi in indoor air are *Penicillium*, *Aspergillus*.
  - The commonest genera of bacteria found in indoor air are *Staphylococci*,
  - *Bacillus* and *Clostridium.*

MICROBIOLOGY OF SOIL:

- Soils contain five major groups of microorganisms. They are bacteria, actinomycetes, fungi, algae and protozoa.
- Among the soil microorganisms, bacteria are most dominant group of organisms.
- All kinds of bacteria are found in the soil. This is because all kinds of organic refuse are disposed off on the soil.
- The actinomycetes population as many as millions per gram of soil is present. The most predominant genera present in the soil are:
  - *Nocardia,*
  - *Streptomyces*
  - *Micromonospora.*

THE NORMAL BACTERIAL FLORA OF HUMANS

Introduction:

- In a healthy animal, the internal tissues, e.g. blood, brain, muscle, etc., are normally free of microorganisms. However, the surface tissues, i.e., skin and mucous membranes, are constantly in contact with environmental organisms and become readily colonized by various microbial species
- The mixture of microorganisms regularly found at any anatomical site is referred to as the normal flora, except by researchers in the field who prefer the term "indigenous microbiota".
- The normal flora of humans consists of a few eukaryotic fungi and protists, but bacteria are the most numerous and obvious microbial components of the normal flora.

Bacteria Commonly Found on the Surface of Human Body

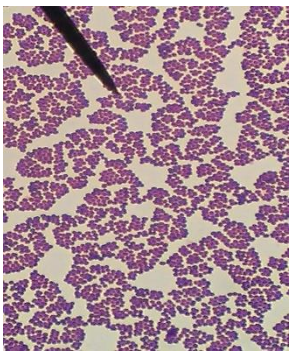
Bacterium	Skin	Conjunctiva	Nose	Pharynx	Mouth	Lower GIT	Ant. Urethra	Vagina
Staphylococcus epidermidis	++	+	++	++	++	+	++	++
*Staphylococcus aureus**	+	+/-	+	+	+	++	+/-	+
Streptococcus mitis				+	++	+/-	+	+
Streptococcus salivarius				++	++			
*Streptococcus mutans**				+	++			
*Enterococcus faecalis**				+/-	+	++	+	+
*Streptococcus pneumoniae**		+/-	+/-	+	+			+/-
*Streptococcus pyogenes**	+/-	+/-		+	+	+/-		+/-
Neisseria sp.		+	+	++	+			+
*Neisseria meningitis**			+	++	+			+
Enterobacteriacae (E.coli)*		+/-	+/-	+/-	+	++	+	+
Proteus sp.		+/-	+	+	+	+	+	+
*Pseudomonas aeruginosa**				+/-	+/-	+	+/-	
*Haemophilus influenzae**		+/-	+	+	+			
*Bacteroides sp.**						++	+	+/-
Bifidobacterium bifidum						++		
Lactobacillus sp.				+	++	++		++
*Clostridium sp.**					+/-	++		
Clostridium tetani						+/-		
Corynebacteria	+		+/-	+/-		+	+	
Mycobacteria				+	+			
Actinomycetes				+	+			
Spirochetes				+	++	++		
Mycoplasmas				+	+	+	+/-	+

++ = nearly 100% + = common (about 25%) +/- = rare (less than 5%) * = Potentional pathogens

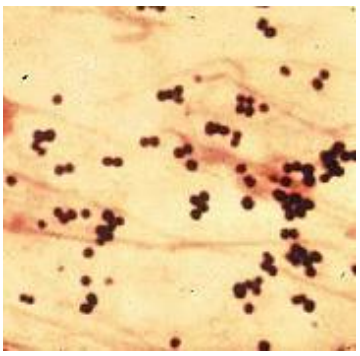
S. epidermidis* and *S. aureus:

The staphylococci and corynebacteria occur at every site listed.

- *Staphylococcus epidermidis* is highly adapted to the diverse environments of its human host.
- *S. aureus* is a potential pathogen.
  - It is a leading cause of bacterial disease in humans e.g endocarditis and skin infection.
  - It can be transmitted from the nasal membranes of an asymptomatic carrier to a susceptible host.



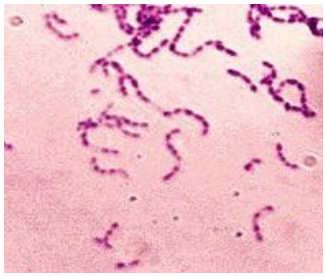
S. epidermidis



S. aureus

Streptococcus mutans:

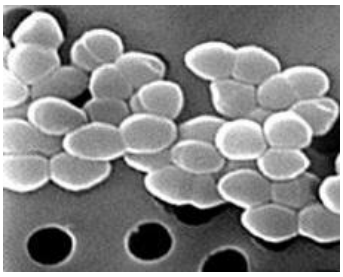
- *Streptococcus mutans* is the primary bacterium involved in plaque formation and initiation of dental caries.
- Viewed as an opportunistic infection, dental disease is one of the most prevalent infectious diseases



Enterococcus faecalis:

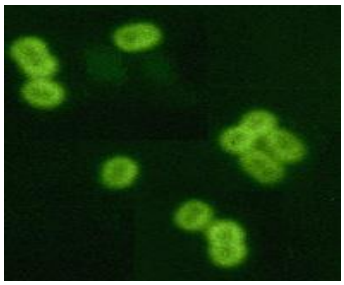
Enterococcus faecalis was formerly classified as *Streptococcus faecalis.*

- The bacterium is a regular a component of the intestinal flora.
- In recent years, *Enterococcus faecalis* has emerged as a significant, antibiotic-resistant, nosocomial pathogen.
- It is vancomycin resistant.

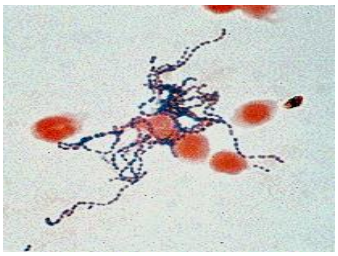


Streptococcus pneumoniae:

- *Streptococcus pneumoniae* is present in the upper respiratory tract of about half the population.
- If it invades the lower respiratory tract it can cause pneumonia.
- *Streptococcus pneumoniae* causes 95 percent of all bacterial pneumonia.



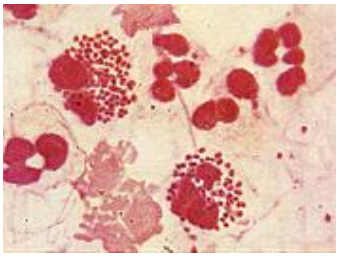
Streptococcus pyogenes:



Streptococcus pyogenes refers to the Group A, Beta-hemolytic streptococci.

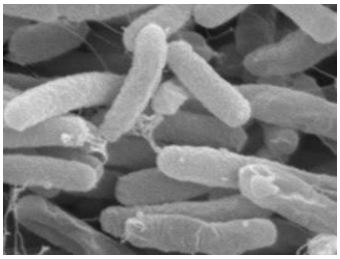
- Streptococci cause tonsillitis (strep throat), pneumonia, endocarditis.
- Some streptococcal diseases can lead to rheumatic fever or nephritis which can damage the heart and kidney.

Neisseria meningitidis:



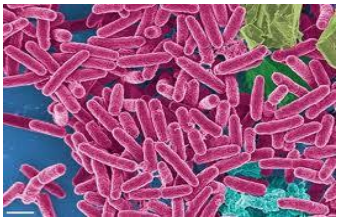
- *Neisseria* and other Gram-negative cocci are frequent inhabitants of the upper respiratory tract, mainly the pharynx.
- *Neisseria meningitidis*, an important cause of bacterial meningitis, can colonize as well, until the host can develop active immunity against the pathogen.

E. coli:



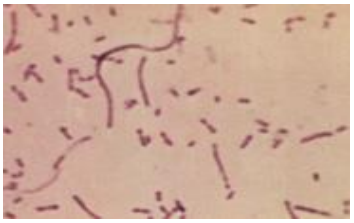
- While *E. coli* is a consistent resident of the small intestine, many other enteric bacteria may reside here as well, including *Klebsiella, Enterobacter* and *Citrobacter*.
- Some strains of *E. coli* are pathogens that cause intestinal infections, urinary tract infections and neonatal meningitis

Pseudomonas aeruginosa:



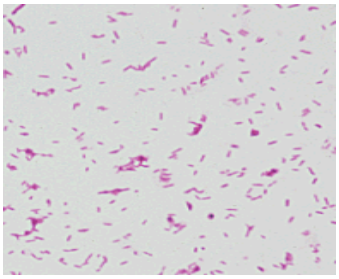
- *Pseudomonas aeruginosa* is the opportunistic pathogen of humans that can invade virtually any tissue.
- It is a leading cause of hospital-acquired (nosocomial) Gram-negative infections, but its source is often exogenous (from outside the host).

Haemophilus influenzae:



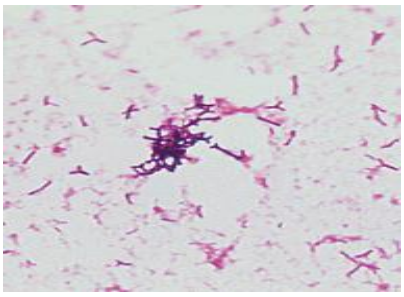
- *Haemophilus influenzae* is a frequent secondary invader to viral influenza and was named accordingly.
- The bacterium was the leading cause of meningitis in infants and children until the recent development of the Hib vaccine.

Bacteroides fragilis:



- The greatest number of bacteria are found in the lower intestinal tract, specifically the colon.
- The most prevalent bacteria are the *Bacteroides*, a group of Gram-negative, anaerobic, non-spore forming bacteria.
- They have been implicated in the initiation colitis and colon cancer

Bifidobacterium bifidum:



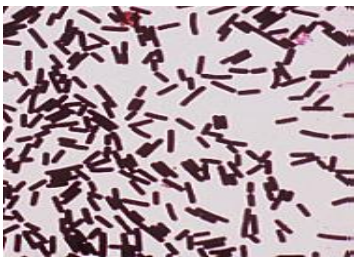
- Bifidobacteria are Gram-positive.
- They have been described as "friendly" bacteria in the intestine of humans.
- *Bifidobacterium bifidum* is the predominant bacterial species in the intestine of breast-fed infants, where it presumably prevents colonization by potential pathogens.
- These bacteria are sometimes used in the manufacture of yogurts and are frequently incorporated into probiotics.

***Lactobacillus* species:**



- Lactobacilli in the oral cavity probably contribute to acid formation that leads to dental caries.
- *Lactobacillus acidophilus* colonizes the vaginal epithelium during child-bearing years and establishes the low pH that inhibits the growth of pathogens.

Clostridium perfringens:



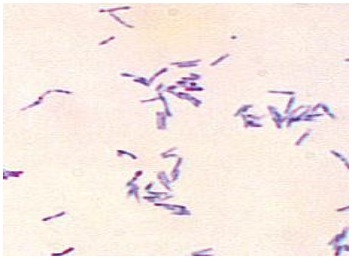
- There are numerous species of *Clostridium* that colonize the bowel.
- *Clostridium perfringens* is commonly isolated from feces.
- *Clostridium difficile* may colonize the bowel and cause "antibiotic-induced diarrhea" or pseudomembranous colitis

Clostridium tetani:



- *Clostridium tetani* is "transiently associated" with humans as a component of the normal flora.
- The bacterium can be isolated from feces in 0 - 25 percent of the population.
- The endospores are probably ingested with food and water, and the bacterium does not colonize the intestine.

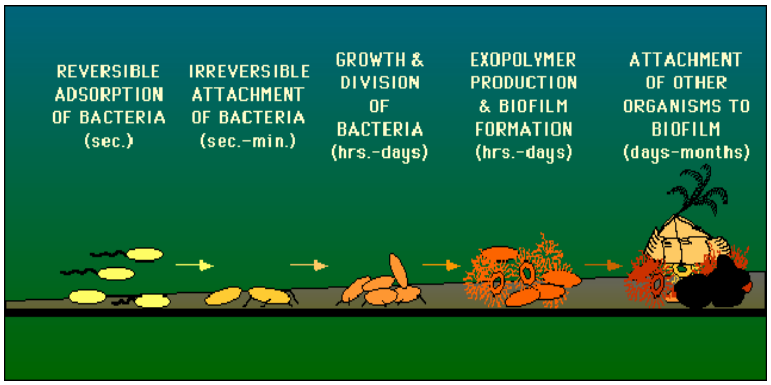
Corynebacterium diphtheriae:



- The corynebacteria, and certain related propionic acid bacteria, are consistent skin flora.
- Some have been implicated as a cause of acne.
- *Corynebacterium diphtheriae, the* agent of diphtheria, was considered a member of the normal flora before the widespread use of the diphtheria toxoid, which is used to immunize against the disease.

Biofilm formation:

- Some of the indigenous bacteria are able to construct biofilms on a tissue surface, or they are able to colonize a biofilm built by another bacterial species.
- Many biofilms are a mixture of microbes, although one member is responsible for maintaining the biofilm and may predominate
- Biofilms usually occur when one bacterial species attaches specifically or nonspecifically to a surface, and then secretes carbohydrate slime (exopolymer) that imbeds the bacteria and attracts other microbes to the biofilm for protection or nutritional advantages.
- The classic biofilm that involves components of the normal flora of the oral cavity is the formation of dental plaque on the teeth. Plaque is a naturally-constructed biofilm, in which the consortia of bacteria may reach a thickness of 300-500 cells on the surfaces of the teeth. These accumulations subject the teeth and gingival tissues to high concentrations of bacterial metabolites, which result in dental disease.



Bacterial Specific Adherence to Host Cells or Tissues:

Some examples of adhesins and attachment sites used for specific adherence to human tissues are described in the table below:

Bacterium	Bacterial Adhesion	Attachment Site
Streptococcus pyogenes	Cell-bound protein (M-protein)	Pharyngeal epithelium
Streptococcus mutans	Cell-bound protein (Glycosyl transferase)	Pellicle of tooth
Streptococcus salivarius	Lipoteichoic acid	Buccal epithelium of tongue
Streptococcus pnemoniae	Cell-bound protein (choline-binding protein)	Mucosal epithelium
Staphylococcus aureus	Cell-bound protein	Mucosal epithelium
Neisseria gonorrhoeae	N-methylphenyl-alanine pili	Urethral/cervical epithelium
Enterotoxigenic *E. Coli*	Type-1 fimbriae	Intestinal epithelium
Uropathogenic *E. Coli*	P-pili (pap)	Upper urinary tract
Bordetella pertussis	Fimbriae (filamentous hemagglutinin)	Respiratory epithelium
Vibrio cholerae	N-methylphenyl-alanine pili	Intestinal epithelium
Treponema pallidum	Peptide in outer membrane	Mucosal epithelium
Mycoplasma	Membrane protein	Respiratory epithelium

THE BENEFITS OF THE NORMAL FLORA

The benefit of the normal flora present in humans are as follows:

1. **The Normal Flora Synthesize and Excrete Vitamins:**
   - In excess of their own needs, which can be absorbed as nutrients by the host. For example, enteric bacteria secrete Vitamin K and Vitamin B_{12}, and lactic acid bacteria produce certain B-vitamins.
   - Germ-free animals may be deficient in Vitamin K to the extent that it is necessary to supplement their diets.

2. **The Normal Flora Prevents Colonization by Pathogens:**
   - By competing for attachment sites or for essential nutrients. This is thought to be their most important beneficial effect, which has been demonstrated in the oral cavity, the intestine, the skin, and the vaginal epithelium.
   - In some experiments, germ-free animals can be infected by 10 *Salmonella* bacteria, while the infectious dose for conventional animals is near 10^6 cells.

3. **The Normal Flora may Antagonize Other Bacteria:**
   - Through the production of substances which inhibit or kill nonindigenous species. The intestinal bacteria produce a variety of substances ranging from relatively nonspecific fatty acids and peroxides to highly specific bacteriocins, which inhibit or kill other bacteria.

4. **The Normal Flora Stimulate the Development of Certain Tissues:**
   - The caecum and certain lymphatic tissues (Peyer's patches) in the GI tract. The caecum of germ-free animals is enlarged, thin-walled, and fluid-filled, compared to that organ in conventional animals.
   - Also, based on the ability to undergo immunological stimulation, the intestinal lymphatic tissues of germ-free animals are poorly-developed compared to conventional animals.

5. **The Normal Flora Stimulate the Production of Cross-reactive Antibodies:**
   - Since the normal flora behave as antigens in an animal, they induce an immunological response, in particular, an antibody-mediated immune (AMI) response.
   - Low levels of antibodies produced against components of the normal flora are known to cross react with certain related pathogens, and thereby prevent infection or invasion.
   - Antibodies produced against antigenic components of the normal flora are sometimes referred to as "natural" antibodies, and such antibodies are lacking in germ-free animals

FUNGI

CHAPTER NO. 07

Dr. Qazi Adnan

FUNGI

Introduction

- Fungus is a member of a large group of eukaryotic organisms that includes microorganisms such as yeasts and molds, as well as more familiar mushrooms.
- Fungi are eukaryotic absorptive heterotrophs.
- There cell wall is made up of chitin.

Classification of Fungi

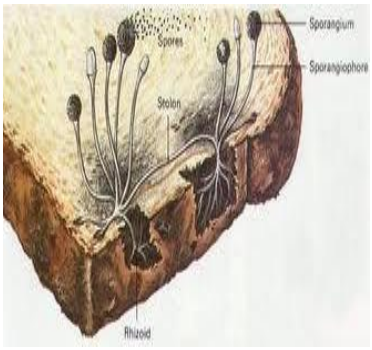
Fungi are classified as follows:

1. Zygomycota
2. Ascomycota
3. Basidiomycota
4. Deuteromcota

1. Zygomycota:

- Also called Zygomycetes
- During their sexual reproduction, zygote formed directly by the fusion of hyphae forms temporary, dormant, thick walled resistant structures called zygospores. Hence the name Zygomycetes.
- Asexual reproduction by spores is common.
- Hyphae are coenocytic.
  Examples: Rhizopus and Mucor etc.





2. Ascomycota:

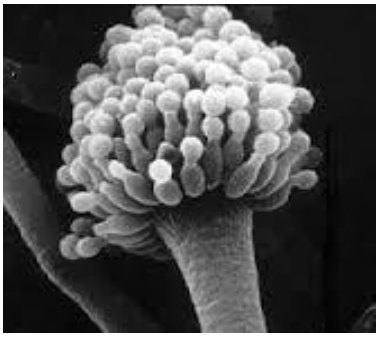
- Also called ascomycetes.
- They produce haploid sexual spores called ascospores by meiosis inside their characteristic sac like structures calles asci (sing. ascus).
- It is the largest group of fungi.
- Hyphae are septate.
  Example: *Blastomyces, histoplasma, microsporum, trichophyton, coccidioides* and yeasts (*saccharomyces* and *candida*)



3. **Basidiomycota:**
   - Also called basidiomycetes.
   - These are named so for their characteristic, club-shaped (hence also called club-fungi) sexual structures called the **basidium** (plural. basidia).
   - These are among the most familiar fungi; edible fungi, devastating plant pathogens etc.
     Example: Mushrooms and *cryptococcus*.



4. **Deuteromycota:**
   - Also called deuteromycetes.
   - This heterogenous group includes all such fungi in which the sexual phase has not been observed.
   - Most of them are related to their sexually reproducing relatives of ascomycetes, however some are related to other two phyla as well.
   - These are also called imperfect fungi.
   - **Example:** Penicillium, aspergillus and fungi causing athlete's foot, ringworm etc.



Terminologies:

A few terminologies related to fungi are explained as below:

- **Yeast**
  Unicellular form that are spherical or ovoid in shape

- **Molds:**
  Complex multicellular microorganisms having variety of specialized structuctures with specific functions

- **Hyphae**
  Thread like tubes containing cytoplasm and organelles of organism

- **Septa**
  Cross walls in hyphae forming individual cells

- **Conidiophore or Sporangiophores**
  Specialized hypha that bears reproductive structures of some molds

- **Conidia or spore**
  Conidia are asexual fungal spores.

- **Diamorphs**
  - Fungi grow as molds in natural environment and in laboratory culture but yeast or yeast like structures in tissues
  - **Example:** *Histoplasma, blastomyces and sporothrix*

MYCOSES

"A disease caused by infection with a fungus is called mycoses."

Types of Mycoses:

There are following types of mycoses:

- Superficial mycoses
- Cutaneous mycoses
- Subcutaneous mycoses
- Systemic mycoses due to primary pathogens
- Systemic mycoses due to opportunistic pathogens

Superficial Mycoses

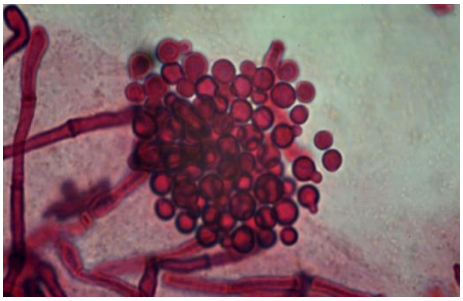
Superficial mycoses are limited to the outermost layers of the skin and hair.

- Do not elicit immune response
- No discomfort
- Cosmetic problems
- Limited to stratum corneum
- **Infections**
  i. Pityriasis versicolor- Affects skin
  ii. Tinea nigra- Affects skin
  iii. Black piedra- Affects Hair
  iv. White piedra-Affects Hair

i. Pityriasis Versicolor:

- *Malassezia furfur* is the causative agent.
- *Malasseszia furfur* is a lipophilic fungus living on the skin as part of the normal flora.
- So, it is cultivated on media containing fatty substances like olive oil.
- Pityriasis versicolor is associated with hyperpigmented or hypopigmented macules usually on the chest and upper back of the patients.
- Pityriasis folliculitis.
- Catheter acquired fungemia in patients receiving total parenteral nutrition containing lipid emulsion.
- **Diagnosis**
  - **Wood's lamp:**
    Wood's light is ultraviolet light at wave length 365 nm.
    The lamp is held four or five inches from the affected skin. If pityriasis versicolor is present on the skin, the affected skin will appear yellowish green in color.
- **Specimen:**
  Skin scrapings from the lesions.
- **Direct Microscopy:**
  - The skin scrapings are mounted in 10-20% KOH.
  - It will show spherical budding cells and short unbranched angular septate hyphae (spaghetti and meat balls).
  - These microscopic features are diagnostic for Malassezia furfur and culture is not necessary.

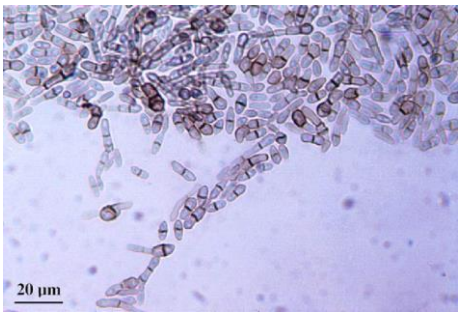
- **Treatment**
  - Daily application of selenium sulfide.
  - Topical or oral azoles are very effective.





ii. Tinea Nigra:

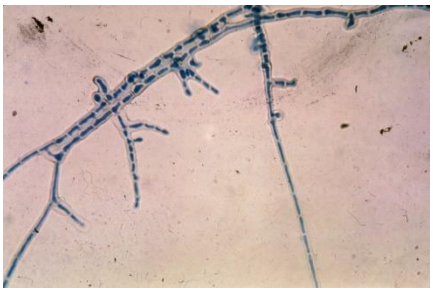
- **Causative agent:** *Exophiala (Hortae) werneckii*
  - It is a saprophytic fungus which occurs in the soil.
  - It is a dematicaeous fungus (contains melanin in its cell wall and appear dark in color under the microscope).
- **Specimen:**
  Skin scrapings from the lesions.
- **Direct Microscopy:**
  - The skin scrapings are mounted in 10-20% KOH.
  - It will show brown to black yeast cells + branched septate hyphae.
- **Treatment:**
  - Keratolytic solutions like salicylic acid.
  - Azoles are also effective.



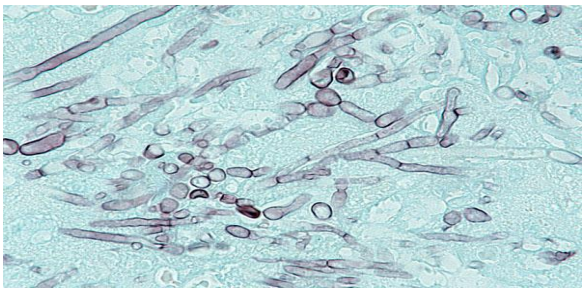

iii. Black Piedra:

- **Causative agent*:*** *Piedraia hortai* is the
- It is a dematicaeous mold.
- It affects the hair like beard, moustache and scalp & pubic hair.
- It causes brown to black hard nodules around the hair shaft.
- **Treatement**
  It is treated by removal of infected hairs + topical antifungal drugs




iv. White Piedra:

- **Causative agent:** *Trichosporon beigelii*
- It is a mold.
- It affects the hair.
- It causes large, white to yellow soft nodules around the hair shafts.
- **Treatment:**
  It is treated by removal of infected hairs + topical antifungal drugs





Cutaneous Mycoses

- Cutaneous mycoses are fungal infections usually confined to the outer layers of skin, hair, and nails, and do not invade living tissues.
- Also known as “ringworm” and tinea (latin “worm”) because of round shape of lesions
- Also called dermatophytes.
- There are 3 Genera:
  1. Trichophyton
  2. Microsporum
  3. Epidermophyton

- **Classification:**
  These can be classified on the basis of Ecology
  i. Anthropophilic
  ii. Zoophilic
  iii. Geophilic

i. Anthropophilic:

- Associated with humans only. Person -to-person transmission through contaminated objects (comb, hat, etc.)
- All three genera i.e. Trichophyton, Microsporum, Epidermophyton

ii. Zoophilic:

- Associated with animals. Direct transmission to humans by close contact with animals.
- Only two genera i.e. Trichophyton and Microsporum

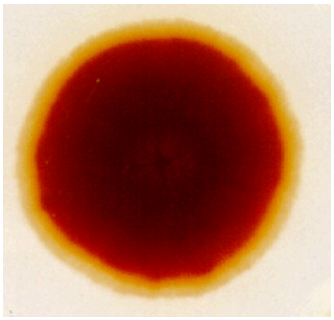
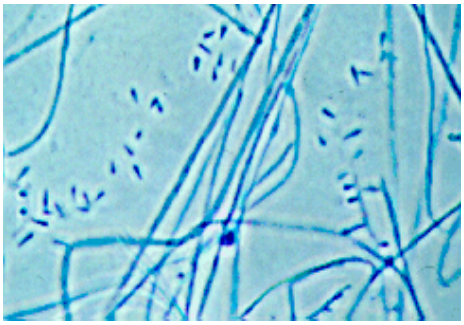
iii. Geophilic:

- Usually found in soil. Transmitted to humans by direct exposure
- Microsporum

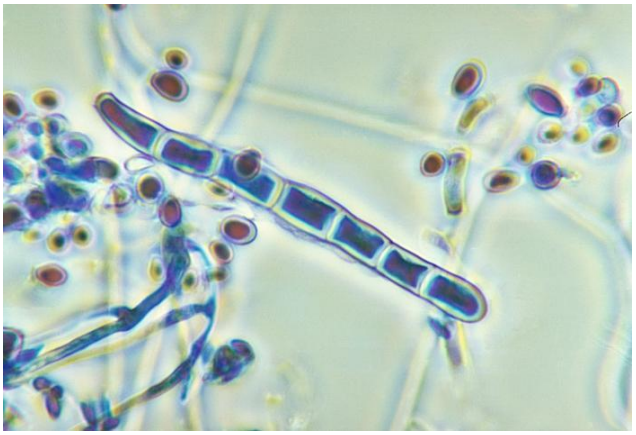
1. Trichophyton:

(19 species)

- The genus *Trichophyton* is characterised by the development of both smooth-walled macro- and microconidia.
- Causes infections of Hair, Skin, Nails
- **Causative agents are**
  - T. rubrum
  - T. violaceum
  - T. mentagrophytes
- On Sabouraud's dextrose agar, colonies are flat to slightly raised, white to cream.

- When sample taken from infection they appear as:



2. Microsporum

(13 species)

- *Microsporum* forms both macroconidia (large asexual reproductive structures) and microconidia (smaller asexual reproductive structures) on short conidiophores.
- Causes infection of skin and hair.
- **Caustive agents are**
  - M. audouinii
  - M. canis
  - M. gypseum

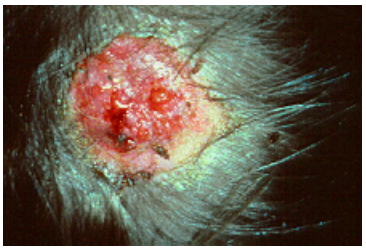
3. Epidermophyton:

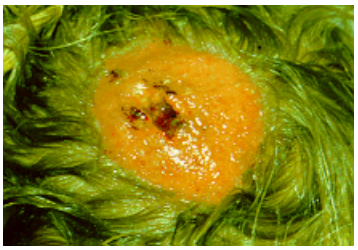
- Causes infection in skin and nails.
- **Causative agent is**
  - *Epidermophyton floccosum*

ANATOMICAL INFECTIONS

1. Tinea Capitis

- Ringworm of scalp and hair
- Tinea capitis refers to dermatophytosis of the scalp. It is caused by tricophyton and microsporum.





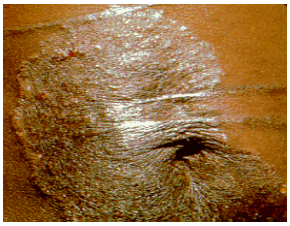

2. Tinea Barbae

- Ringworm of beard region.
- Tinea of the beard caused by *T. rubrum*



3. Tinea Corporis

- Ringworm of the smooth skin of the body


Tinea corporis caused by *T. rubrum*


Tinea corporis caused by *M.canis*

4. **Tinea Cruris**
   - Starts in groin area ("Jock itch")
   - It occurs more commonly in males and is usually due to spread of the fungus from the feet. Thus, the usual causative agents are *T. rubrum, T. interdigitale* and *E. floccosum*.

5. **Tinea Pedis**
   - Classically interdigital (" Athlete's foot")
   - Substrates like carpet and matting that hold skin scales make excellent vectors. Thus, transmission of dermatophytes like *Trichophyton rubrum, T. interdigitale* and *Epidermophyton floccosum* is usually via the feet. In this site infections are often chronic and may remain subclinical for many years only to become apparent when spread to another site, usually the groin or skin.
   - Figure 5.1 shows Tinea pedis caused by *T. rubrum*
   - Figure 5.2 shows tinea pedis caused by *E. floccosum*.
   - Figure 5.3 shows vesicular type tinea pedis caused by *T. interdigitale*.
   - Figure 5.4 shows tinea pedis caused by *T. rubrum*



Figure 5.1



Figure 5.2

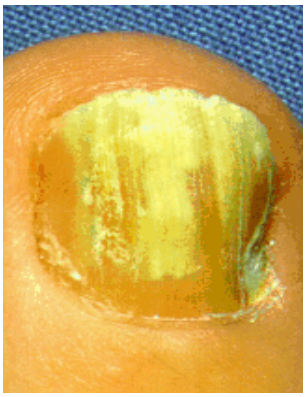


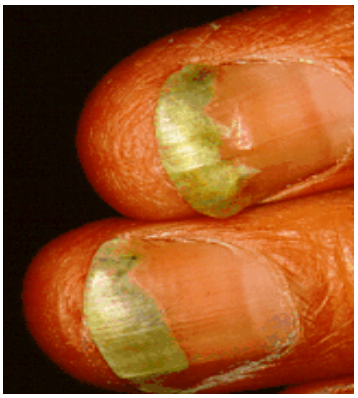
Figure 5.3



Figure 5.4

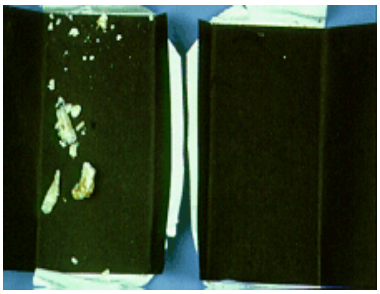
6. **Tinea Unguium**
   - Infection of finger and toe nails.
   - *Trichophyton rubrum* and *T. interdigitale* are the dominant dermatophyte species involved.





Laboratory Diagnosis

- **Clinical Material**
  Skin Scrapings, nail scrapings and epilated hairs
- The laboratory needs enough specimen to perform both microscopy and culture. Routine turn around times for direct microscopy should be less than 24 hours, however culture may take several weeks.





- **Direct Microscopy**
  - KOH mount of infected skin scales (figure 1) and nail material (figure 2) showing typical dermatophyte hyphae breaking up into arthroconidia. Invasion by *M. gypseum* (figure 3) KOH mount of an infected hair showing an invasion caused by *T. tonsurans* (figure 4)



Figure 1

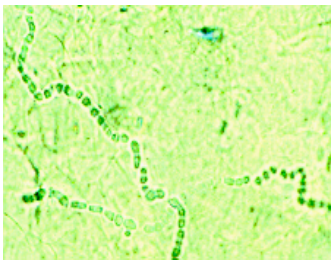


Figure 2

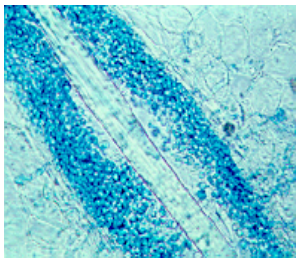


Figure 3

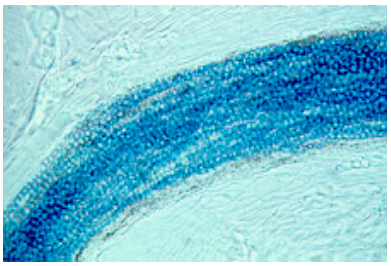


Figure 4

- **Culture**
  Mixed culture of *T. violaceum* and *T. tonsurans* from a case of endothrix Tinea capitis



Subcutaneous Mycoses

- *Subcutaneous mycoses* involve the dermis, subcutaneous tissues, muscle, and fascia.
- These infections are chronic and can be initiated by piercing trauma to the skin, which allows the fungi to enter.
- These infections are difficult to treat and may require surgical interventions.
- There are following types of subcutaneous mycoses:
    1. Sporotrichosis
    2. Chromoblastomycosis
    3. Mycetoma
    4. Zygomycosis

1. Sporotrichosis

- Sporotrichosis is primarily a chronic mycotic infection of the cutaneous or subcutaneous tissues and adjacent lymphatics characterized by nodular lesions which may ulcerate
- **Causative agent:**
  Sporothrix schenckii is a fungus that can be found world-wide in the environment.
    - It is found growing as filamentous hyphae.
    - In host tissue it is found as a yeast.
    - The transition between the hyphal and yeast forms is temperature dependent making *S. schenckii* a thermally dimorphic fungus.

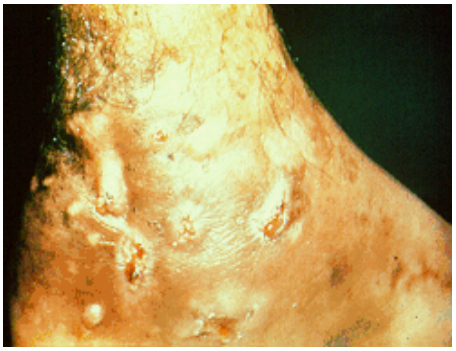



2. Chromoblastomycosis:

- Chromoblastomycosis is a chronic fungal infection of the skin and the subcutaneous tissue caused by traumatic inoculation of a specific group of dematiaceous fungi.
- **Causative Agent:**
  Cladophialophora carrionii, Phialophora verrucosa




3. Mycetoma:



- Mycetoma is a chronic, progressively destructive morbid inflammatory disease usually of the foot but any part of the body can be affected.
- Infection is most probably acquired by traumatic inoculation of certain fungi or bacteria into the subcutaneous tissue.
- **Causative agents:**
  Acremonium sp., Aspergillus nidulans, Madurella grisea, Madurella mycetomatis

4. Zygomycosis:



- Zygomycosis is a rare infection caused by a class of fungi called Zygomycetes.
- **Causative agents:**
  Mycocladus, Basidiobolus , Conidiobolus,, Mucor sp., Rhizomucor, Rhizopus oryzae, Rhizopus sp.

Systemic Mycoses Due to Primary Pathogens

- Systemic mycoses due to primary pathogens originate primarily in the lungs and may spread to many organ systems.
- Organisms that cause systemic mycoses are inherently virulent.
- Generally, primary pathogens that cause systemic mycoses are dimorphic
  1. Histoplasmosis
  2. Coccidioidomycosis
  3. Blastomycosis
  4. Paracoccidioidomycosis

1. Histoplasmosis:



- It is an infection by a fungus found in the droppings of birds and bats in humid areas.
- It is not serious if confined to the lungs but can be fatal if spread throughout the body.
- **Causative agent**
  Histoplasma capsulatum

2. Coccidioidomycosis:

- Coccidioidomycosis is an infection usually caused by. inhaling the spores (“seeds”) of either Coccidioides. immitis or Coccidioides posadasii fungi.
- **Causative agent**
  Coccidioides immitis





3. **Blastomycosis:**
   - Blastomycosis (also known as "North American blastomycosis", "blastomycetic dermatitis", and "Gilchrist's disease") is a fungal infection of humans and other animals, notably dogs and occasionally cats, caused by the organism Blastomyces dermatitidis.
   - **Causative agent**
     Blastomyces dermatitidis



4. **Paracoccidioidomycosis:**
   - Paracoccidioidomycosis (PCM) (also known as "Brazilian blastomycosis", "South American blastomycosis", "Lutz-Splendore-de Almeida disease" and "paracoccidioidal granuloma") is a fungal infection caused by the fungus *Paracoccidioides brasiliensis*.
   - **Causative agent**
     Paracoccidioides brasiliensis



SYSTEMIC MYCOSES DUE TO OPPORTUNISTIC PATHOGENS

- Systemic mycoses due to opportunistic pathogens are infections of patients with immune deficiencies who would otherwise not be infected.
  - Examples of immunocompromised conditions include AIDS, alteration of normal flora by antibiotics, immunosuppressive therapy, and metastatic cancer.
  - Examples of opportunistic mycoses include Candidiasis, Cryptococcosis and Aspergillosis

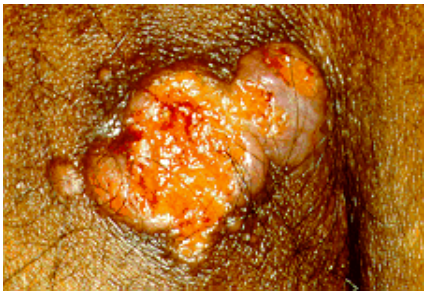
1. Candidiasis
2. Cryptococcosis
3. Aspergillosis

1. **Candidiasis:**
   - Candidiasis is a primary or secondary mycotic infection caused by members of the genus *Candida*



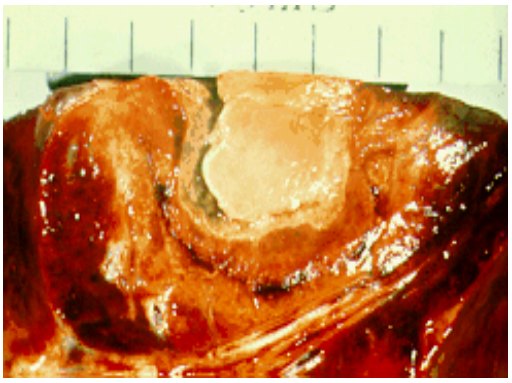


2. **Cryptococcosis:**
   - Infestation with a yeast like fungus, resulting in tumors in the lungs and sometimes spreading to the brain.
   - **Causative agents:**
     Cryptococcus albidus, Cryptococcus laurentii, Cryptococcus neoformans





3. **Aspergillosis:**
   - Aspergillosis is the name given to a wide variety of diseases caused by infection by fungi of the genus *Aspergillus*.
   - The majority of cases occur in people with underlying illnesses such as tuberculosis, or chronic obstructive pulmonary disease, but with otherwise healthy immune systems.
   - **Causative agents:**
     Aspergillus flavus, Aspergillus fumigatus, Aspergillus nidulans, Aspergillus niger



Treatment:

- Antifungal drugs are used to treat mycoses. Depending on the nature of the infection, a topical or systemic agent may be used. Photochemotherapy or photopheresis is a technique used at medical centers for the treatment of mycosis fungoides
- An example of antifugal is fluconazole, which is the basis of many over-the-counter antifungal treatments. Another example is amphotericin B, which is more potent. It is used in the treatment of the most severe fungal infections that show resistance to other forms of treatment and it is administered intravenously
- Drugs to treat skin infections are Tolnaftate, an over the counter topical; Ketoconazole, especially used to treat tinea versicolor and other dermatophytes; Itraconazole; Terbinafine, Griseofulvin, commonly used for infections involving the scalp and nails.
- Yeast infections in the vagina, caused by candida albicans, can be treated with medicated suppositories whereas skin yeast infections are treated with medicated ointments

Prevention:

- Keeping the skin clean and dry, as well as maintaining good hygiene. Because fungal infections are contagious, it is important to wash after touching other people or animals with fungal disease. Sports clothing should also be washed after use.

FUNGAL CULTURE PROCESS

Fungal culture process consists of following steps:

- Specimen collection and transportation
- Direct examination of specimen
- Selection and inoculation of media
- Evaluation of fungal growth
- Serological testing
- Antifungal susceptibility testing

1. **Specimen Collection:**

   In this procedure following steps are performed:

   - Specimen types collected from area most likely infected
   - Use sterile technique
   - Keep specimen moist
   - Label container properly
   - Transport right away
   - Process right away

2. **Direct Examination:**
   - Provides preliminary report
   - Observe yeast phase of dimorphic
   - Gives clues to identify causative agent
   - Inoculate special media
   - May require more than one direct examination method
   - Saline wet mount
   - Lactophenol cotton blue wet mount
   - 10% KOH preparation
   - Gram stain
   - Acid fast stain
   - Indian ink stain
   - Calcofluor white stain
   - Wright's stain
   - Gomori Methenamine Silver stain
   - Periodic Acid Schiff stain

3. **Specimen Processing:**
   1. **Safety**
      - Tube media preferred over plate media
      - Work in safety hood
      - Wear gloves and lab coat
      - Autoclave specimens and media
      - Disinfect work area daily
   2. **Primary isolation media**
      - Goal: isolate potential pathogens
      - Use non-selective and selective media
      - Proper ingredients
      - Incubation temperature
      - Incubation time
      - Incubation atmosphere

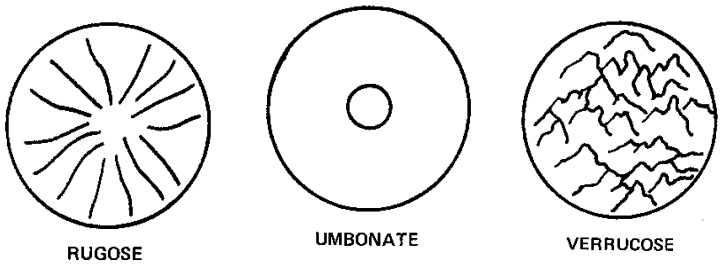
Non-selective Media:
- Sabouraud dextrose agar
- Brain heart infusion (BHI) with/without 5% blood and 1% glucose

Selective Media:
- Mycosel agar
- Inhibitory mold agar
- Dermatophyte test medium

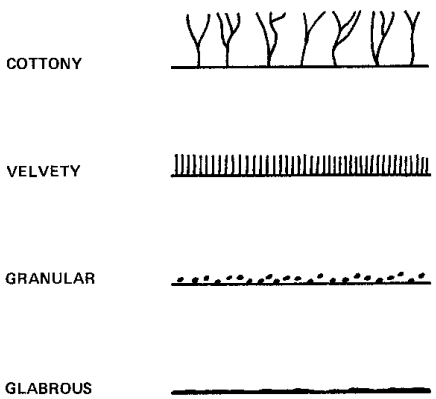
Subculture / Identification Media:
- Neutral Sabouraud dextrose agar (Emmon's)
- Cornmeal-Tween 80 agar
- Niger seed agar (Birdseed agar)
- Tween 80 / Oxgall / caffeic acid agar
- Potato dextrose agar

Examination of Culture Growth:
- Potential pathogens
  - Slow growers
  - Growth on Mycosel
  - Color: dull buff, brown, mousy gray
  - Dimorphic
- Growth rate
  - Rapid growers: 1-5 days
  - Intermediate growers: 6-10 days
  - Slow growers: >10 days

Appearance of Colony Morphology:
- Rugose
- Umbonate
- Verrucose
- Flat



Texture of Colony Morphology:
- Cottony
- Velvety
- Glabrous
- Granular



Microscopic Morphology:

- Definitive means of identification
- Evaluate:
- Shape
- Method of production
- Arrangement of conidia/spores
- Size and color of hyphae

Microscopic Techniques:

- Tease mount
- Scotch tape preparation
- Slide culture

Serological Diagnosis:

- Immunodiffusion
- Complement fixation
- Latex agglutination

Antifungal Susceptibility:

- Determine appropriateness
    - Standardization of testing
    - Methods
    - Predictability in vivo
- Antifungal agents

PROTOZOA

CHAPTER NO. 08

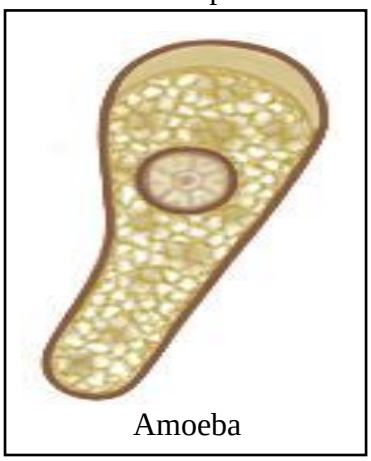
Dr. Qazi Adnan

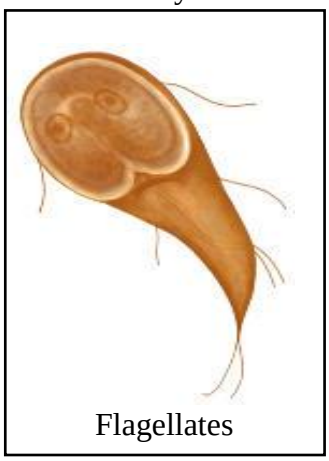
PROTOZOA

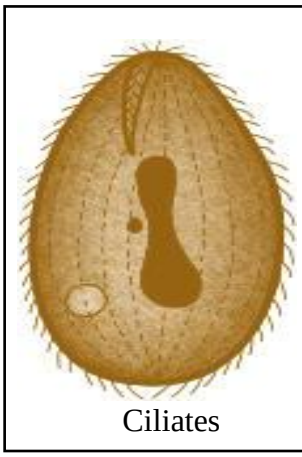
"Protozoa are eukaryotic organisms (with a membrane-bound nucleus) which exist as structurally and functionally independent individual cells."

Biodiversity

- Four main groups of protozoa are recognized on the basis of their locomotion using specialized subcellular and cytoskeletal features


Amoeba


Flagellates


Ciliates


Sporozoa

1. **Sarcodina (Amoebae):**
   - Amoebae use pseudopodia (singular: pseudopodium) to creep or crawl over solid substrates.
   - Pseudopodia (or 'false feet') are temporary thread-like or balloon-like extensions of the cell membrane into which the protoplasm streams.
   - Similar amoeboid motion has been observed in cells of many life-forms, especially phagocytic cells (e.g. human macrophages).

2. **Mastigophora (Flagellates):**
   - Flagellates use elongate flagella (singular: flagellum) which undulate to propel the cell through liquid environments.

3. **Ciliates:**
   - Ciliates use numerous small cilia (singular: cilium) which undulate in waves allowing cells to swim in fluids.
   - Cilia are 'hair-like' extensions of the cell membrane similar in construction to flagella but with interconnecting basal elements facilitating synchronous movement.
   - Ciliated cells are found in specialized tissues and organs in many other higher life-forms (e.g. human bronchial epithelial cells).

4. **Sporozoa:**
   - Sporozoa ('spore-formers') were originally recognized not on the basis of their locomotion, but because they all formed non-motile spores as transmission stages.
   - Recent studies, however, have shown that many pre-spore stages move using tiny undulating ridges or waves in the cell membrane imparting a forward gliding motion, but the actual mechanisms involved are not yet known.

Location	Species	Mode of transmission	Disease
Skin (Blood and tissue)	*Leishmania tropica* *L. braziliensis*	Sand fly	cutaneous leishmaniasis mucocutaneous leishmaniasis
Intestine (Intestinal tract)	*Cryptosporidium spp* *Giardia lamblia* *Cyclospora cayetanensis* *Microsporidia* *Entamoeba histolytica*	Ingestion of cysts in food	cryptosporidiosis gardiasis cyclosporiasis microsporidiosis amebiasis
Liver	*Entamoeba* *Leishmania*		
Blood	*Plasmodium vivax, P. ovale* *P. malariae* *Trypanosoma gambiense*	Anopheles mosquito Tsetse fly	malaria sleeping sickness
CentraL Nervous System	*Toxoplasma gondii* *Entamoeba* *Plasmodium* *Trypanosoma*	Ingestion of cyst in raw meat	toxoplasmosis amebae malaria trypanosomes

Diseases Caused by Protozoans:

Some of the diseases caused by the protozoans are as follows:

1. Amebiasis
    a. Primary Amebic Meningoencephalitis
2. Giardiasis
3. Trichomoniasis
4. Balantidiasis
5. Toxoplasmosis
6. Malaria
7. Cryptosporidiosis
8. Pneumocystosis
9. Leishmaniasis

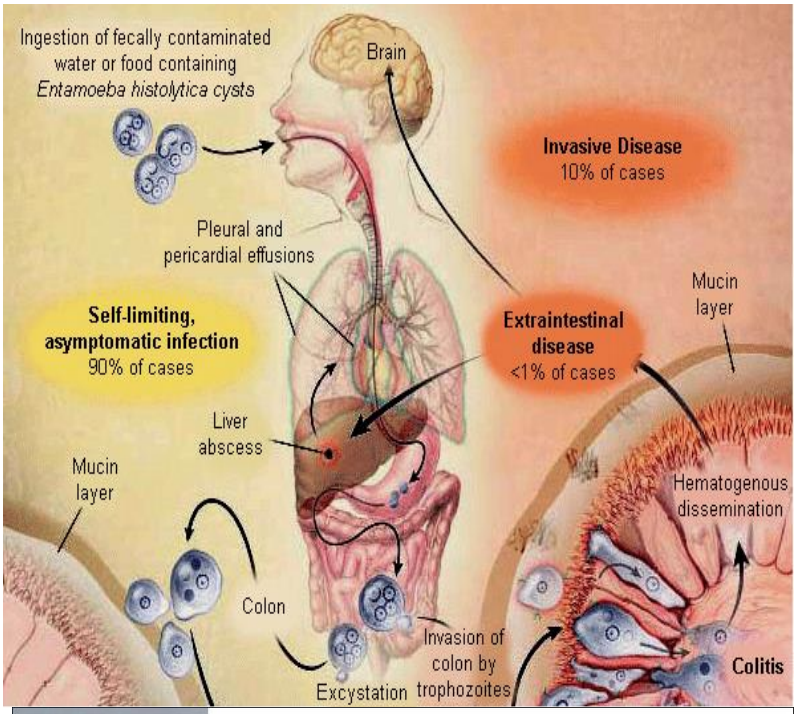
1. **Amoebiasis:**

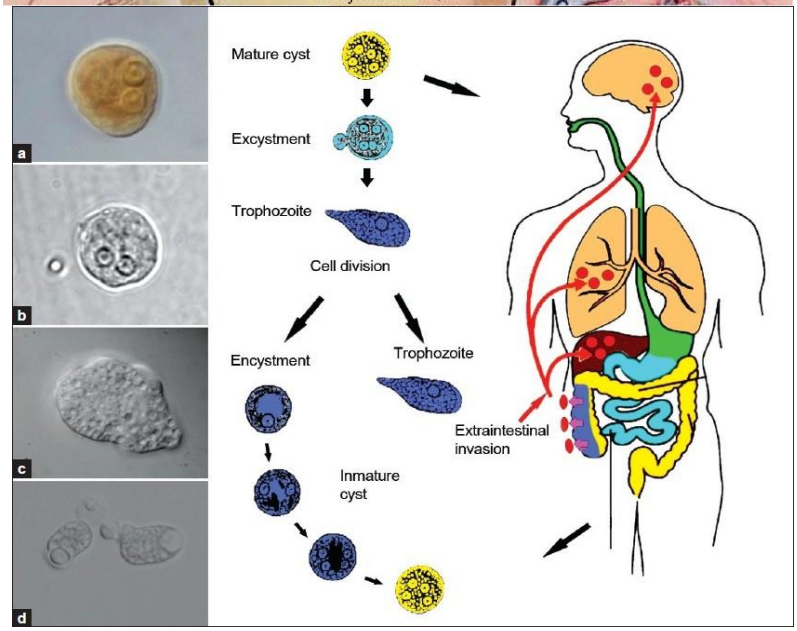
Causative agent: *Entamoeba histolytica*

- A parasitic amoeba; class Sarcodina
- Transmitted via contaminated food & water
- Cysts are shed in the feces
- When ingested: Excystation occurs in the intestine
- Trophozoites grow & reproduce in the intestinal tract

Symptoms

- Abdominal pain
- Little diarrhea but often blood in the stool ("amoebic dysentery")
- Possibly of intestinal ulceration; perforation; infection of internal organs such as liver & lungs





2. **Trichomoniasis:**

Causative agent: *Trichomonas vaginalis*

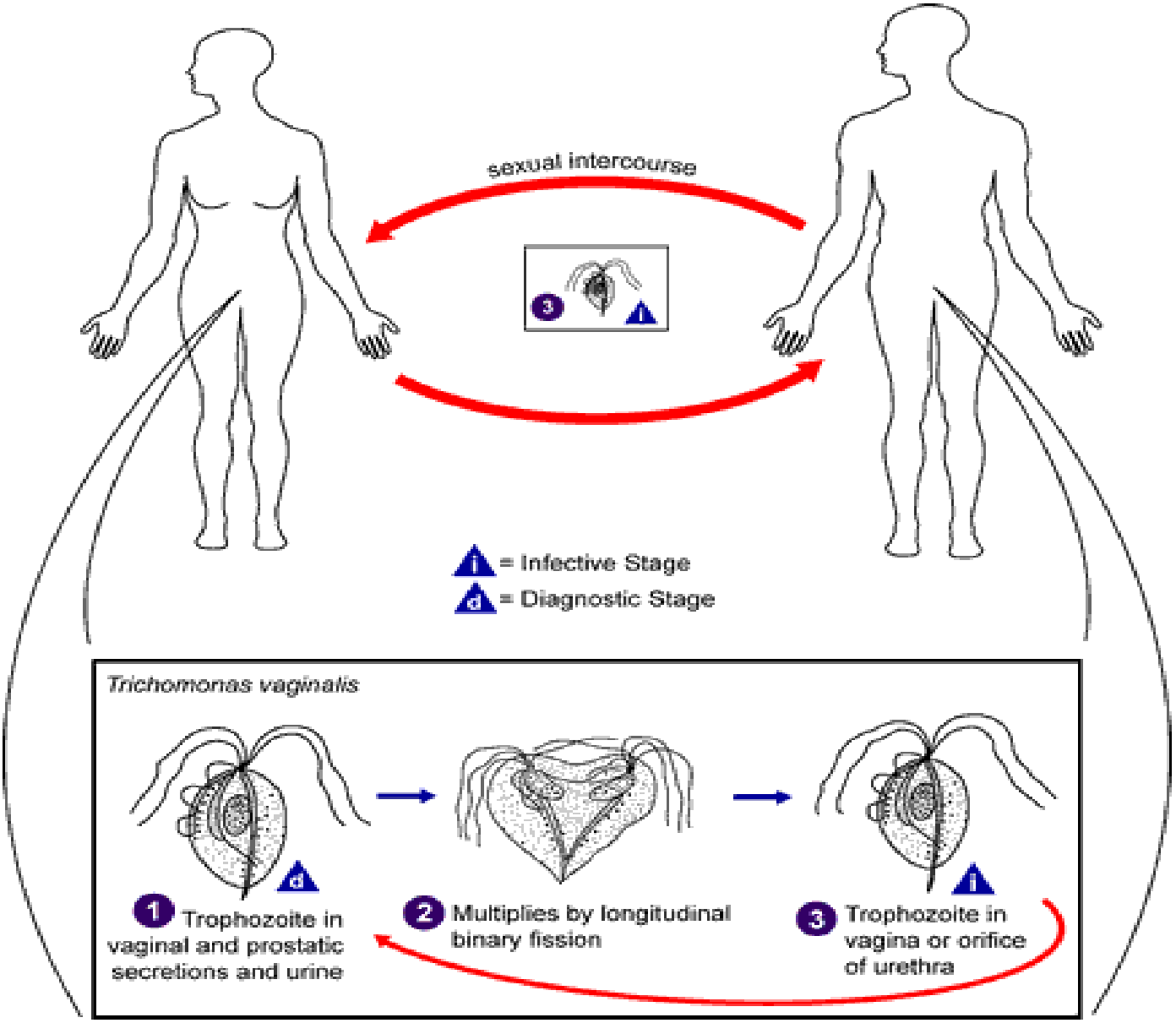
- Flagellate of the class Mastigophora
- No cyst stage

Mode of Transmission:

- Transmitted via sexual contact.

Symptoms:

- Vaginal discharge, which can be white, gray, yellow, or green, and usually frothy with an unpleasant smell.
- Vaginal spotting or bleeding.
- Genital burning or itching.
- Genital redness or swelling.
- Frequent urge to urinate.
- Pain during urination or sexual intercourse.



3. **Giardiasis:**

Causative agent: *Giardia lamblia*

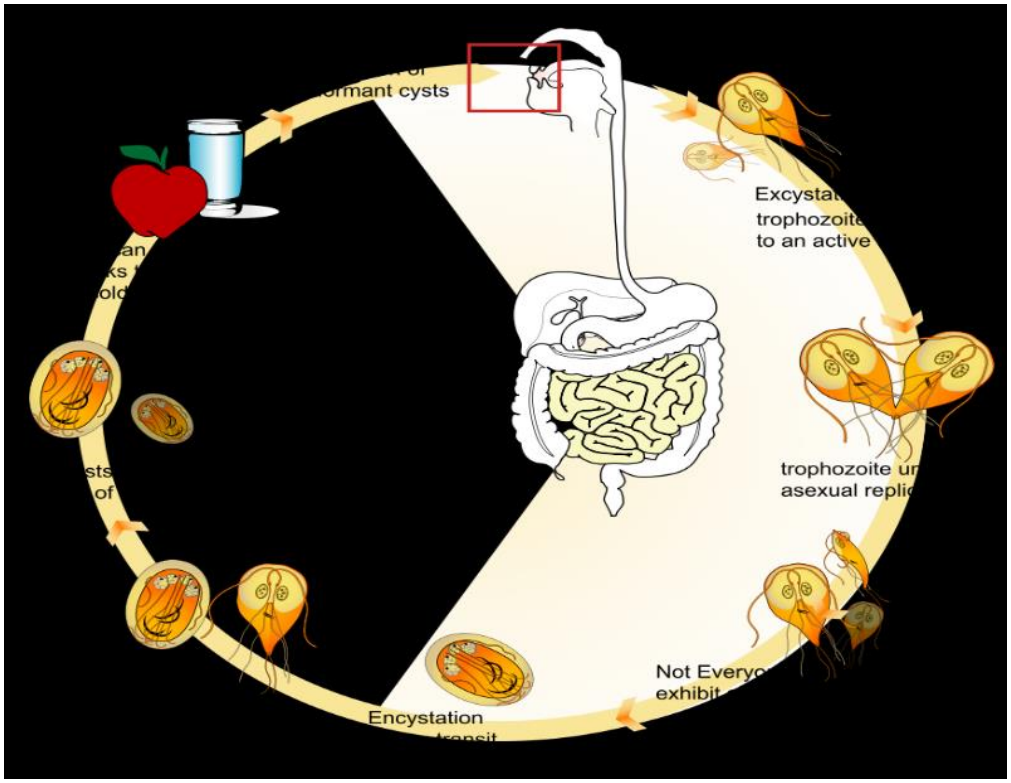
- A flagellate of class Mastigophora.
- The trophozoite has four pairs of flagella & two nuclei; giving it a face-like appearance.
- Grows in the intestinal tract.
- Cysts are shed in the feces.

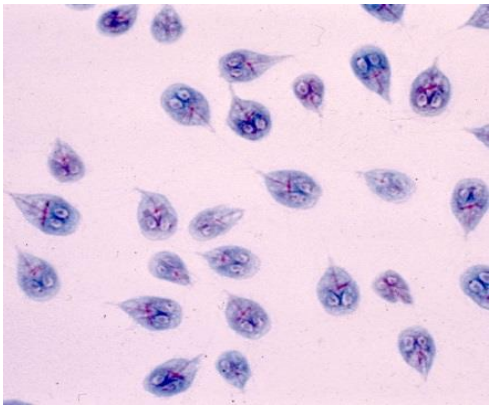
Mode of Transmission:

- Transmitted via contaminated water & food.

Symptoms:

- Foul-smelling profuse diarrhea.
- Sometimes chronic.
- Often misdiagnosed.
- Diagnosis via microscopic examination.
- Enterotest capsule.





4. **Balantidiasis:**

Causative agent: *Balantidium coli*

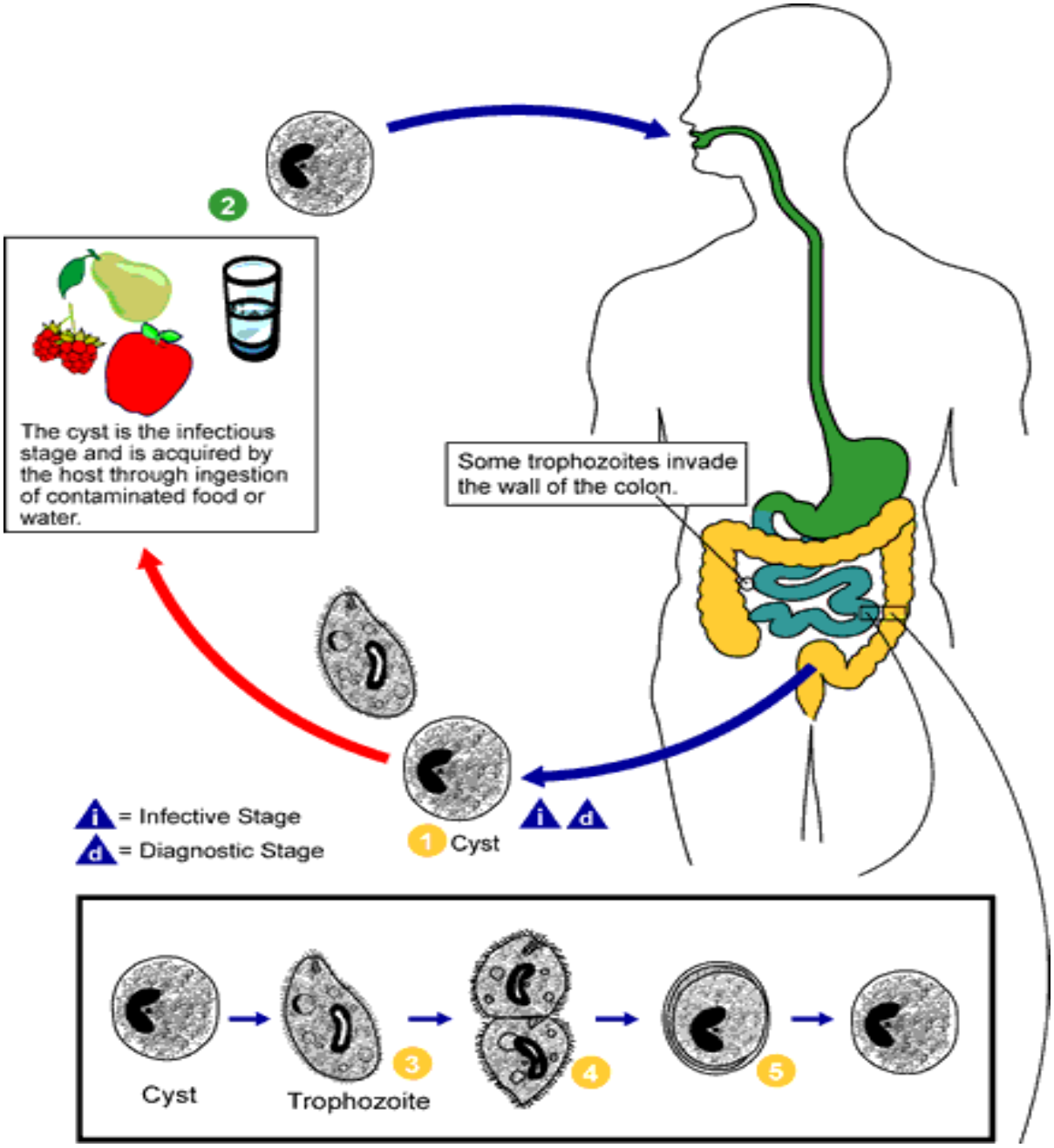
- Ciliated, in Class Ciliophora
- Trophozoites grow in the intestinal tract
- Cysts are shed in the feces and may remain embedded in intestinal walls, causing chronic infections

Mode of Transmission:

- Contaminated water & food

Symptoms:

- Ulceration in intestines
- Profuse diarrhea



5. **Toxoplasmosis:**

Cause: *Toxoplasma gondii*

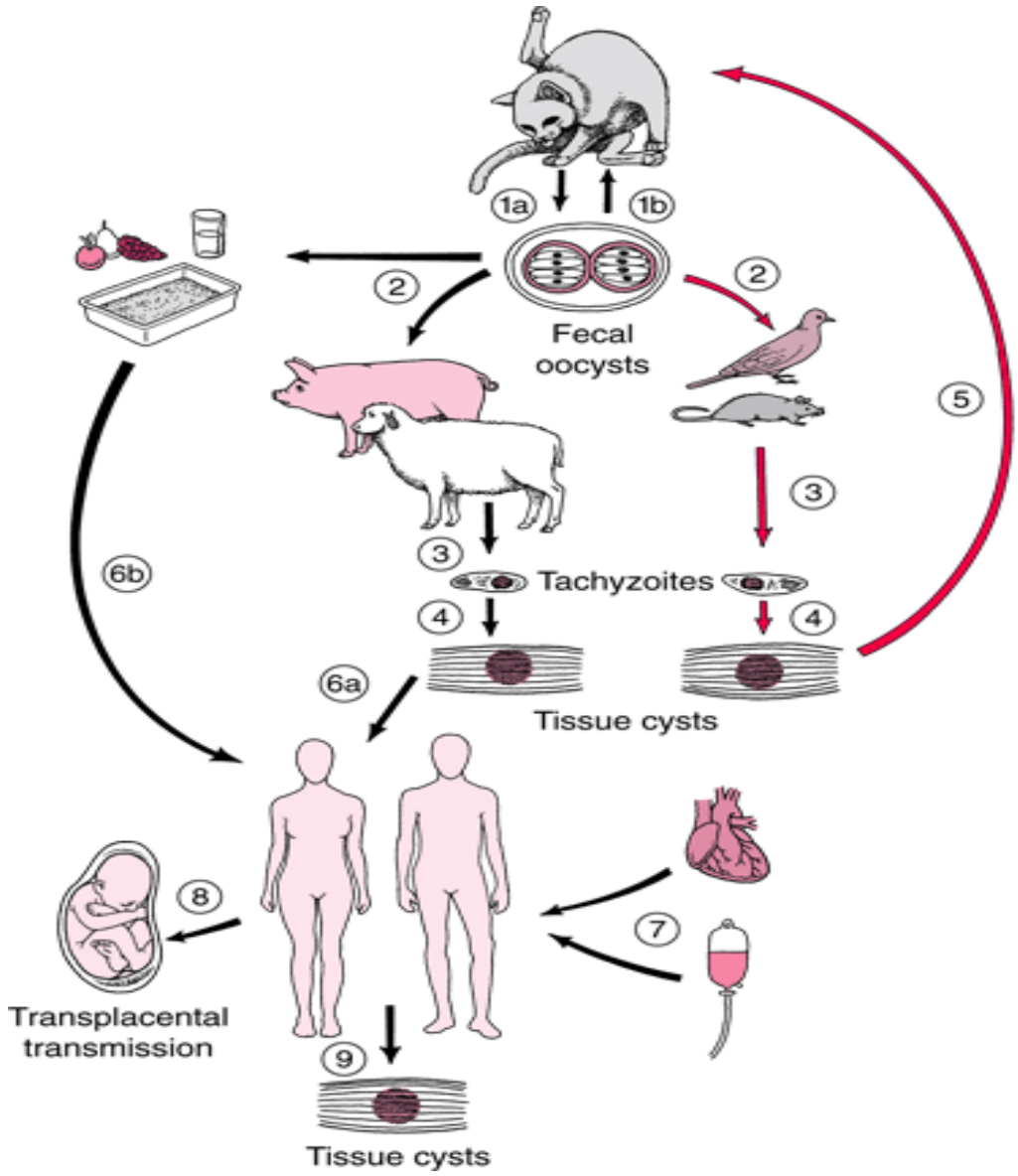
- In Class Sporozoa
- Invasion of several areas of the body, including internal organs & muscle tissue
- Possibly the most prevalent protozoan parasite of humans

Mode of Transmission:

- Transmitted via eating contaminated meat
- Cysts are found in the meat, especially of grazing animals such as cattle
- Cats can transmit the disease through their feces

Symptoms:

- Lymph node swelling; flu-like symptoms
- Generally mild in non-immunosuppressed persons
- Severe infections in immunosuppressed persons
- Pregnant women are in danger of fetal damage or miscarriage



6. **Malaria:**

Cause: *Plasmodium* species

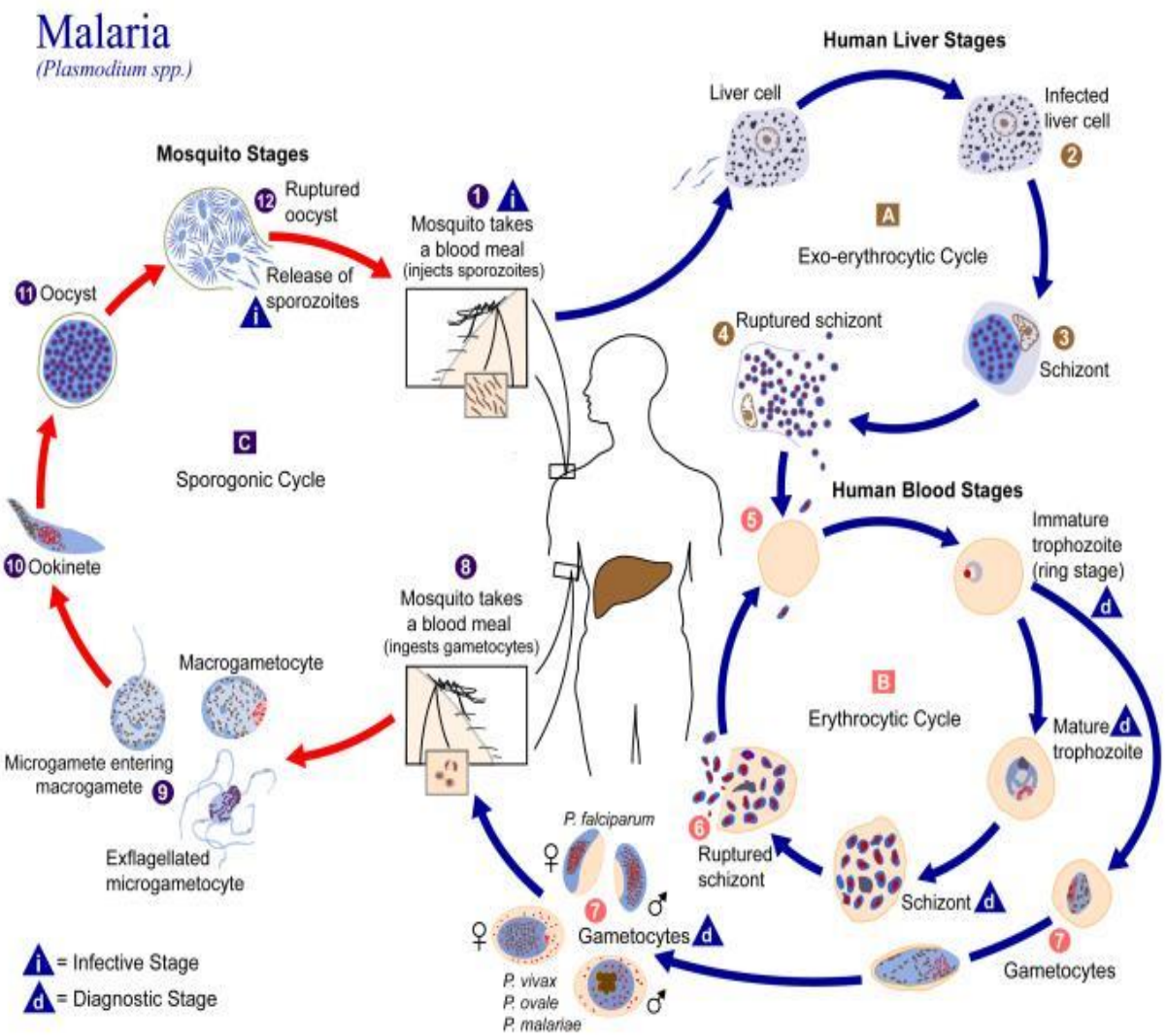
- In Class Sporozoa
- Complex life cycle requiring two hosts: a mammal (human) and the mosquito *Anopheles*

Mode of Transmission:

- The asexual merozoite stage infects human blood from the bite of a mosquito
- The parasite invades erythrocytes & reproduces.

Symptoms:

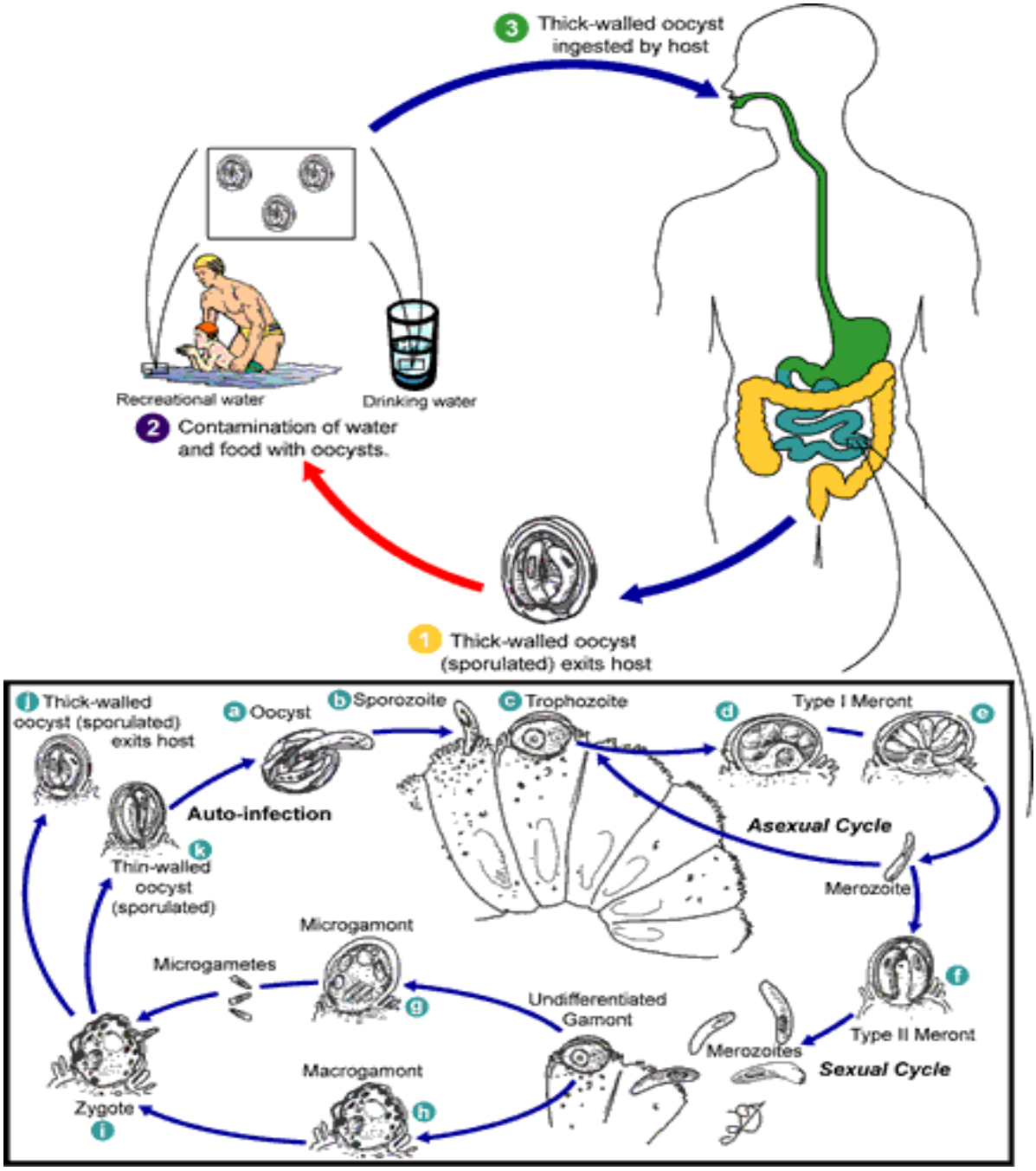
- Erythrocytes break open & release the parasite; this causes the characteristic fever, chills, anemia, weakness, hemorrhaging
- Intense fever & chills about 48 - 72 hr after exposure, due to the simultaneous rupture of so many erythrocytes
- The sexual stage of *Plasmodium* develops in the human host and is picked up by a mosquito
- The sexual stage is completed in the salivary gland of the mosquito to complete the cycle
- Treatment by quinine & other antimalarial drugs.



7. **Cryptosporidiosis**:
   Cause: *Cryptosporidium* species
   - In class Sporozoa
   - A common protozoan parasite in humans

   Mode of Transmission:
   - Contaminated water

   Symptoms:
   - Mild diarrhea in non-immunosuppressed persons
   - Severe diarrhea in immunosuppressed persons



8. **Pneumocystosis:**

Cause: *Pneumocystis carinii*

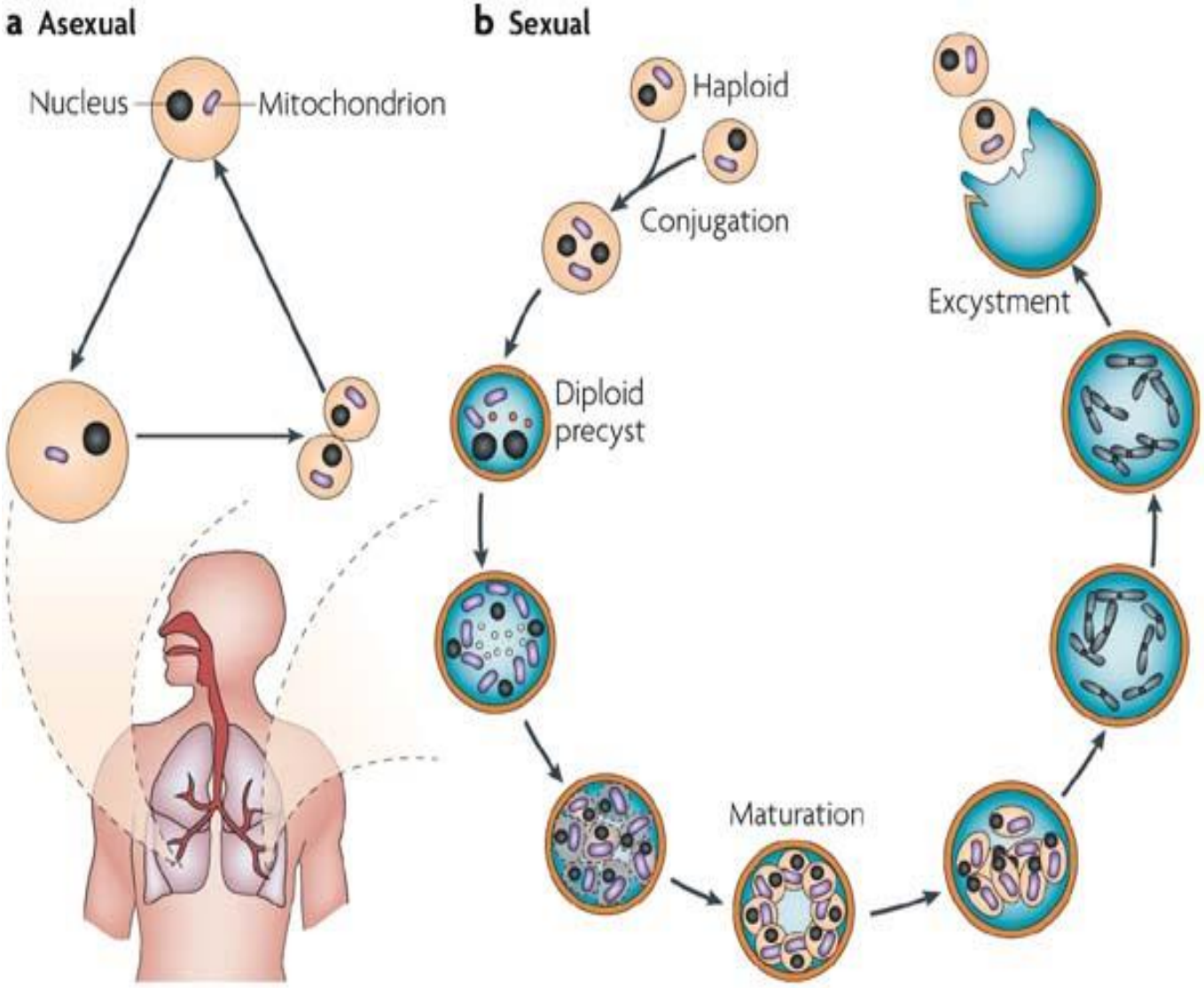
- Usually classified as a protozoan of class Sporozoa, but recent evidence suggests that it may be a fungus.

Mode of Transmission:

- Transmitted by airborne contact

Symptoms:

- Usually asymptomatic or with mild respiratory symptoms
- Causes severe pneumonia in immunocompromised persons
- A major secondary infection & cause of death in AIDS patients



9. Leishmaniasis:

Cause: *Leshmania*

Mode of Transmission:

- Transmitted through bites of infected phlebotomine sandflies which feed on blood to produce eggs.

Types:

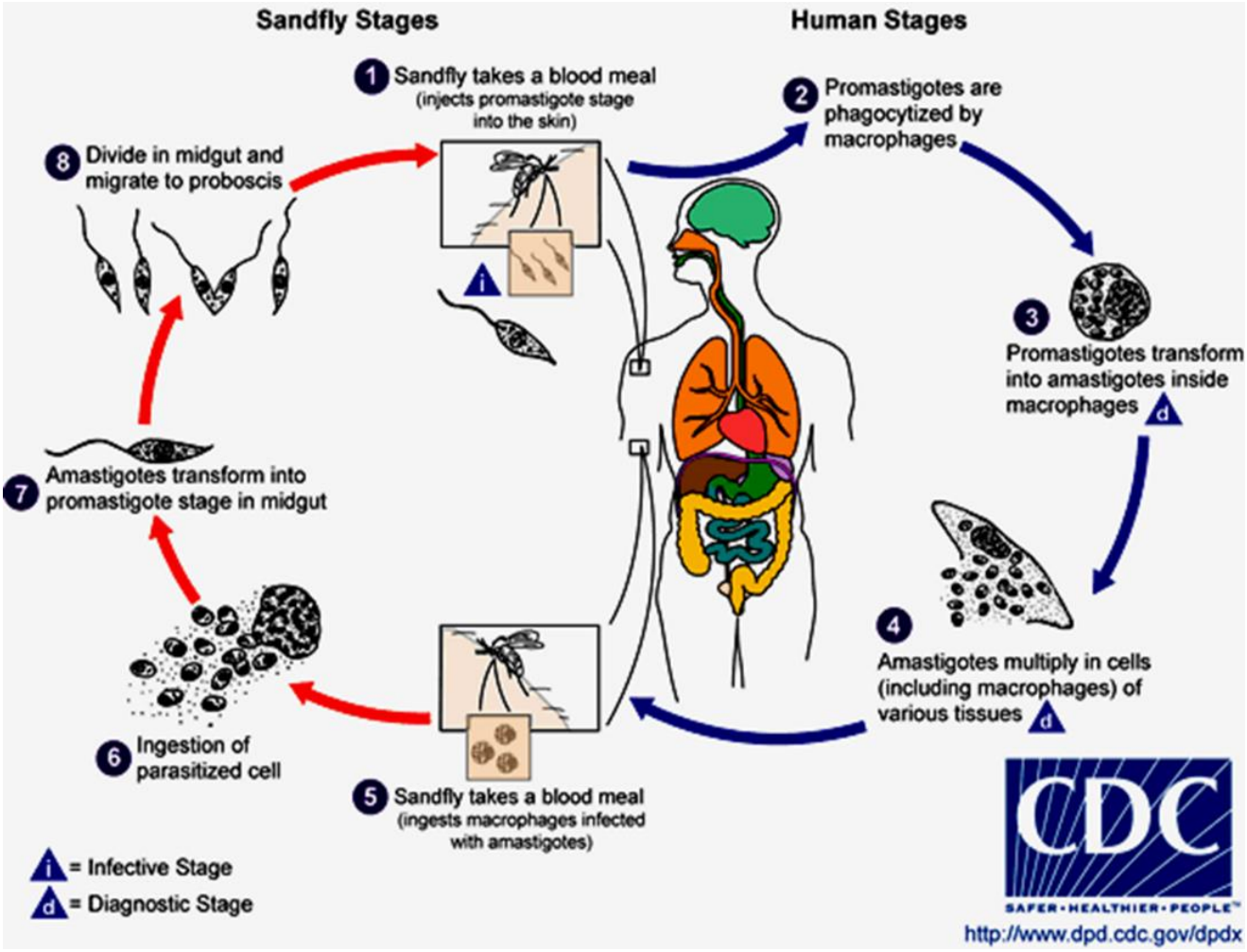
Leshmanais is of further two types:

- **Cutaneous leishmaniasis** affects the skin and mucous membranes.
- **Systemic or visceral leishmaniasis** affects the entire body. Is fatal if untreated.

Symptoms:

- Symptoms of cutaneous leishmaniasis depend on where the lesions are located and may include:
- Breathing difficulty.
- Skin sores, which may become a skin ulcer that healsvery slowly.
- Stuffy nose, runny nose, and nosebleeds.
- Swallowing difficulty.





FERMENTATION

CHAPTER NO. 09

Dr. Qazi Adnan

FERMENTATION

"The chemical breakdown of a substance by bacteria, yeasts, or other microorganisms, typically involving effervescence and the giving off of heat is called as fermentation."

- Anaerobic enzymatically controlled biological process in which organic compounds are converted into simple ones
- Effervescence takes place along with release of heat
- But now it is defined as large scale cultivation of microorganisms whether aerobic or anaerobic (without O2)

Types of Fermentation:

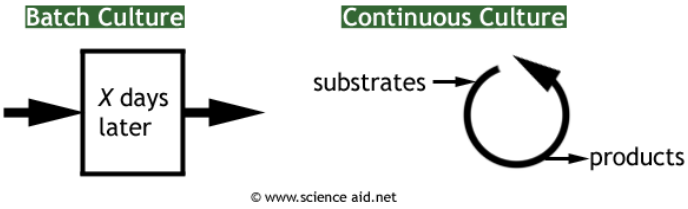
- ❖ Aerobic (in presence of O2)
- ❖ Anaerobic (in absence of O2)

- ❖ **Aerobic Fermentation:**
  - Adequate aeration
  - adequate supply of sterile air
  - Antibiotics, enzymes, vitamins.

- ❖ **Anaerobic Fermentation:**
  - In anaerobic fermentation, a provision for aeration is usually not needed.
  - Lactic acid, ethanol, wine

Types of Fermentation Processes:

- Fermentation could be of two following process types
  1. Batch mode
  2. Fed batch mode (continuous fermentation)



1. Batch fermentation:

- Most fermentations are batch processes
- Nutrients and the inoculum are added to the sterile fermenter and left to get on with it!
- Anti-foaming agent may be added.
- Once the desired amount of product is present in the fermenter the contents are drained off and the product is extracted.
- After emptying, the tank is cleaned & prepared for a new batch.

2. Continuous fermentation:

- Some products are made by a continuous culture system.
- Sterile medium is added to the fermentation with a balancing withdrawal of broth for product extraction.

ANTIBIOTICS BY FERMENTATION

- Of all the microbial products manufactured commercially, antibiotics are the most important.
- Antibiotics are chemical substances produced by microorganisms to kill or inhibits the growth of other microorganisms.
- They are used in the treatment of infectious diseases.

Production of Antibiotics

Antibiotics:

" Substance produced by one microorganism which in low concentration inhibit the growth of other microorganisms."

Sources:

1-Microorganisms
2-Semi-synthetic
3-Synthetic

1. **Microorganisms:**
   - E.g. Bacitracin and polymyxin from *Bacillus* species.
   - Streptomycin tetracyclines etc. from *Streptomyces* species.
   - Gentamicin from *Micromonospora purpurea.*
   - Penicillins and cephalosporins from certain genera (*Penicillium acremonium*) family Aspergillaceae.

2. **Semi-synthetic**:
   - Chemical synthesis that uses compounds isolated from natural source.
   - Usually done through fermentation process.
   - Product after fermentation is modified through chemical process to produce bulk antibiotics or Active Pharmaceutical Ingredient (API).
   - The bulk antibiotic or API is formulated into required dosage form e.g. tablets vials injections etc.
   - There are 2 stages for Semi-synthetic process
     - **Primary manufacture**: Stages up to API
     - **Secondary manufacture**: Stages from API to formulated product.

3. **Synthetic:**
   - Chemotherapeutic or antimicrobial agents of synthetic origin.
     E.g. Chloramphenicol is usually produced by Synthetic process.

Steps involved in Antibiotic preparation:

The steps involved in the preparation of antibiotics are:

1. **The organism:**
   - The original organism is required for the production of antibiotics.

2. **Inoculum preparation:**
   - The aim is to develop a pure inoculum in sufficient volume & fast growing phase so that high population density is quickly achieved.
   - Adequate oxygen is provided in the form of sterile air and temperature is controlled at desired level.

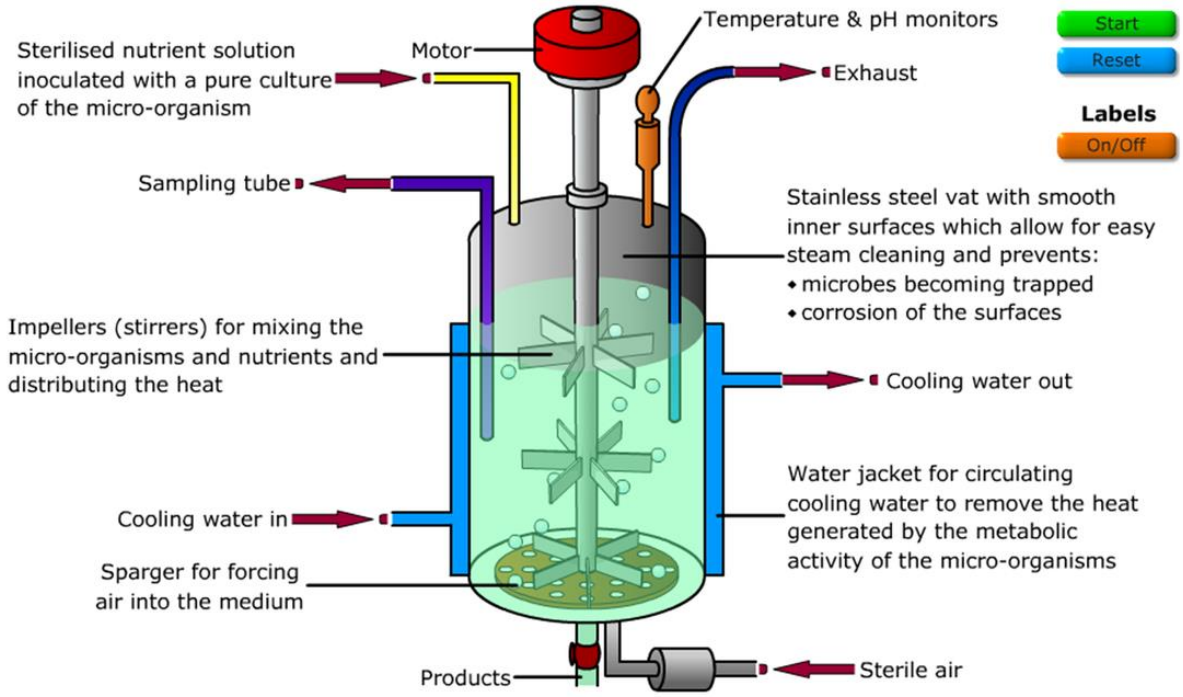
3. **Fermenter :**
   An apparatus used for fermentation process.

4. **Control of Fermentation:**
   Following are the conditions required for controlled fermentation.
   - **Batched medium:**
     The batch medium contains nitrogen sources and additional nutrients such as calcium magnesium, sulphate, phosphate trace metals.
   - **Fed nutrients :**
     Sucrose or glucose, molasses starch hydrolysate and rice etc.
   - **Termination.**

FERMENTER:

"An apparatus that maintains optimal conditions for the growth of microorganism used in large scale fermentation."

- A typical fermenter is a closed vertical cylindrical stainless steel vessel with convexly dished ends and of 25-250m^3 capacity .
- Its height is usually two to three times in diameter.



Process that takes place in fermenter:

- **Oxygen supply:**
  - Impellers and baffles assist the transfer of oxygen and promotes mixing of cells and nutrients.
  - Impellers are mounted on a rotating vertical stirrer shaft which is mechanically driven by a poweful electric motor.
- **Temperature control:**
  - Heat transfer is achieved by circulating chilled water through banks of pipes inside vessel.
  - Cooling water system is also used to cool batched medium.

- **Defoaming agents and instrumentation:**
  - Instrumentation provides display of important variables e.g temperature, pH & 0_2 consumed etc.
  - An analysis of exhaust gases provides physiological information.
- **Media addition:**
  - Media addition by fed batch process takes place.
- **Transfer and sampling system:**
  - Aseptic system are used to transfer the inoculum to vessel
  - Asepsis is ensured by engineering design and by steam.
  - Sampling is essential to monitor amount of growth and running level of key nutrients.

Some Antibiotics produced by Microorganisms

Antibiotic	Producing Microorganism
Cephalosporin	*Cephalosporium acrimoneum*
Chloramphenicol	*Streptomyces venezuelae*
Erythromycin	*Streptomyces erythreus*
Griseofulvin	*Penicillium griseofulvin*
Penicillin	*Penicillium griseus*
Streptomycin	*Streptomyces griseus*
Tetracycline	*Streptomycin aureofaciens*
Gentamycin	*Micromonospora pupurea*

Production of Penicillin By Fermentation

MOA of Penicillin:

- All penicillin like antibiotics inhibit synthesis of peptidoglycan, an essential part of the cell wall.
- They do not interfere with the synthesis of other intracellular components.
- These antibiotics do not affect human cells because human cells do not have cell walls.

Spectrum of Activity:

- Penicillins are active against Gram positive bacteria
- Some members (e.g. amoxicillin) are also effective against Gram negative bacteria but not *Pseudomonas aeruginosa*

Production of Penicillin:

- Penicillin was the first important commercial product produced by an aerobic, *submerged* fermentation
- First antibiotic manufactured in bulk.
- Used as input material for some semi synthetic antibiotics.
- It is fermented in a batch culture.

Primary Metabolites:

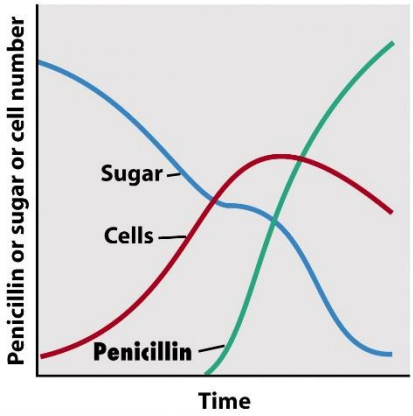
- Primary metabolites are produced during active cell growth

Secondary Metabolites:

- Secondary metabolites are produced near the onset of stationary phase.

Like all antibiotics, penicillin is a secondary metabolite, so is only produced in the stationary phase

Commercial Production of Penicillin:



Industrial Production of Antibiotic- Penicillin

- The industrial production of penicillin was broadly classified in to two processes namely,
  1. **Upstream processing**
  2. **Downstream processing**

1. Upstream Processing:

- Upstream processing encompasses any technology that leads to the synthesis of a product.
- Upstream includes the exploration, development and production.
- Inoculum Preparation:
  - The medium is designed to provide the organism with all the nutrients that it requires.
  - Inoculation method- submerged technique
  - Spores -major source of inoculum
- **Raw Materials:**
  - **Carbon Sources:**
    Lactose acts as a very satisfactory carbon compound, provided that is used in a concentration of 6%. Others such as glucose & sucrose may be used.
  - **Nitrogen Sources:**
    Corn steep liquor (CSL)
    Ammonium sulphate and ammonium acetate can be used as nitrogenous sources.
  - **Mineral Sources:**
    Elements namely potassium, phosphorus, magnesium, sulphur, zinc and copper are essential for penicillin production. Some of these are applied by corn steep liquor.
    Calcium can be added in the form of chalk to counter the natural acidity of CSL
    PAA- precursor
- **Fermentation Process:**
  - The medium is inoculated with a suspension of conidia of Penicillium chrysogenum.
  - The medium is constantly aerated and agitated, and the mould grows throughout as pellets.
  - After about seven days, growth is complete, the pH rises to 8.0 or above, and penicillin production ceases

2. Downstream Process:

- The extraction and purification of a biotechnological product from fermentation is referred to as downstream processing.

Stages in Downstream Processing:

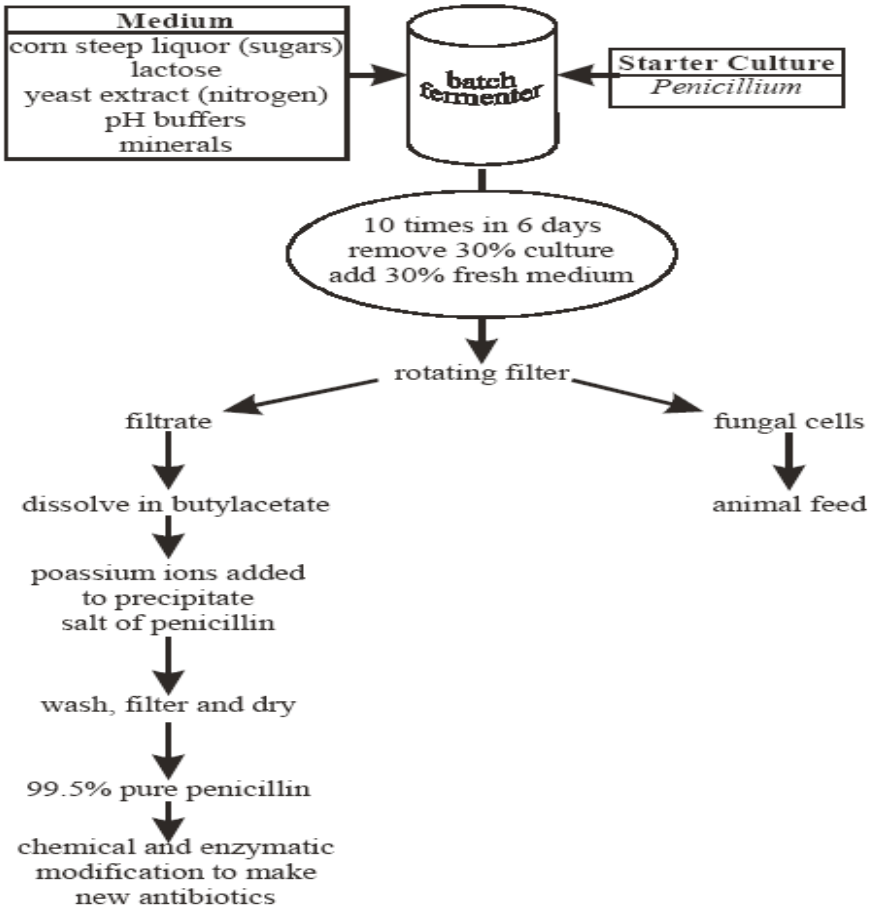
- **Removal of cells**
  - The first step in product recovery is the separation of whole cells and other insoluble ingredients from the culture broth by technique such as filtration and centrifugation.
- **Isolation of Benzyl Penicillin:**
  - The PH is adjusted to 2-2.5 with the help of phosphoric or sulphuric acids.
  - This step has to be carried out quickly for penicillin is very unstable at low pH values.
- Antibiotic is then extracted back into an aqueous buffer at a pH of 7.5, the partition coefficient now being strongly in favor of the aqueous phase. The resulting aqueous solution is again acidified & re-extracted with an organic solvent.
- These shifts between the water and solvent help in the purification of penicillin.
- The treatment of the crude penicillin extract varies according to the objective, but involves the formation of an appropriate penicillin salt.
- The solvent extract recovered in the previous stage is carefully extracted back with aqueous sodium hydroxide.
- This is followed by charcoal treatment to eliminate pyrogens and by sterilization.
- Pure metal salts of penicillin can be safely sterilized by dry heat, if desired. Thereafter, the aqueous solution of penicillin is subjected to crystallization.

3. Further Processing:

- For parental use, the antibiotic is packed in sterile vials as a powder or suspension.
- Searching tests (ex: for purity, potency) are performed on the appreciable number of random samples of the finished product.

The Main Stages of Penicillin Production are:

1. A medium of corm steep liquor (a by product of starch manufacture), yeast extract and other substrates are added in the fermenter.
2. After 40 hours, Penicillin begins to be secreted by the fungus.
3. The mould mycelium (cell matter) is filtered from the harvested product.
4. Penicillin is extracted in the organic solvent; butylacetate, in which it dissolves.
5. Potassium salts are added and a penicillium precipitate is formed. This is washed and dried.



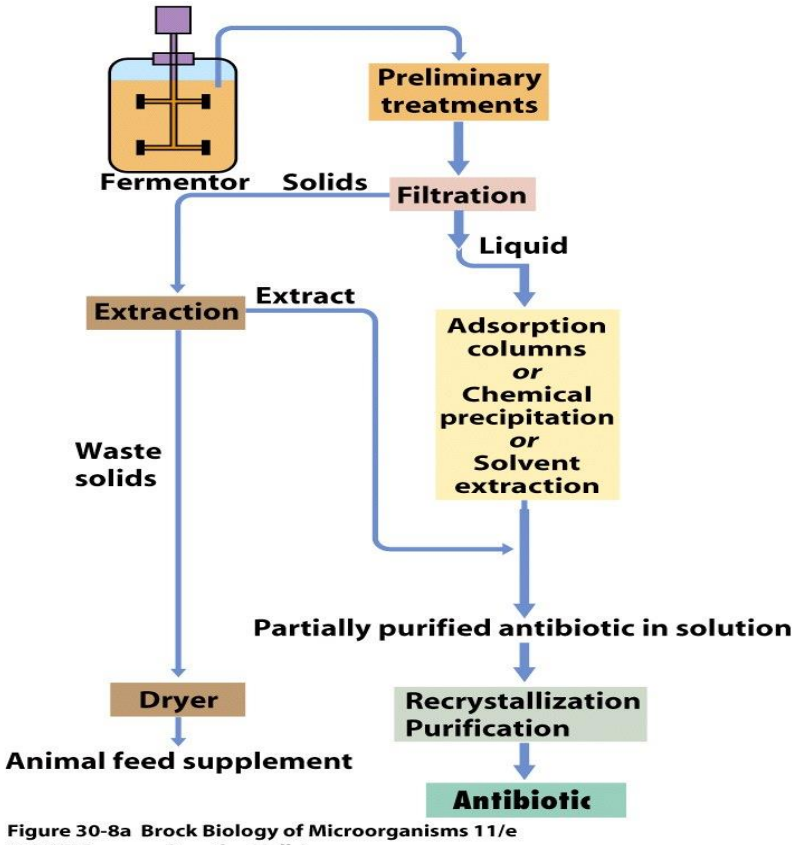


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Production of Penicillin V:
- Phenoxy methyl penicillin
- Addition of different Acyl groups to the medium.
- Phenoxyacetic acid as precursor instead of phenyl acetic acid.

Penicillin:
- Organisms: *Penicillium chrysogenum*
- Fed nutrient: glucose

Streptomycin:
- Organism: *Streptomyces griseus*
- Aerated submerged fermentation.
- Carbon source: starch dextrin & glucose
- Nitrogen source: soyabean meal & corn steep liquor

Tetracyclines:
- Organism: *Streptomyces aureofaciens*

Gentamicin:
- Organism: *Micromonospora purpurea*

Cephalosporin:
- Organism: *Cephalosporium acremonium*

Chloramphenicol:
- Organism: *Streptomyces venezuelae*

Erythromycin:
- Organism: *Streptomyces erythreus*

Griseofulvin:
- Organism: *Penicillium griseofulvin*

Regards: Noor Mobin